Asset Allocation Through Retirement in New Zealand: A Simulation and Optimization Approach

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Abstract

This report examines lifecycle asset allocation for retirement in New Zealand, with a focus on the decumulation phase and its implications for advisers and policy design. We develop and test a simulation-optimisation framework calibrated to New Zealand institutional settings, including New Zealand Superannuation and KiwiSaver, using Monte Carlo simulation of investor outcomes and gradient-based policy optimisation. Across model settings, we find that rising-equity glidepaths can emerge as optimal in aggregate, but that glidepaths are generally an ineffective policy class for heterogeneous retiree populations. Wealth-responsive and jointly time-wealth-responsive policies consistently outperform generic glidepaths, which suggests that investor age alone is an informationally sparse basis for retirement allocation decisions. Glidepaths perform well mainly when they are tightly tailored to a known individual profile. We also find that recommended allocations are highly sensitive to modelling assumptions, especially the return-generation process and inflation dynamics, with materially different asset mixes across geometric-Brownian and block-bootstrap generators. Taken together, the analysis provides a strong framework for future work on asset allocation policy design, and supports personalised strategy frameworks over one-size-fits-all age-based rules.

1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (hereafter, Consilium). Consilium is a New Zealand-based financial services company that provides tools and consulting services to financial advisers and their clients in areas including investment management, portfolio construction, risk management, and financial planning. Consilium also offers managed funds and portfolio solutions designed to support advisers in meeting a wide range of client needs. The aim of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with particular focus on retirees in the New Zealand context. Our objective is to produce insights and recommendations that may help Consilium better support its tools, services, and clients in making informed and effective decisions about asset allocation through retirement. In this report, we present the findings of our research, which includes a review of the relevant literature, the development and implementation of a simulation and optimization framework for evaluating asset allocation strategies, and the analysis of results to derive practical implications for investors and financial advisers.

Although this research is conducted in partnership with Consilium, the views expressed in this paper are the author's and do not necessarily reflect the official policy or position of Consilium. The author retains full academic independence in the design, analysis, and reporting of this work. The author also retains full ownership of the intellectual property associated with this research, while granting Consilium a non-exclusive licence to use the findings internally and share them with clients and partners. No confidential or proprietary methods or information provided by Consilium has been disclosed in this study.

We thank Consilium for their support and collaboration in this research project, and for the feedback that has helped shape the direction and focus of this work. We would also like to acknowledge the support of Ian Perera from the New Zealand Society of Actuaries (NZSA) for his interest and feedback on this research. Any errors or omissions in this report are the responsibility of the author alone.

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2 Introduction

Asset allocation is the process of determining how to distribute an investor's wealth across different investments so that the portfolio meets the investor's objectives and risk tolerance. In general, sources of higher asset returns are associated with higher risks (e.g., (Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007)). The challenge is to find the right balance between risk and return, taking into account the investor's time horizon, financial goals, and personal circumstances. Lifecycle asset allocation further complicates this problem by introducing the idea that asset allocation should change over time in response to the investor's evolving needs and circumstances. This is particularly relevant for retirement planning, where the investor's financial situation and risk tolerance may change significantly as they transition from working life to retirement, and as they experience changes in health, longevity, and spending needs.

The global asset-backed pension market is now extremely large, at over 69.8 trillion USD at the end of 2024 (OECD, 2025). Western countries are facing a demographic shift with an ageing population, which has significant implications for retirement planning and asset allocation. In New Zealand, the retirement landscape is shaped by the interaction of public and private retirement savings, with New Zealand Superannuation providing a universal pension to eligible retirees, and KiwiSaver serving as a voluntary, work-based savings scheme. The combination of these two systems creates unique challenges and opportunities for retirement planning in New Zealand, particularly in terms of how individuals should allocate their assets to achieve their desired retirement outcomes. Treasury projections indicate substantial long-run pressure from population ageing (The Treasury, 2025). This has lead to a growing interest in the role of KiwiSaver in retirement planning, and the need for better guidance on how to manage retirement savings effectively (Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025).

Broadly speaking, the investor's lifecycle can be divided into two phases: accumulation and decumulation. During the accumulation phase, the investor is typically working and saving, building up their wealth over time. When they transition to retirement, they enter the decumulation phase, where they draw down on their accumulated wealth to fund their spending needs. This report will focus primarily on the decumulation phase, but the accumulation phase is also relevant for understanding the optimal trajectory of risk over the lifecycle, and so we will spend some time discussing the literature on both phases, especially as it relates to the shape of the optimal glidepath (the trajectory of risk over time).

This report will also principally focus on the optimal allocation of capital to three broad asset classes: equities, bonds, and cash. Although some investors and advisors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes. Therefore, allocation to these asset classes is arguably the primary driver of a portfolio's long-term behaviour. Furthermore, the types of investment products that are typically used for retirement investing in New Zealand (e.g., KiwiSaver, managed funds, and other pooled investment vehicles) are usually structured around these broad asset classes, rather than individual securities. The investor's typically excerice little control over the specific securities held in their portfolio, and instead choose from a limited set of pre-defined strategies that allocate to different amounts of "Growth" (i.e equities) and "Income" (i.e. Bonds, Cash) assets as a proxy for the riskiness of the portfolio. These asset classes will be constructed similarly to the way they are in Consilium's model portfolios, with Growth assets consisting of a diversified mix hedged and unhedged global equities, and Bonds and Cash consisting of a diversified mix of hedged global bonds and domestic bonds and cash.

The goal of this report is to provide a review of the literature on lifecycle asset allocation and the retirement settings in New Zealand. We will then develop a simulation and optimization framework to evaluate different retirement asset allocation strategies. This simulation and optimisation framework is the key contribution of this report, and will allow us to test the performance of different strategies under a range of assumptions and scenarios. We will demonstrate the flexibility of this framework by applying it a hypothetical New Zealand retiree household, and use it to evaluate the performance of different asset allocation strategies. Finally, we will

2.1 The Lifecylce Model of Wealth and Lifetime Asset Allocation

A foundational concept in this area is the lifecycle model of wealth. Developed by Modigliani and Brumberg (2005) and expanded by Merton (1969), the lifecycle model posits that individuals make consumption and saving decisions over their lifetime to smooth consumption in response to changes in income, wealth, and spending needs. In essence, the lifecylce model suggests that individuals should save during their working years but only insofar as they need to meet their consumption needs in retirement, and that they should draw down on their accumulated wealth in retirement to fund their spending needs. The optimal wealth "trajectory" over the lifecycle is therefore a hump-shaped curve, with wealth peaking around the point of retirement and then declining as the individual draws down on their savings to fund their spending needs. The perfect lifecycle model is a theoretical construct, and in practice, individuals face a range of uncertainties and constraints that can affect their ability to save and spend optimally over their lifetime. These include uncertainty about future income, health, longevity, and investment returns, which we will return to later in this report. This lifecycle model introduces us to two key risks that are particularly relevant for retirement planning. The first is longevity risk, which is the risk that an individual will outlive their savings and be unable to fund their spending needs in retirement. The second is the opposite risk, which is the risk that an individual will die before they have fully consumed their accumulated wealth. In the first case, the individual may face financial hardship in later life, while in the second case, the individual may have endured a lower standard of living than they could have enjoyed had they been able to spend more of their wealth during their lifetime.

A well performing allocation strategy is one that maximises the investor's expected welfare, which is a function both of the expected return and the protection against adverse outcomes. Where the investor's needs are fixed, Merton (1969) shows that the optimal risky-asset share is constant through time. There are many simplifying assumptions in this result, but probably the most significant for lifecycle investing is that the investor's spending needs are fixed proportion of the investor's wealth, such that the investor adjusts their spending in response to changes in wealth. This is a strong assumption, and it is often not true in practice. In reality, this is not true as diferential longevity risk, health shocks, and changing preferences all mean that spending needs are uncertain and evolve over time. Futhermore, this fixed-proportion of wealth does not allow the investor to optimally consume according to the lifecycle model, which wants to see the investor optimally draw down on their wealth to zero at the end of their life.

Conventional wisdom suggests that the optimal risky-asset share should decline with age, as the investor's wealth peaks and they approach retirement. An explaination for this is that the investor's risk tolerance declines with age, as they have less time to recover from potential losses. Bodie et al. (1992) arrives at this conclusion through a lifecycle model of wealth and consumption, which incorporates the idea of human capital. Human capital is the present value of an individual's future labor income, and Bodie et al. (1992) treats human capital as a bond-like asset, which implies that younger households can rationally hold more a larger proportion of their portfolio in risky, high-return assets, and then gradually reduce this proportion as they approach retirement age (and their human capital declines). Based in a understanding of these principles, a large number of intuitively appealing, easy-to-communicate age-based rules have emerged that have become the industry standard for retirement investing. These include the various age-in-bonds rules (e.g. the 120-minus-age rule), and target-date funds which are investment products that automatically adjust the asset allocation over time according to a predetermined schedule. This asset allocation trajectory over a lifetime is often referred to as a "glidepath". This idea that risk should decline with age and that the glidepath should be declining is widely accepted in the financial industry, and has been developed into a myriad of financial products, typically known as "target-date funds" (TDFs). A local example is the SuperLife Age Steps strategy, which automatically adjusts the allocation to Growth and Income assets based on the investor's age, with the proportion of Growth assets declining as the investor gets older. The asset allocation is shown in Figure 1.

Although widely recommended, the optimal shape of the glidepath is a subject of active debate in the literature, with different studies reaching different conclusions about whether risk should decline, rise, or remain constant over time, and whether the optimal strategy differs between the accumulation and decumulation

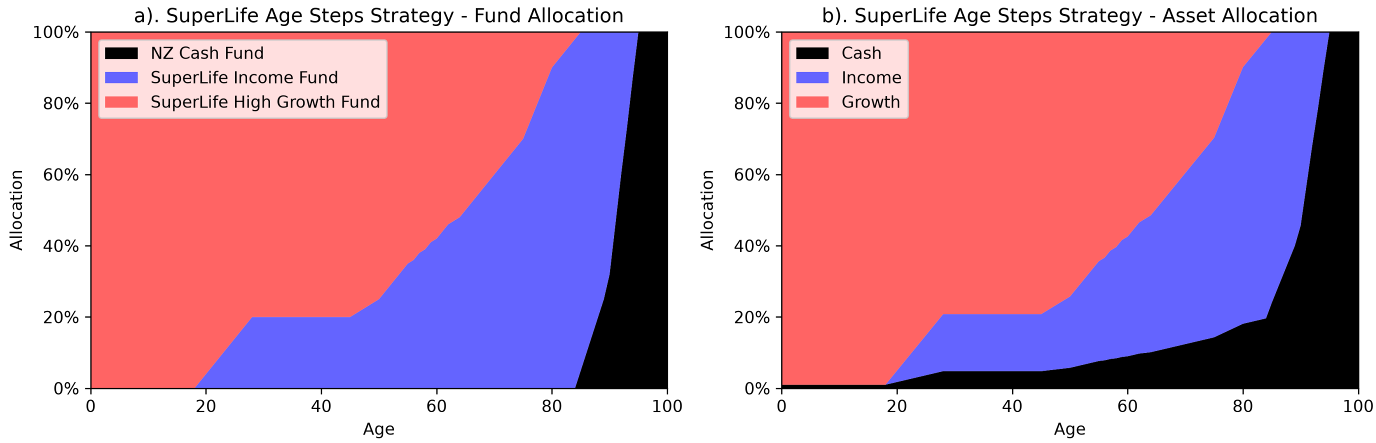

Figure 1: SuperLife's Age Steps target strategy as per their Statement of Investment Policy and Objectives (SIPO), showing the allocation to different funds (a) and the resulting allocation to different asset classes (b). Source: SuperLife (2026).

phases. What emerges from the literature is a complex and unresolved debate about the optimal shape of the glidepath, with different studies reaching different conclusions about whether risk should decline, rise, or remain constant over time, and whether the optimal strategy differs between the accumulation and decumulation phases. There are essentially five main shapes that have been proposed for the optimal glidepath (see Figure **??**):

- Declining equity (DE): the conventional wisdom that risk should decline with age, often implemented as a simple age-based rule.
- Rising equity (RE): the contrarian view that risk should actually increase with age, particularly at the start of retirement, to capture the higher expected returns of equities when they are most needed.
- U-shaped: declining equity during the accumulation phase, followed by rising equity during retirement.
- Inverted U-shaped: rising equity during the accumulation phase, peaking at retirement, followed by declining equity during retirement.
- Static allocation: the view that a constant equity share throughout life is optimal, and that the shape of the glidepath is less important than the overall level of risk taken.

The standard industry recommendation is that investors decrease their risk exposure as they age, continuing to reduce risk even after retirement (DE). This approach appears in investment textbooks (e.g Bengen, 1996; Bodie et al., 2023) and is widely recommended by financial advisers and retirement planning tools (Choi, 2022). A review of U.S target date funds (TDFs) by Dolvin et al. (2010) find that most follow some form of declining-risk structure, often resembling a 120-minus-age equity rule. For a domestic example, take the SuperLife Age Steps allocation profile shown in Figure 1. At one point a declining-risk glidepath was considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (see R. Shiller, 2005; R. J. Shiller, 2006).

However, declining-risk glidepaths have been challenged by a substantial body of academic work, and the post-retirement period in particular is a site of active and unresolved disagreement in the literature. Responding to the aforementioned Bush-era proposal, R. Shiller (2005) and R. J. Shiller (2006) find that declining-equity glidepaths often fail to outperform a 100% equity strategy. Further work by Estrada (2014, 2015) finds that declining-risk strategies tend to underperform constant-risk alternatives or even contrarian rising-risk strategies (RE, U-shaped) across a wide range of countries and historical periods.

The reasons for this disagreement are complex, but a key fault seems to lie line in the choice of methodology for generating return scenarios and evaluating strategies. Different studies use different approaches, including historical rolling periods, Monte Carlo simulation with IID returns, and block bootstrap methods that preserve time-series dynamics. These methodological choices have a significant impact on the results and conclusions drawn about the optimal glidepath shape. Studies generating return scenarios using Monte Carlo simulation with IID returns tend to find support for U-shaped glidepaths with rising equity in retirment. This means a declining-risk strategy during the accumulation phase with reduced equity exposure around the point of

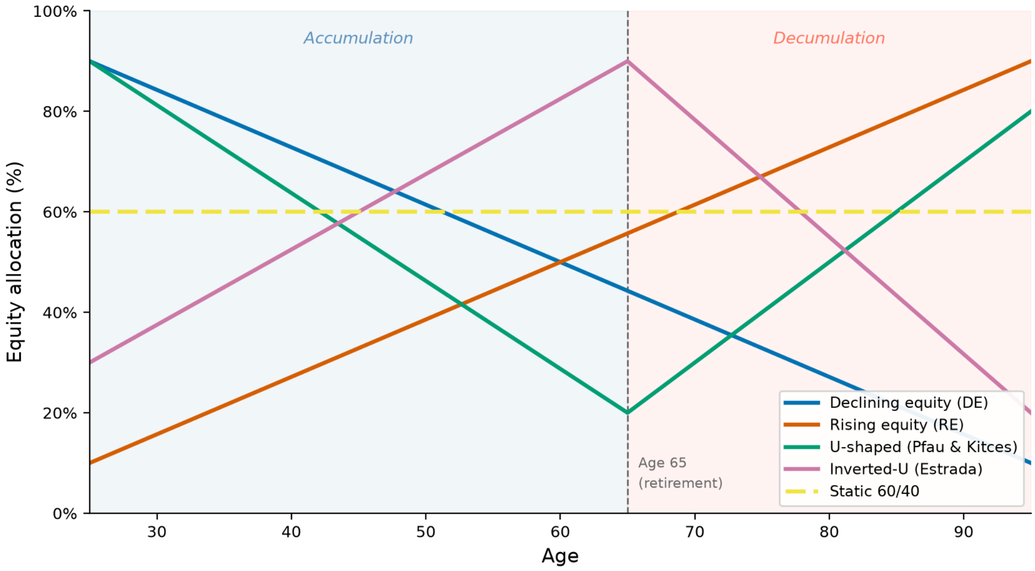

Figure 2: Illustration of different proposed glidepath shapes. The x-axis represents age, with the vertical dashed line indicating the point of retirement. The y-axis represents the equity share of the portfolio. The different lines represent the different proposed glidepath shapes: declining equity (DE), rising equity (RE), U-shaped, inverted U-shaped, and static allocation.

retirement, and then rising again during the retirement/decumulation phase. Pfau and Kitces (2014) make the case for rising equity during retirement using Monte Carlo simulation. Across 10,000 simulated retirements with lognormal IID returns, 121 glide path combinations (varying start and end equity exposure from 0% to 100%), they find that rising equity during retirement reduces both the probability and magnitude of portfolio failure. The optimal starting allocation is 20–40% equity, rising to 60–80% by end of retirement. In their analysis, a 30–60% rising path beats a static 60/40 on failure rate and failure magnitude. They report that this holds across three capital market assumption sets and argue that retirement outcomes are almost entirely determined by the first 15 years. They argue that a RE strategy during retirement limits the damage from a bad early period and then systematically accumulates equities at depressed prices through the downturn.

The rationale is that retiring with a conservative asset allocation limits the damage from a bad market crash. This is because, as in the lifecycle model, the investor is assumed to be at "peak-accumulation" at the point of retirement, and so the portfolio is at its largest in dollar terms. A market crash at this point would have a large impact on the portfolio, and a conservative allocation would limit the damage. This market-risk mechanism is often called "sequence-of-returns risk" in which the investor experiences a stochastically "unlucky" sequence of negative returns early in retirement.

Proponents of a U-shaped glidepath further argue that the same logic applies in reverse during retirement: starting conservatively at the point of retirement limits the damage from a bad early period, while systematically accumulating equities at depressed prices through a downturn allows the portfolio to capture the eventual recovery when growth is most needed. Anarkulova et al. (2025) use a block-bootstrap methodology applied to a dataset of monthly real returns across 39 developed countries from 1890 to 2023, drawing on over 31,800 country-months of data and preserving the time-series and cross-sectional structure of historical returns. Under block bootstrap, the optimal allocation is approximately 100% static equity throughout the lifecycle. The sole exception is a brief tactical allocation to bills at the point of retirement, which does resemble a kind of aggressive U-shaped glidepath, but they argue that this is a special case, driven by the rigidity of a fixed real 4% withdrawal rule (based on Bengen, 1994) and when spending flexibility is restored, this exception disappears and the all-equity result holds at every age. They also argue that the conventional preference for bonds is substantially overstated by return-generation decisions in the literature. They compare optimal glidepaths under IID return sampling and block bootstrap, and find that the optimal bond allocation is higher under IID assumptions. They argue that this is because the long-horizon properties of bonds are much less favourable than their short-horizon statistics suggest. At 30-year horizons, bond variance *increases* relative to the one-year level (variance ratio 2.30), whereas equity variance *decreases* (variance ratio 0.75). Bonds also carry a correlation of -0.78 against inflation, which they argue destroys real purchasing power in precisely the environments most harmful to retirees. In essence, Anarkulova et al. (2025) argue that IID assumptions overstate the risks of sequence-of-return risk. In reality, timeseries effects mean the sequence is actually autocorrelated and mean-reverting. In this environment, the higher expected returns of equities are more reliably captured over long horizons and the risks of bonds are more pronounced.

Further challenging the conventional wisdom, Estrada (2014) find that lifecycle strategies that move against the declining-equity convention can outperform them. Using historical asset-return data across 19 countries over a 110-year period, they show that strategies that increase equity exposure as retirement approaches can reduce failure rates relative to conventional declining-equity glidepaths, although they also generate greater dispersion in terminal wealth. Their companion paper, Estrada (2015), applies the same historical dataset and rolling-window methodology to the decumulation phase, testing 81 rolling 30-year retirement periods with a 4% real withdrawal rule. In this setting, the result reverses: declining-equity strategies deliver both lower failure rates and higher terminal wealth than rising-equity strategies in every specification. This stands in contrast to Pfau and Kitces (2014), whose Monte Carlo simulations with IID returns conclude that a rising equity glidepath is optimal during retirement.

The divergence is not merely a disagreement over the sign of the glidepath; it reflects a more fundamental methodological split. Estrada's findings are based on historical rolling periods, whereas Pfau and Kitces rely on Monte Carlo simulations with IID return draws. In this respect, the two studies are not simply asking the same question under different parameter values. They are evaluating different return-generating processes, and those processes impose very different assumptions about the timing, persistence, and recovery of shocks. The important point is that, despite the disagreement over the optimal direction of movement in equity exposure during retirement, both strands of evidence tend to support a static allocation as a robust baseline. In both Estrada's historical framework and Pfau and Kitces' simulation framework, a roughly 60/40 portfolio is close to optimal, and in both cases it performs materially better than many more complex dynamic rules.

This conclusion is reinforced by Blanchett (2007), who tests 43 glidepaths across 1,071 scenarios using bootstrapped monthly return data from 80 years of US history. His results show that static allocations have the lowest failure probability in all but one scenario. The best static portfolio by raw failure probability is 100% equity, while the best risk-adjusted portfolio is a 60/40 stocks-bonds allocation under his "Success to Variability Ratio". The broader implication is that, once risk is measured in a realistic way, glidepath dynamics may matter far less than the overall level of equity exposure.

Adding further complexity to the debate, Duarte et al. (2021) use deep reinforcement learning to solve a lifecycle model with both accumulation and decumulation phases, incorporating shocks to health, longevity, and spending needs. Their model features autocorrelated return dynamics, and they find that the optimal equity allocation at retirement is approximately 60%. Target-date funds, which typically land at 30–40% equity in retirement, are therefore too conservative in their framework. However, they also find that the post-retirement allocation is broadly flat rather than strongly rising or falling, and that a declining trajectory of equity exposure over the lifecycle is generally optimal. This again points away from a strongly path-dependent glidepath and toward a more stable long-run equity share.

The practical significance of these findings is also highly context-dependent. Blanchett (2007) shows that the choice of glidepath matters little when withdrawal rates are moderate and retirement horizons are short. At a 4% withdrawal rate over a 20-year period, the range of failure probabilities across all 43 strategies is only 1.56 percentage points. By contrast, the spread widens sharply under more adverse assumptions: at a 6% withdrawal rate over 40 years, it reaches 62.45 percentage points. This suggests that the glidepath debate is most consequential in aggressive drawdown scenarios, while in more moderate settings the difference across strategies may be relatively small.

The larger lesson is that disagreement in the literature is driven as much by methodology as by economics. Historical rolling periods preserve the actual structure of return sequences, including recession clustering and mean reversion, while IID Monte Carlo simulations impose a much simpler and more symmetric return process. Block bootstrap methods, such as those used by Anarkulova et al. (2025), sit between these extremes by preserving some time-series dependence while generating a larger synthetic sample. In each case, the resulting policy recommendations change materially. Anarkulova et al. (2025) find that a near-100% equity strategy is optimal under block bootstrap, whereas Pfau and Kitces (2014) recommend a rising glidepath under IID simulation, and Estrada (2015) favour declining equity under historical rolling periods. These differences arise partly from return-generation assumptions, but also from the choice of objective function. Failure-rate minimisation, terminal wealth maximisation, risk-adjusted success ratios, and expected utility with explicit risk aversion each shift the optimum in different ways. As a result, no single glidepath emerges as universally dominant across methods.

Taken together, the literature suggests that the optimal glidepath is highly sensitive to assumptions about returns, spending, and risk preferences. Yet a more robust conclusion emerges: the overall equity share appears to matter more than the precise shape of the path, and static or near-static allocations often perform surprisingly well. This is especially true when one considers the practical limitations of implementing complex glidepaths in real retirement portfolios. In that sense, a simple and stable allocation may be superior to a more elaborate age-based rule, even if the literature remains divided on the precise direction of any age-related adjustment.

Finally, the specification of the investor's spending and income needs is a critical driver of optimal policy. In discussing the paper by Anarkulova et al. (2025), Felix (2025) argues that the real risk to investors is not the risk of portfolio failure per se, but the risk of being unable to adjust spending in response to market conditions.

2.2 Consilium's Position

Consilium does not have an official position on the optimal glidepath shape nor does it currently offer any specific guidance or products that implement a particular glidepath strategy. Consilium's approach to retirement planning is focused on providing tools and insights that help financial advisers and their clients make informed decisions about asset allocation and withdrawal strategies during retirement. However, an adviser using Consilium's official products and recommendations could construct a glidepath strategy themselves by combining the recommendations from two different areas of Consilium's offering: the Model Portfolios and the Goal-Positioning Software (GPS). For this reason, much of the analysis in this report will be contextualised with reference to these two areas of Consilium's offerings.

"Model Portfolios" are predesigned asset allocation strategies for different risk profiles following Consilium's proprietary asset allocation methodology, which is based primarily on the Fama-French three-factor model (Fama & French, 1993). There are 42 model portfolios across five families of strategies: Classic, SRI, Synergy Classic, Synergy SRI, and Synergy PIE. These portfolios are designed to cater to a range of investor preferences and risk tolerances, with varying allocations to Growth and Income assets. A glidepath approach could be constructed from these portfolios by tatically switching between them at different ages, but currently there is no official recommendation or guidance on how to do this, and the portfolios themselves are not explicitly designed to be used in this way. The availability of portfolios across risk levels by family is shown in Table 1.

Table 1: Availability of Consilium model portfolios across risk levels by family

Risk Level (Growth/Income)	**Synergy PIE**	**Synergy Classic**	**Synergy SRI**	**SRI**	**Classic**
20/80	-	x	x	x	x
30/70	x	x	x	x	x
40/60	x	x	x	x	x
50/50	x	x	x	x	x
60/40	x	x	x	x	x
70/30	-	x	x	x	x
80/20	x	x	x	x	x
90/10	-	x	x	x	x
98/02	x	x	x	x	x

The Goal-Positioning Software (GPS) is a tool that helps advisers and their clients set and track progress toward financial goals, including retirement. It provides personalised projections of future wealth and income based on the client's current and expected contributions, asset allocation (to one of the model portfolios), and other factors. The GPS allows users to simulate different scenarios and see how changes in their savings rate, asset allocation, or other assumptions might affect their ability to meet their retirement goals. Central to the GPS is a Monte Carlo simulation engine that generates a large number of potential future return paths based on the historical performance of the model portfolios, and uses these simulations to estimate the probability of achieving the client's retirement goals under different scenarios. The GPS does not currently have any specific features or recommendations related to glidepath strategies, but it does allow the user to change their allocation at any point in time, which could be used to implement a glidepath strategy if the user chooses to do so. It is likely that some advisers and clients have used the GPS to implement glidepath strategies, but there is no official guidance or recommendation on how to do this, and the GPS does not provide any specific analysis or insights on the optimal glidepath shape.

The GPS is beset by the same modelling choices that drive disagreement in the literature, and so it is important to understand how these choices affect the recommendations it produces. For example, the GPS uses a Monte Carlo simulation engine with IID asset returns generated through Brownian Motion, which may lead to different conclusions about optimal glidepath shape compared to a block bootstrap approach that preserves time-series dynamics. Additionally, the performance of the GPS is lacking, and it is currently unable to compute more than around 1,000 simulated return paths in a reasonable time frame, which limits the robustness of its recommendations. This is an area where improvements could be made and while it is not the focus of this research, we will note any areas of our approach that could be implemented in the GPS to improve its performance and robustness.

2.3 Problem Statement and Research Questions

The review above reveals that the literature has not converged on a single answer to even the most basic question of optimal retirement allocation, and that disagreements are substantially driven by modelling choices rather than fundamental differences in the underlying economic problem. The choice of return-generation methodology (IID Monte Carlo, historical rolling periods, or block bootstrap), the choice of risk or objective measure (failure probability, terminal wealth, expected utility), and the specification of the spending rule (fixed real withdrawal vs. flexible spending) each independently shift the apparent optimal policy. A secondary but practically important question is whether the glidepath shape matters at all compared with the more fundamental question of whether bonds should be held in retirement portfolios at the levels currently recommended.

Concretely, we wish to answer the following questions:

1. Are conventional age-based declining-risk glidepaths optimal under standard assumptions? And does that result survive under NZ-calibrated return dynamics and spending paths that decline in real terms with age?

2. How sensitive are "optimal" recommendations to modelling assumptions (return-generation process, IID vs. block bootstrap, risk measure, spending rule)?

3. What (if any) asset allocation strategies outperform fixed asset allocation strategies, and how do they perform under different modelling assumptions?

We make three contributions relevant to the New Zealand context and Consilium's position on this topic. First, we review the New Zealand-specific literature on retirement and retirement policy, and outline the needs and behaviour of New Zealand retirees. Second, we develop a method for simulating lifetime wealth and spending paths for a representative New Zealand retiree, calibrated to the institutional context of New Zealand Superannuation and KiwiSaver. A secondary goal of this is to improve on the current model used by Consilium in the GPS and other areas. Third, we will develop an optimistation framework to create dynamic asset allocation strategies and compare their performance under different modelling assumptions.

3 Retirement in New Zealand

The New Zealand-specific literature on retirement concerns remains limited, so we draw extensively on international evidence. Understanding the institutional context is essential for interpreting this evidence appropriately. New Zealand's retirement system combines a universal public pension (New Zealand Superannuation) with voluntary private savings through KiwiSaver. New Zealand Superannuation provides all residents aged 65 and over with a baseline income regardless of work history or assets, and adjusts annually for inflation. KiwiSaver allows voluntary contributions from employees, matched by employers and the government. Together, these mechanisms provide a safety net while encouraging personal saving (Ministry of Social Development, 2022; Te Ara Ahunga Ora: Retirement Commission, 2021).

KiwiSaver was introduced in 2007, so most current retirees have little or no accumulated balance. Currently, 40 percent of New Zealanders aged 65 rely entirely on New Zealand Superannuation, with a further 20 percent holding minimal additional savings (Retirement Commission Te Ara Ahunga Ora, 2022). As the scheme matures and government policy encourages higher contribution rates (Luxon, 2025), future cohorts will retire with substantially larger balances. This demographic shift increases the need for robust guidance on asset allocation and withdrawal strategies during retirement.

Given these trends, the Australian experience becomes increasingly relevant. Australia's retirement system combines a means-tested public pension with mandatory superannuation contributions of 12 percent of income (Australian Taxation Office, n.d.; Swoboda, 2014). While institutional differences remain, the growing role of KiwiSaver makes the two systems more comparable than in the past.

3.1 New Zealand Superannuation Policy Settings

New Zealand Superannuation (NZ Super) is a universal, non-means-tested public pension paid from age 65, funded from general taxation rather than a contributory insurance scheme. It forms the guaranteed income floor underpinning all retirement planning in New Zealand. Entitlement is not conditional on prior employment, investment, or KiwiSaver participation; a recipient's private wealth or income has no effect on the payment rate.

Eligibility

The primary eligibility criteria are age and residency. A person becomes entitled upon turning 65, provided they have been both resident and present in New Zealand for a sufficient period since age 20. For those born on or before 30 June 1959, the requirement is 10 years total, of which 5 must have accrued since age 50. For those born on or after 1 July 1977, the total requirement is 20 years (with 5 since age 50), reflecting a graduated increase introduced by the Social Assistance (Residency Qualification) Legislation Act 2018. Special provisions apply to recognised refugees and to periods spent in the Cook Islands, Niue, and Tokelau (New Zealand Parliament, 2001, s. 8).

Payment Rates

Rates are set in Schedule 1 of the New Zealand Superannuation and Retirement Income Act 2001 and adjusted each 1 April by Order in Council. The April 2026 rates (before deduction of tax under code M) are listed in Table 2.

Table 2: NZ Super rates from 1 April 2026 (before tax). Source: Work and Income New Zealand (2026a, 2026b).

Rate type	Per week	Per fortnight	Per year
Single living alone	$647.37	$1,294.74	$33,663
Single sharing	$595.57	$1,191.14	$30,970
Couple (per person)	$492.14	$984.28	$25,591
Couple (combined)	$984.28	$1,968.56	$51,182

After deduction of standard tax (code M), the fortnightly after-tax rates are approximately $1,110 for a single person living alone and $1,708 combined for a couple, equivalent to roughly $28,900 and $44,400 per year respectively (Work and Income New Zealand, 2026a, 2026b).

Indexation and the Wage Floor

The Act mandates annual adjustment for CPI inflation (so the rate cannot decline in real terms) subject to a binding wage floor. The combined after-tax couple rate must remain between 66 and 72.5 percent of net average ordinary-time weekly earnings, as measured by the Quarterly Employment Survey (New Zealand Parliament, 2001, s. 16). In practice, wages have grown faster than CPI in recent decades, meaning the wage floor has been the binding constraint. The single living alone rate is fixed by statute at 65 percent of the couple combined rate (after tax), and the single sharing rate at 60 percent. This legislated floor is a key feature distinguishing NZ Super from defined-benefit pension schemes that index only to prices.

Policy Adjustments and Reform Debate

Despite ongoing fiscal pressure from population ageing, NZ Super's core parameters have remained largely unchanged since 2001. The most significant recent amendment was the gradual increase in the residency qualifying period from 10 to 20 years for those born after 1 July 1977, phased in from 2018. The eligibility age of 65 has not changed; a proposal to raise it to 67 was advanced and subsequently abandoned. A 2022 government review (the Retirement Review Issues Paper, conducted by Te Ara Ahunga Ora) examined a range of possible adjustments including age, means-testing, and residency, without proposing binding changes (Ministry of Social Development, 2022).

Supplementary Benefits

Several additional transfers are available to eligible retirees:

- **Winter Energy Payment:** a universal supplement paid automatically to NZ Super recipients from 1 May to 1 October each year. The 2026 rate is \$20.46 per week for single people and \$31.82 per week for couples and people with dependants (Work and Income New Zealand, 2026a).
- **Accommodation Supplement:** a means- and asset-tested housing cost subsidy available to renters, boarders, and mortgage holders across all age groups including retirees. The amount varies by accommodation costs and location zone, with higher rates in Auckland and Wellington. Eligible retirees with moderate housing costs may receive up to several hundred dollars per fortnight (Work and Income New Zealand, 2026a).
- **SuperGold Card:** a concession card issued automatically to NZ Super recipients, providing discounts on public transport (including free off-peak travel on most regional services), selected retail, and other services (Ministry of Social Development, 2022).
- **Residential Care Subsidy:** an asset-tested subsidy for long-term residential aged care (rest homes and dementia units). As at 2022, a single person is generally required to contribute their own assets up to a threshold of around \$240,000 before the subsidy takes effect, with the family home exempt for a defined period if a spouse, dependent, or carer remains in residence (Ministry of Social Development, 2022).

Together, NZ Super plus the Winter Energy Payment provide a reliable inflation-linked income floor for all retirees.

3.2 Patterns of Spending Among Retirees in New Zealand and Abroad

3.2.1 Spending by Age

The Stats NZ Household Economic Survey (HES) provides the most comprehensive data on New Zealand retirement spending patterns. Two recent reports are particularly relevant. Le and Richardson (2023) analyse expenditure patterns by age across New Zealand households, while Retirement Income Interest Group (2024) focus specifically on spending through retirement and its implications for drawdown strategies.

Using HES data for the year ended June 2019, Le and Richardson (2023) find that expenditure declines sharply at retirement and continues to fall thereafter. Mean household expenditure is \$64,700 for ages 65–74, then 28% lower for ages 75–84, and 44% lower for ages 85+. Expenditure declines across most categories, with the largest falls in mortgage repayments, air travel, and recreation. Unusually, healthcare expenditure also declines with age in New Zealand, likely reflecting the public funding of most core medical services. Retirement Income Interest Group (2024) estimate the real decline at roughly 2% per year after age 65.

The HES data cannot, however, distinguish voluntary from involuntary spending reductions. Retirees may reduce expenditure because they become less active and have genuinely lower needs, or because they face binding financial constraints. Without longitudinal data tracking individuals over time, these explanations cannot be separated cleanly. The observed decline likely reflects both channels.

Understanding the source of this decline is central to evaluating welfare. If spending falls primarily due to financial constraints, optimal strategies should aim to sustain pre-retirement consumption levels. If it reflects voluntary lifestyle adjustment, forcing consumption to remain high may be neither necessary nor welfare-improving. This distinction also affects risk tolerance: financially constrained retirees may need safer portfolios, while those adjusting voluntarily may tolerate more risk.

As we noted earlier, beginning with the work of Modigliani and Brumberg (2005) and later extended by Ando and Modigliani (1963) and others, the Life-Cycle Hypothesis (LCH) establishes a theoretical, or "ideal", pattern of saving and consumption over the life course. Under the LCH, individuals are expected to smooth consumption over time, in other words, they seek to maintain a broadly stable standard of living throughout life. The LCH predicts that individuals will save during their working years, accumulating wealth to fund consumption in retirement (but only insofar as doing so does not unnecessarily reduce pre-retirement consumption), and then draw down that wealth during retirement to maintain living standards, ultimately exhausting it near the end of life.

Empirically, there appear to be two related but distinct departures from this benchmark. The first is that measured spending often falls at retirement and continues to decline thereafter. This is the phenomenon usually described as the "retirement consumption puzzle". The second is that many retirees do not decumulate their wealth as the model predicts, and a substantial share even continue to accumulate wealth during retirement. The two phenomena are related, but they are not identical: the first concerns the path of consumption, while the second concerns the path of wealth.

One possible explanation for the retirement consumption puzzle is that spending falls because retirees are financially constrained. On this view, households would prefer to maintain pre-retirement consumption, but insufficient savings, unexpected health shocks, or other adverse events compel them to cut spending. However, the separate empirical regularity of weak decumulation complicates this explanation. If lower spending were primarily the result of binding financial constraints, we would expect retirees to draw down wealth aggressively when possible. In practice, many do not. In a survey of Canadian retirees, King (1982) finds that the majority of respondents not only failed to draw down on their wealth, but that many actually accumulated more wealth during retirement. This finding has been corroborated by a number of subsequent studies in many different countries and time periods (e.g. De Nardi et al., 2010; Love et al., 2009; Palumbo, 1999). A paper by the Grattan Institute in Australia (Grattan Institute, 2019) finds that fewer than half of Australian retirees draw down wealth at all, and that over 40% accumulate wealth during retirement. Moreover, it finds that most Australian retirees are not financially constrained, and that retirees report feeling more financially secure than working-age households.

For the spending puzzle specifically, how much of the observed decline at retirement is driven by financial constraints versus voluntary lifestyle adjustment remains an open question, but it does not appear that financial constraints are the primary driver. Consumption expenditure is an imperfect proxy for welfare, particularly when higher spending is driven by medical needs or lower spending reflects voluntary lifestyle adjustment. While Barrett and Brzozowski (2010) find that households retiring unexpectedly (and are thus unprepared) due to adverse health shocks do experience sharper spending declines than those retiring as planned, which is consistent with involuntary constraints driving some of the observed decline, the majority of the observed decline may be voluntary. Retirees spend less on commuting, work clothing, and purchased meals (Aguiar & Hurst, 2008; Battistin et al., 2009), and may substitute market purchases with home production, reducing measured expenditure while maintaining consumption (Becker, 1965; Hurd & Rohwedder, 2003). These mechanisms can mask substantial cross-household heterogeneity in retirement outcomes. Consistent with this interpretation is the fact that the largest expenditure declines observed in the HES data by Le and Richardson (2023) are in discretionary categories such as transport and recreation, while spending on essentials such as food and housing remains comparatively more stable. Furthermore, concessions made to elderly households, such as lower prices for certain goods and services (such as those provided by the SuperGold card in New Zealand), may allow retirees to maintain consumption while reducing measured expenditure.

Cross-country comparisons highlight the role of institutional context. Banks et al. (2016) find that U.K. retirees reduce real consumption by 2.2% per year versus 1.4% per year in the U.S. This divergence is driven largely by healthcare financing: U.S. retirees face high out-of-pocket costs that rise with age, while U.K. retirees benefit from comprehensive public coverage. When healthcare spending is excluded, consumption trajectories are more similar across countries.

Australian evidence also shows spending declines of 0.5–1% per year in real terms (Grattan Institute, 2019). Unlike New Zealand's cross-sectional HES data, the Grattan Institute analysis uses longitudinal data tracking individuals over time. The decline persists in both cross-sectional and longitudinal samples, suggesting it reflects genuine behavioural change rather than cohort effects. As Retirement Income Interest Group (2024) note, the weight of evidence indicates that spending reductions primarily reflect voluntary adjustment to changing lifestyle needs rather than binding constraints. From our perspective, this means that it is reasonable to model retirement spending as declining in real terms with age.

3.2.2 Non-Decumulation, Bequests, and Precautionary Savings

The second empirical departure from the LCH is non-decumulation, the fact that many retirees do not run down their savings as the model predicts, and that a notable share die with substantial wealth intact. Financial planners often describe this as a form of risk, namely that a retiree may preserve so much wealth that they die having "underconsumed" relative to what was feasible. A common explanation for this phenomenon is that retirees intentionally preserve wealth to leave bequests to their heirs (e.g. Hurd, 1987). In this view, retirees are not underconsuming, but are instead consuming optimally while also deriving utility from leaving a bequest to their heirs. However, the empirical evidence for intentional bequest motives is weak. Alonso-García et al. (2022) find that Australian and Dutch retirees have little to no intentional bequest motive outside provision for a surviving spouse, and that the observed pattern of non-decumulation is more consistent with saving for uncertain future costs, such as aged care or health shocks. This motive is often referred to as a "precautionary" motive, and is distinct from a bequest motive in that the utility of leaving a bequest is not derived from the act of leaving a bequest itself, but rather from the desire to avoid the risk of being unable to cover future costs.

Precautionary motives are well-supported empirically. De Nardi et al. (2010) show that exposure to uncertain health and long-term care costs explains much of the observed non-decumulation in the United States. Similarly, Banks et al. (2016) find that healthcare costs drive higher late-life spending among U.S. retirees relative to those in the U.K. with broader public coverage. New Zealand's system is mixed: most core medical services are publicly funded, but long-term care such as assisted living and memory care can involve substantial out-of-pocket costs (New Zealand Government, n.d.). Retirement Income Interest Group (2024) conclude that precautionary motives in New Zealand are likely weaker than in the United States, but stronger than in countries such as France and Germany with more comprehensive public provision, placing New Zealand alongside the United Kingdom in a mixed-funding model. However, aged care remains a potentially significant late-life expense.

3.2.3 Spending by Household Composition and Wealth

Household composition and housing tenure are among the strongest predictors of retirement spending levels and subjective wellbeing in New Zealand. Le and Richardson (2023) find that single-person retired households spend around $30,700 per year on average, compared with $65,100 for couples. Notably, spending increases more than proportionately with household size: the average couple spends more than double the amount of a single-person household, implying weaker economies of scale (in New Zealand at least) than other studies, such as (Anarkulova et al., 2025), have suggested in other contexts. Le and Richardson (2023) attribute the disproportionate gap primarily to differences in discretionary spending: couples devote substantially more to transport and recreation and culture, the two largest discretionary categories. Single-person households living alone are also more likely to be renters or to carry a mortgage, further constraining their discretionary budget, while couples are more likely to own outright.

Homeownership status has a particularly pronounced effect on both spending and wellbeing. Among retired New Zealand households, approximately 66 percent own their home outright, 14 percent carry a mortgage, and 14 percent rent (Le & Richardson, 2023). Outright owners report the highest subjective wellbeing of any tenure group and spend around $55,800 per year, of which housing costs form a relatively small share. Renters spend considerably less in total, at around $41,400 per year, but pay approximately $13,100 of this in rent annually, leaving a significantly lower discretionary budget. Renters also report the lowest wellbeing among all tenure types, even after controlling for income. This suggests that the financial and psychological burden of ongoing housing costs in retirement is substantial and is not fully compensated by lower total spending. This wellbeing gap may have interesting consequences for asset allocation decisions, especially as it pertains to whole-life allocation and spending strategies. One of the peculiarities of the New Zealand KiwiSaver system is that it allows accumulated savings to be withdrawn for the purchase of a first home. Balancing the competing goals of homeownership and retirement savings is a unique challenge for New Zealand investors, and one that may have important implications for their lifetime asset allocation decisions. For instance, if utility gained from homeownership is sufficiently high, it may justify depleting the investor's retirement savings while young to achieve homeownership, even if this results in a lower savings balance at retirement. Conversely, if buying a house depletes the investor's retirement savings to a degree that the excess burden of housing costs would otherwise have been more than offset by the additional savings balance at retirement, it may be optimal to delay homeownership and instead prioritise retirement savings.

The importance of homeownership is expected to increase over time. Retirement Income Interest Group (2024) note that homeownership among New Zealanders aged 65 and over is projected to fall from around 80 percent today to approximately 60 percent by 2048, as younger cohorts face higher barriers to entry in the housing market. This shift implies that expenditure benchmarks calibrated on the current cohort of retirees (who benefit from unusually high ownership rates) will understate spending requirements for future generations, particularly with respect to housing costs.

3.2.4 Expenditure Benchmarks for New Zealand Retirees

The most widely used empirical benchmarks for New Zealand retirement spending are the annual Retirement Expenditure Guidelines published by the NZ Fin-Ed Centre at Massey University (Matthews, 2025). These are derived from the Statistics New Zealand Household Economic Survey (HES) and reported at two tiers: a "No Frills" budget based on the second income quintile of retired HES households, reflecting a basic standard of living with few luxuries, and a “Choices” budget based on the fourth quintile, reflecting a more comfortable lifestyle with some discretionary spending. The 2025 figures, adjusted to 30 June 2025, are shown in Table 3.

Table 3: New Zealand Retirement Expenditure Guidelines, 2025. Weekly spending, with after-tax NZ Super rates for comparison. Source: Matthews (2025).

	One-person household		Two-person household	
Budget	Metro	Provincial	Metro	Provincial
No Frills ($/week)	705	581	937	1,061
Choices ($/week)	791	772	1,780	1,243

For the No Frills provincial single-person household, NZ Super alone covers approximately 89 percent of spending, implying a required annual drawdown of only around $3,200 from private savings. For the No Frills metro single, the gap is somewhat larger at around $9,700 per year. Couples at the Choices tier, particularly in metro areas, face substantially higher gaps; the Choices metro couple budget of approximately $92,600 per year implies a drawdown requirement of around $51,000 annually above NZ Super. However, this upper estimate should be interpreted cautiously, since the fourth income quintile of HES retired households likely includes a material proportion of households still receiving employment or investment income, rather than pure retirees drawing down accumulated savings.

A central limitation of these guidelines is that they are constructed from cross-sectional data and therefore implicitly treat retirement spending as fixed in real terms throughout retirement. The HES samples households of all retirement ages at a single point in time, which means the guidelines effectively average the higher spending of the more numerous newly retired with the lower spending of older cohorts, without capturing the within-person spending trajectory over time (Retirement Income Interest Group, 2024). If real spending declines by approximately 2 percent per year after age 65 (as both Retirement Income Interest Group (2024) and Le and Richardson (2023) find) then the guidelines will overstate the savings balance required at age 65 for any given income target.

Retirement Income Interest Group (2024) quantify this directly. Using a replacement-rate methodology, they calculate that reaching a total income target of $56,000 per year in real terms for a single retiree, on top of NZ Super, over a retirement from age 65 to 90, requires a savings balance of $605,000 under a constant-real-spending assumption. If instead nominal income is assumed to remain flat (equivalent to a 2 percent real decline when inflation is also 2 percent), the required balance falls to $375,000, a reduction of 40 percent. An intermediate assumption of a 1 percent real annual decline yields a balance of $480,000. These calculations assume a 3.5 percent net investment return, which is not particularly high and is broadly consistent with the long-run real return on a relatively conservative portfolio.

For the purposes of this study, we treat the expenditure guidelines as an empirical anchor for age-65 spending levels. We calibrate initial retirement spending to the No Frills and Choices benchmarks as the lower and upper bounds for a representative retiree, and model real consumption as declining at approximately 2 percent per year thereafter. NZ Super is treated as a guaranteed, inflation-linked income floor from age 65. Under this calibration, the required drawdown from financial assets diminishes progressively as retirement spending falls toward the level already covered by NZ Super. This pattern has material implications for both the required savings balance at retirement and the optimal asset allocation strategy during decumulation.

4 Methods

The crux of our analysis is the ability to simulate realistic wealth trajectories for individual investors. While this is not, in itself, a novel contribution, the specific features of our framework are designed to address the questions in this report. Our aim is to develop a flexible, modular framework that can be extended beyond these questions to support ongoing research and policy analysis for Consilium and others. Although our immediate focus is the optimality of age-based glidepaths in the New Zealand retirement context, the framework is intended to be broadly applicable across lifetime asset-allocation problems, including whole-life allocation and spending strategy design, the role of housing and other non-financial assets, and interactions between retirement saving and other life goals such as homeownership. This section describes the general structure of the framework, while calibration details and report-specific assumptions are presented later. We also outline potential extensions in the appendix, including explicit treatment of housing and related homeownership decisions.

Simulating lifetime wealth trajectories has two components: methods for investor behaviour, and methods for market conditions. The former includes the asset-allocation strategy, spending rule, and risk measure used to evaluate outcomes. The latter includes the return and inflation generation methodology (IID Monte Carlo, historical rolling periods, or block bootstrap). In this section, we describe the structure of our framework, how it accommodates these choices, and how we calibrate it to the New Zealand context.

4.1 Lifecycle Simulation Framework

With the exception of the continuous-time approach pioneered by Merton (1969), the majority of the literature on simulating lifetime wealth trajectories uses discrete-time approaches that model the investor's wealth at regular intervals, such as annually or monthly (e.g. Anarkulova et al., 2025; Estrada, 2014; Mäkinen & Toivanen, 2024). This approach is far easier to implement and makes fewer assumptions about the underlying return process than the continuous-time approach, which relies on strong assumptions about the return process (for example, that returns are normally distributed and independent over time) that are not supported by empirical evidence.

In this framework, we evolve the investor's wealth over time according to a simple recursive transition equation that accounts for returns, contributions, and withdrawals. For the transition from time t to $t+1$, the investor's wealth is updated as follows in Equation 1:

$$w_{t+1} = g_{t+1}w_t + \pi_{t+1} \tag{1}$$

Where w_t is the wealth at time t, g_{t+1} is the growth factor on the portfolio between time t and $t+1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t+1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Assuming that the portfolio is rebalanced at the end of each period, the growth factor can be calculated as follows in equation 2:

$$g_{t+1} = 1 + \mathbf{a}_t^{\top}\mathbf{r}_{t+1} \tag{2}$$

Where $\mathbf{a}_t$ is the asset allocation vector at time t, with each element $a_{i,t}$ representing the proportion of wealth invested in asset class i, and $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t+1$. Both $\mathbf{a}_t$ and $\mathbf{r}_{t+1}$ are vectors of length k, where k is the number of asset classes in the portfolio.

In a Monte Carlo simulation, we generate a large number of possible investor wealth trajectories by simulating returns across multiple paths. The returns are stored in a tensor $\mathcal{R}$ of dimension $N \times T \times k$, where N is the number of simulated paths, T is the number of time periods, and k is the number of asset classes. Similarly, the allocation policy can be represented as a tensor $\mathcal{A}$ of dimension $N \times T \times k$, where each element $A_{i,t,j}$ represents the allocation to asset class j at time t for simulation path i. In our nomenclature, asset allocation policies are functions or sets of rules that determine the allocation vector $\mathbf{a}_t$ at each time t. By representing the allocation policy as a tensor, we can accommodate a wide range of policies, including those that depend on path-specific state variables such as wealth or age. For example, a simple age-based glidepath can be represented by setting all paths to have identical allocations at each time, while a more complex policy that adjusts allocations based on the investor's current wealth can be represented by allowing the allocations to vary across paths at each time step.

At each time $t+1$, we extract the $N \times k$ matrix $\mathbf{R}_{:,t+1}$ containing returns at time $t+1$ across all paths and asset classes, and the $N \times k$ matrix $\mathbf{A}_{:,t}$ containing allocations at time t for all paths. The growth factors for all paths are then computed in vector form as shown in equation 3:

$$\mathbf{g}_{t+1} = \mathbf{1}_N + \sum_{j=1}^{k} \mathbf{R}_{:,t+1,j} \odot \mathbf{A}_{:,t,j} \tag{3}$$

Where $\mathbf{1}_N$ is an $N \times 1$ vector of ones, $\mathbf{R}_{:,t+1,j}$ and $\mathbf{A}_{:,t,j}$ are the $N \times 1$ column vectors for asset class j, $\odot$ denotes element-wise multiplication, and $\mathbf{g}_{t+1}$ is the resulting $N \times 1$ vector of growth factors across all paths. Equivalently, this can be expressed as $\mathbf{g}_{t+1} = \mathbf{1}_N + \text{rowsum}(\mathbf{R}_{:,t+1} \odot \mathbf{A}_{:,t})$, where the element-wise product is summed across the asset class dimension. The wealth evolution across all paths then follows equation 4:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \tag{4}$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t, and $\odot$ denotes element-wise (Hadamard) multiplication. The wealth across all paths over all time periods can be represented as an $N \times T$ matrix, $\mathbf{W}$, where each row corresponds to a single simulated path of wealth over time.

Note that the system of equations above is agnostic to the return-generation process. The returns tensor $\mathcal{R}$ is a flexible input that can be generated using any methodology. This allows us to test the robustness of our findings across different assumptions about return dynamics, such as using historical returns, bootstrapped returns, or model-simulated returns. Additionally, $\mathbf{a}_t$ and $\boldsymbol{\pi}_{t+1}$ can be specified as functions of time, wealth, or other state variables, allowing for a wide range of asset allocation and spending strategies to be simulated within the same framework. This flexibility is a key strength of our approach, as it allows us to explore a broad set of strategies and assumptions without needing to fundamentally alter the underlying simulation mechanics. It is on the construction of these functions and the calibration of the return-generation process that we will focus in the next sections.

Implications for Consilium: Matrix Operations and Computational Efficiency

The effort taken to express the model in matrix form is not only for mathematical clarity; it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (for example, `numpy` (Harris et al., 2020)) to vectorise calculations across the matrix's longest dimension (the path dimension). This means that we can compute the wealth evolution across all paths virtually simultaneously using efficient matrix operations, rather than iterating through each path sequentially with for-loops. This vectorised approach can take advantage of modern CPU architectures and optimised libraries to perform computations much faster than a non-vectorised implementation, which can substantially accelerate simulation when path counts and horizon lengths are large. Compared to the GPS, which takes up to 30 seconds to simulate fewer than 1,000 paths over a 30-year horizon, our matrix-based implementation can simulate 500,000 paths over the same horizon in under a second on a standard laptop. This dramatic improvement in computational efficiency enables us to explore a much wider range of strategies with much more rigour than would be possible with the GPS tool, and also opens the door to more complex strategies that may require more computational resources, such as those that involve path-dependent allocations or more sophisticated risk measures.

Refactoring the GPS tool to use matrix operations would be a significant improvement in its computational efficiency and flexibility. It would allow Consilium to support more complex strategies and larger simulations, which could significantly enhance the tool's value for financial advisers and their clients.

4.2 Modelling Contributions and Withdrawals

The flow of wealth into and out of the portfolio is critical not only for the evolution of the system, but also for the evaluation of outcomes. During retirement, when our investor is dissaving to fund consumption, how much the investor withdraws, and is able to withdraw, is a key determinant of welfare. Modelling contributions and withdrawals in a realistic way is therefore essential for simulating realistic wealth trajectories and evaluating the performance of different asset allocation strategies. There have been many approaches to modelling and planning withdrawals in the literature and financial advice space.

Our framework consists of essentially three components: the withdrawal rule, which specifies how much the investor desires to withdraw in each period; the consumption floor, which specifies the minimum amount the investor must withdraw to meet essential needs; and income, which specifies guaranteed inflows from sources such as NZ Super. The withdrawal rule is the most commonly discussed component in the literature, and there are many different rules that have been proposed. However, on its own, it is not sufficient to model the withdrawal behaviour of a representative retiree, as it does not account for the fact that retirees have minimum spending needs that must be met regardless of market returns, and that these needs are not captured by any of the simple rules. Income is also important, as it provides a guaranteed source of inflow that can help to meet essential needs and reduce the risk of depleting wealth too quickly.

Net inflows, π_t, are modelled by equation 5 as the difference between the income, y_t, and realised consumption, $\hat{c}_t$.

$$\pi_t = y_t - \hat{c}_t \tag{5}$$

Note that we constrain $\pi_{t+1} \geq -w_t$, such that the investor cannot withdraw more than their current wealth balance (that is, cannot go into debt). This is a realistic constraint for a typical retiree. While retirees may have access to credit or other sources of funds that could allow them to temporarily withdraw more than their current wealth balance, this is not a sustainable strategy and would likely lead to financial distress in the long run.

4.2.1 Inflation

Central to any model of consumption is the treatment of inflation. In the context of retirement planning, inflation is a critical factor that erodes the purchasing power of money over time and therefore has a significant impact on the real value of withdrawals and the sustainability of retirement savings. Inflated dollar amounts can also be a significant source of confusion, as retirees and planners may have different perceptions of what constitutes a reasonable withdrawal amount depending on whether they are thinking in "real" (inflation-adjusted) or "nominal" (actual dollar) terms.

In our framework, we denote the rate of inflation at time t as ι_t, and the cumulative inflation factor from time 0 to time t as $s_t = \prod_{i=1}^{t}(1+\iota_i)$. Real dollar amounts can be converted to nominal dollars by multiplying some initial real amount by the cumulative inflation factor. At $t=0$ (the point of retirement), real and nominal dollar amounts are the same, but as time progresses, the real value of a given nominal amount will decline if there is positive inflation. For example, if a retiree desires to maintain a constant real consumption level of $50,000 per year, and cumulative inflation over the first three years of retirement is 10%, then the nominal withdrawal amounts in year three would need to be $55,000 to maintain the same real purchasing power.

While there are technically many different types of inflation that affect different parts of a retiree's overall consumption bundle differently, for the purposes of this analysis we treat inflation as a single aggregate factor (linked to the consumer price index) that affects the overall purchasing power of the retiree's withdrawals. This is a simplification, but it is a common approach in the literature and is sufficient for our purposes here.

Implications for Consilium: Inflation

Another major weakness of the GPS tool is that it only accounts for inflation in a very limited way. The tool allows the user to specify an assumed inflation rate, but this is a constant rate that is applied uniformly across all time periods and does not account for the stochastic nature of inflation or its potential correlation with asset returns. This means that the GPS tool cannot capture the impact of unexpected inflation shocks on the real value of withdrawals or the sustainability of retirement savings, which is a critical aspect of retirement planning.

There are very good reasons to be wary of this approach. As we have seen over the 2020-2024 period, inflation can be highly volatile. Additionally, as noted by Anarkulova et al. (2025), inflation can be correlated with asset returns, particularly due to the fact that inflation is often addressed by central banks increasing interest rates, which can have a negative impact on asset returns. Monetary policy has a complex relationship with both fixed income and equity returns. For example, rising interest rates can lead to lower bond prices and higher yields, which can reduce the value of fixed income investments. Rising interest rates can also lead to lower equity prices, as they increase the cost of borrowing for companies and reduce the present value of future cash flows. By not accounting for the stochastic nature of inflation and its potential correlation with asset returns, the GPS tool may provide an overly optimistic view of the sustainability of retirement savings and the real value of withdrawals, particularly in periods of high or volatile inflation.

4.2.2 Consumption Floors and Minimum Desired Consumption

While simple withdrawal rules provide a useful starting point for modelling retirement spending, they do not account for the fact that retirees have minimum spending needs that must be met regardless of market returns.

To model this more realistically, we specify a minimum real spending requirement or floor, $\underline{c}_t$, that must be met in each period. This gives us the minimum desired consumption at time t, denoted c_t, as the maximum of the desired consumption from the withdrawal rule, c_t^*, and the consumption floor, $\underline{c}_t$, as shown in equation 6:

$$c_t = \max\left(c_t^*, \underline{c}_t\right) \tag{6}$$

This consumption floor is designed so that it can vary over time, or even with respect to other state variables such as wealth or health status. For example, the floor could be set to reflect the empirical expenditure benchmarks for New Zealand retirees, which vary by household composition and housing tenure. The floor could also be adjusted for inflation or other factors that affect the cost of living over time. Random shocks to the floor could also be introduced to reflect unexpected expenses, such as health shocks or changes in housing costs. This can be modelled as shown in Equation 7:

$$\underline{c}_t = \underline{c}_0 \times s_t \tag{7}$$

Where $\underline{c}_0$ is the initial consumption floor at age 65, and s_t is the cumulative inflation factor at time t since retirement.

4.2.3 Income

While certain types of assets pay dividends or interest that can be considered as income, we treat these as part of the portfolio returns rather than as a separate income stream. Income within our framework refers to exogenous sources of income that are not derived from the portfolio. Within a New Zealand retirement context, the most important source of exogenous income is NZ Super, which provides a guaranteed, inflation-linked income floor from age 65. We denote income at time t as y_t.

NZ Super provides a guaranteed, inflation-linked income floor from age 65. The income received from NZ Super is calculated within our framework as follows in Equation 8:

$$y_t = y_0 \times s_t \tag{8}$$

Where y_0 is the initial NZ Super payment for the household at age 65 (see Table 2), and s_t is the cumulative inflation factor at time t since retirement.

Extensions to the Framework: Modelling Other Sources of Income and the Whole Life Cycle

While our current framework focuses on retirement and the decumulation phase, it can be easily extended to model the whole life cycle, including the accumulation phase and other sources of income. For example, we could model labour income during the accumulation phase, which would allow us to simulate the entire life cycle of wealth accumulation and decumulation. We could also model other sources of income during retirement, such as part-time work, rental income from property, or annuity payments. This would provide a more comprehensive view of the retiree's financial situation and allow us to evaluate how different asset allocation strategies interact with these various sources of income throughout the life cycle.

In a whole-life cycle model, a good model of labour income is critical, as it is the primary source of wealth accumulation for most individuals and one that is also highly heterogeneous across individuals and simulation paths. Modelling labour income, and particularly shocks to labour income such as unemployment or disability, would allow us to capture the risks that individuals face during the accumulation phase and how these risks interact with their retirement savings and asset allocation decisions. This income, and the shocks to it, would also have correlations with other economic variables, such as asset returns and inflation. This model is well outside the scope of the work presented here, but it is a natural extension of the framework we have developed and would be a valuable area for future research.

4.2.4 Realised Consumption

Because our retirees are forbidden from withdrawing more than their current wealth balance, it is possible that the desired minimum consumption, c_t, exceeds the available resources. In such cases, the actual consumption is limited by the wealth balance and any exogenous income, such as NZ Super. The realised consumption at time t, denoted $\hat{c}_t$, is therefore given by the minimum of the desired consumption and the total available resources, as shown in equation 9:

$$\hat{c}_t = \min\left(c_t, w_t + y_t\right) \tag{9}$$

Net contributions to the portfolio are then given by:

$$\pi_t = -\hat{c}_t + y_t \tag{10}$$

This realised consumption is one of the most important outputs of our simulation, as it reflects the actual spending that the retiree is able to achieve given their wealth trajectory and the withdrawal rules. It is also a critical input for evaluating the performance of different asset allocation strategies, as it directly affects the retiree's welfare and the sustainability of their retirement savings. By modelling realised consumption in this way, we can capture the trade-offs that retirees face in balancing current consumption against future needs, and we can also evaluate how different asset allocation strategies perform under different withdrawal rules and income scenarios.

Implications for Consilium: Modelling Consumption and Income

The GPS currently only outputs the investor's wealth trajectory over time, and does not model or output the actual consumption that the retiree is able to achieve given their wealth trajectory and any exogenous income. This is a significant limitation, as it means that the GPS cannot directly evaluate the welfare implications of different asset allocation strategies, since welfare is ultimately determined by the retiree's consumption rather than their wealth.

This weakness is in part due to the limited and inflexible way that the GPS models withdrawals, which does not allow for the kind of dynamic withdrawal strategies described above, and also does not account for the fact that retirees have minimum spending needs that must be met regardless of market returns. While it has a fairly faithful implementation of the NZ Super rules, it does not integrate this income into the overall model of consumption and wealth evolution in a way that allows for the kind of dynamic interactions described above.

4.3 Asset Allocation Policies

The asset allocation policy is the key decision variable in our analysis, and the one that we seek to optimise. Here, a "policy" refers to a function or set of rules that determine the allocation vector $\mathbf{a}_t$ at each time t. The policy can be as simple as a fixed allocation that does not change over time, or it can be a complex, dynamic strategy that adjusts allocations based on the investor's current wealth, age, or other state variables. We express the policy as a function that maps from the current state of the system (for example, time, wealth, or other state variables) and a set of policy settings to the asset allocation vector $\mathbf{a}_t$. We denote this policy function as $\mathcal{A}$, which takes as input the current state and "policy settings", denoted θ, and outputs the allocation vector $\mathbf{a}_t$ for each time t and simulation path. The specific form of the policy function will depend on the particular class of policies we are testing, but the key point is that the goal of finding an optimal asset allocation strategy can be framed as an optimisation problem over the policy parameters θ, where we seek to find the set of parameters that maximises our chosen risk measure of the resulting wealth trajectories.

Our approach borrows heavily from the control-matrix approach of Mäkinen and Toivanen (2024), which provides a flexible and interpretable way to represent a wide range of asset allocation policies, including those that depend on path-specific state variables such as wealth or age. Another notable example of a flexible policy representation is the use of neural networks, as proposed by Duarte et al. (2021), which can capture complex, non-linear relationships between state variables and asset allocations. However, while neural networks offer a high degree of flexibility, the policy settings they produce are not easily interpretable. Because an optimal solution is likely to be linked to the return-generation process we use, we want to be able to understand the structure of the optimal policy and how it responds to different state variables. The control-matrix approach allows us to do this, as the policy settings are directly interpretable as allocations for different combinations of the state variables fed into it.

In the control-matrix approach, the policy settings θ take the form of a matrix or tensor where each dimension corresponds to a different state variable that the policy can depend on. It is worth spending some time discussing how we see the control-matrix approach fitting into the broader landscape of asset allocation policy representations in the literature, as our interpretation of this approach will also inform how we implement and interpret the results of our optimisation. The dimensions of the control matrix correspond to the state variables on which the policy can depend. In Mäkinen and Toivanen (2024), the control matrix has two dimensions, time and wealth. This allows for policies that adjust allocations based on both the investor's current age (or time since retirement) and current wealth level. This is an extension of the more traditional age-based glidepath, which only allows dependence on time, and that is itself an extension of a fixed allocation strategy, which allows no dependence on state variables.

To show you what we mean by this, consider the following example. A fixed allocation strategy can be represented as a policy function that does not depend on any state variables, and simply returns the same allocation vector $\mathbf{a}$ at each time t and for each simulation path. In this world, the policy function can be represented as a constant function:

$$\mathcal{A}(\theta) = \mathbf{a} \quad \text{for all } t \text{ and all paths} \tag{11}$$

In a two-asset portfolio, θ would simply be the allocation to the risky asset, with the allocation to the safe asset determined residually, e.g. $\mathcal{A}(\theta) = (a, 1-a)$. For a portfolio with N asset classes, θ would be a vector of length $N-1$ representing the allocations to $N-1$ of the asset classes, with the allocation to the remaining asset class determined residually as one minus the sum of the other allocations. We denote this transformation as

$$\varphi(\boldsymbol{\theta}) = \boldsymbol{a} = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_{N-1} \\ 1 - \sum_{i=1}^{N-1} \theta_i \end{pmatrix} \tag{12}$$

This means that the allocation vector $\mathbf{a}_t$ must satisfy the following constraints:

$$\sum_{j=1}^{k} a_{j,t} = 1 \quad \text{and} \quad a_{j,t} \geq 0 \quad \text{for all } j \tag{13}$$

By extension, our policy settings θ must satisfy the following constraints to ensure that the resulting allocations are valid:

$$\sum_{i=1}^{N-1} \theta_i \leq 1 \quad \text{and} \quad \theta_i \geq 0 \quad \text{for all } i \tag{14}$$

Traditional glidepath strategies vary asset allocations with age (or equivalently, time since retirement). In this case, the policy function can be represented as a function of time:

$$\mathcal{A}(\theta, t) = \boldsymbol{a}_t \tag{15}$$

Where $\boldsymbol{a}_t$ is the allocation vector at time t, which is determined by the policy settings θ. In a very simple case, θ could be a $T \times (k-1)$ matrix where each row corresponds to a time period and each column corresponds to an asset class (except for the safe asset, which is determined residually). This is essentially the form of the policy function used in Anarkulova et al. (2025), where discrete asset allocations are specified for each time period, and the optimal policy is found by searching over the space of possible allocations at each time period. In this case, $\mathcal{A}$ becomes:

$$\mathbf{a}_t = \mathcal{A}(\theta, t) = \varphi(\theta_t) \tag{16}$$

A key insight from Mäkinen and Toivanen (2024) is that you can reduce the parameter space of the policy function (and by extension, extend the policy function to continuous state variables) by using interpolation to determine the asset allocation for time periods that fall between the defined grid points. For example, if we have discrete allocations defined at time periods 0, 5, and 10, we can use linear interpolation to determine the allocation at time period 3 by interpolating between the allocations at time periods 0 and 5. This allows us to have a more parsimonious parameterisation of the policy function, while still allowing for a high degree of flexibility in the shape of the asset allocation strategy over time.

Formally, suppose we have policy settings θ defined at a discrete set of time nodes $\{t_0, t_1, \ldots, t_M\}$, where θ is now a matrix of dimension $M \times (k-1)$ with each row θ_i specifying the risky asset allocations at node t_i. For any time t such that $t_i \leq t < t_{i+1}$, the interpolated allocation is given by:

$$\mathbf{a}_t = \mathcal{A}(\theta, t) = \varphi\left((1-\alpha)\theta_i + \alpha\theta_{i+1}\right) \tag{17}$$

where the interpolation weight α is computed as:

$$\alpha = \frac{t - t_i}{t_{i+1} - t_i} \tag{18}$$

This linear interpolation ensures that the policy varies smoothly between node points while requiring only $M \times (k-1)$ parameters rather than $T \times (k-1)$ if allocations were specified independently at every time period.

The transformation φ is then applied to the interpolated risky allocations to obtain the full allocation vector including the safe asset.

Our policy function specifies the allocation to each asset class at each time t for each simulation path, and is denoted by the operator $\mathcal{A}$. New Zealand investors cannot leverage or short-sell, so we restrict our attention to policies that only allow for long positions in the available asset classes. Because short-selling and leverage are not allowed, we can reduce the dimensionality of the problem by treating one of the asset classes as the "safe" asset, with the allocation to this asset class determined residually as one minus the sum of the allocations to the other asset classes. For example, if we have two asset classes (a risky asset and a safe asset), we can represent the allocation to the risky asset as a_t and the allocation to the safe asset as $1 - a_t$. This reduces the number of decision variables in our optimisation problem and simplifies the search for optimal policies.

This is our interpretation of a glidepath strategy, which is a policy that specifies the asset allocation as a function of time. A linear glidepath can be represented by θ with two nodes (or rows), one at the point of retirement ($t = 0$) and one at the terminal age ($t = T$), with the allocations at these nodes representing the initial and final allocations, respectively. More complex glidepaths can be represented by adding more nodes and allowing for non-linear interpolation between them. Adding more nodes allows for more complex shapes. By varying the number of nodes, we can also determine the sensitivity of the optimal policy to the granularity of the time dimension, which can provide insights into how important it is to have a finely-tuned glidepath versus a more simple one.

Figure 3 illustrates this interpolation process for a three-asset portfolio (stocks, bonds, and cash) with three time nodes at $t = 0, 15, 30$. Panel (a) shows the policy settings matrix θ, where each row specifies the allocations to stocks and bonds at a discrete time node, with the cash allocation computed residually via φ. Panel (b) shows the resulting continuous allocation surface obtained by linear interpolation, with the vertical dashed lines marking the discrete nodes where θ is defined.

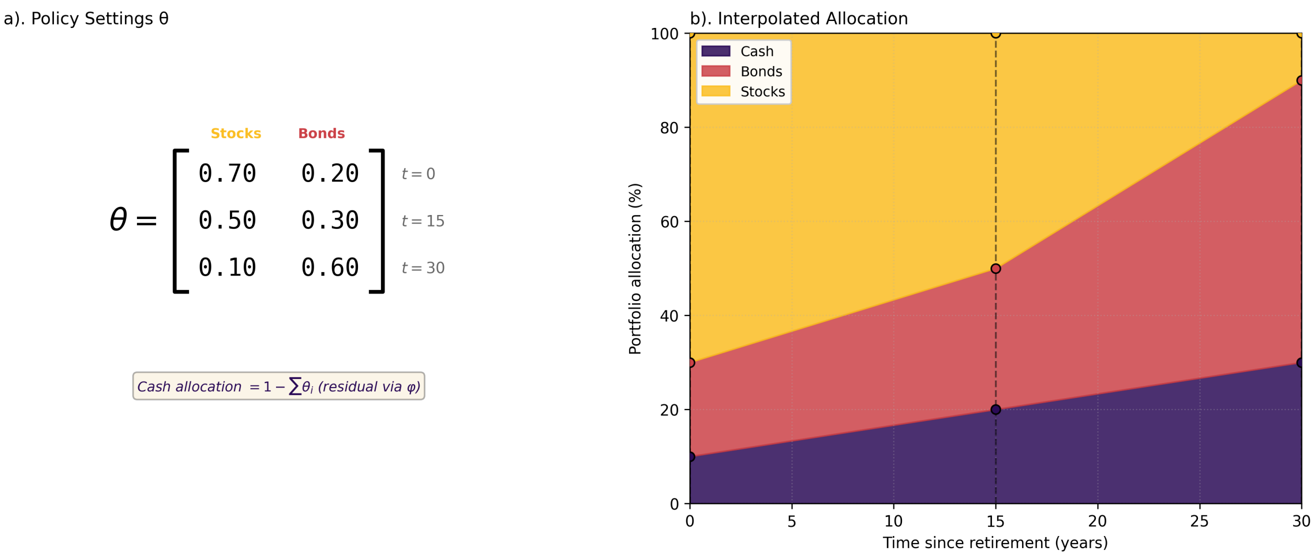

Figure 3: Linear interpolation for a time-based allocation policy with three assets and three time nodes.

This approach also allows us to construct a wealth-responsive glidepath, which, rather than being a function of time, is a function of the investor's current wealth level. In this case, the policy function can be represented as a function of wealth:

$$\mathcal{A}(\theta, w_t) = \boldsymbol{a}_t \tag{19}$$

And, similar to above, θ is a matrix of parameters that specify the allocations at discrete wealth levels or nodes ($\{w_0, w_1, \ldots, w_M\}$), with interpolation used to determine the allocation for wealth levels that fall between the defined grid points.

The final form of the policy function that we will consider is one that allows for dependence on both time and wealth, which we can represent:

$$\mathcal{A}(\theta, t, w_t) = \boldsymbol{a}_t \tag{20}$$

This directly corresponds to the control-matrix approach of Mäkinen and Toivanen (2024), where the policy settings θ are now a three-dimensional tensor of dimension $M_t \times M_w \times (k - 1)$, defined on a discrete grid of time nodes $\{t_0, t_1, \ldots, t_{M_t}\}$ and wealth nodes $\{w_0, w_1, \ldots, w_{M_w}\}$. Each element $\theta_{i,j}$ specifies the risky asset allocations when time is at node t_i and wealth is at node w_j.

For any time t and wealth w such that $t_i \leq t < t_{i+1}$ and $w_j \leq w < w_{j+1}$, the interpolated allocation is given by bilinear interpolation:

$$\mathbf{a}_t = \mathcal{A}(\theta, t, w) = \varphi\left((1-\alpha_t)(1-\alpha_w)\theta_{i,j} + (1-\alpha_t)\alpha_w\theta_{i,j+1} + \alpha_t(1-\alpha_w)\theta_{i+1,j} + \alpha_t\alpha_w\theta_{i+1,j+1}\right) \tag{21}$$

where the interpolation weights are computed as:

$$\alpha_t = \frac{t - t_i}{t_{i+1} - t_i}, \quad \alpha_w = \frac{w - w_j}{w_{j+1} - w_j} \tag{22}$$

This bilinear interpolation ensures that the policy varies smoothly across both time and wealth dimensions while requiring only $M_t \times M_w \times (k-1)$ parameters. The transformation φ is then applied to the interpolated risky allocations to obtain the full allocation vector including the safe asset. For time or wealth values that fall outside the bounds of the defined grid (that is, $t < t_0$, $t \geq t_{M_t}$, $w < w_0$, or $w \geq w_{M_w}$) we apply nearest-node extrapolation, returning the allocation specified at the closest boundary node rather than extrapolating beyond the grid.

To illustrate this more concretely, consider Figure 4, which shows a simple control matrix for a two-asset portfolio. The matrix has six time steps (0 to 5) and five wealth levels (0 to 4), giving a 6 by 5 grid. Each cell corresponds to a risky-asset allocation (with the remainder allocated to the safe asset). For instance, if an investor is at time step 2 with wealth level 3, the policy returns the allocation in that cell. For time or wealth values between grid points, we use bilinear interpolation to determine the allocation, which allows for a smooth transition between different allocation strategies across the state space. For example, if an investor is at time step 2.5 with wealth level 3.5, we would interpolate between the four surrounding cells to determine the appropriate allocation.

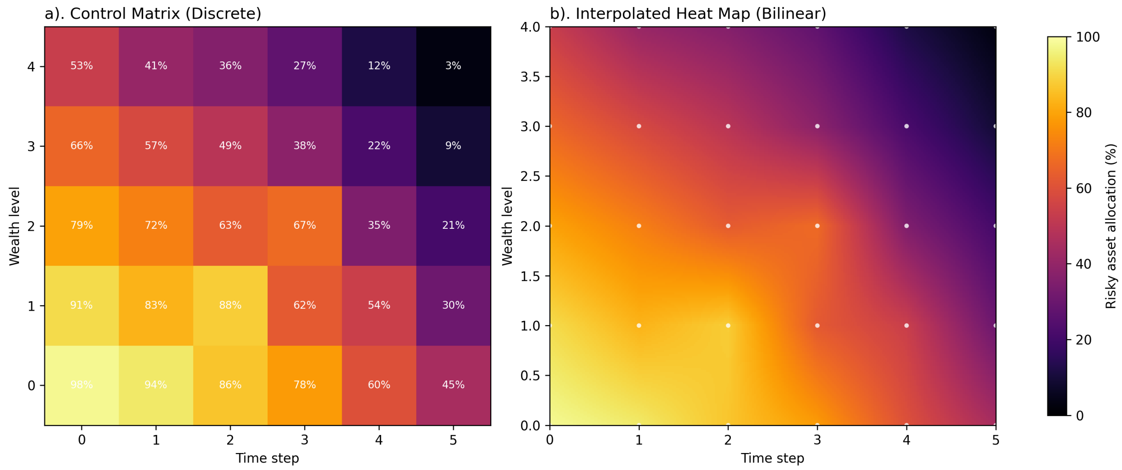

Figure 4: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

A key concept in this framework is that the dimensions of θ correspond to the different state variables that the policy can depend on. Here we have extended it to two dimensions, time and wealth, which allows for policies that can adjust allocations based on both the investor's current age and their current wealth level. It is also possible to extend this approach to allow for dependence on additional state variables, such as health status or macroeconomic indicators, by adding additional dimensions to the control matrix. These can be both discrete and continuous, and the use of interpolation allows us to have a parsimonious parameterisation of the policy function while still allowing for a high degree of flexibility in the shape of the asset allocation strategy across different state variables. Compared to a neural network approach (as in Duarte et al. (2021)), the control-matrix approach allows for a more interpretable policy representation, as the parameters directly correspond to allocations at specific points in the state space, and the interpolation provides a clear mechanism for how the policy varies across different states. However, it is probably less flexible and will have a higher number of parameters than a neural network at higher dimensions, which is a trade-off that we are willing to make for the sake of interpretability in this analysis.

4.4 Methods and Data for Simulating and Bootstrapping Market Conditions

Central to our analysis is the ability to simulate a large number of realistic market condition trajectories, particularly asset class returns and inflation. As previously noted, there are two broad approaches to simulating returns: synthetic return generation using a parametric model, and bootstrapping from historical returns.

> **Three Market Condition Generators**
>
> Throughout this report, we employ three distinct market condition generators that produce simulated returns, inflation, and volatility regimes. Each generator creates return paths through different mechanisms, all capturing full market dynamics—returns, inflation, and cross-asset correlations—rather than returns alone.
>
> **Synthetic Markets (IID GBM)**: A parametric generator based on Consilium's long-run return expectations and historical volatilities. Generates return paths assuming independent, normally-distributed returns with no time-series dependence or volatility clustering.
>
> **Historical Markets (Full Period)**: A block-bootstrap generator using asset returns and inflation figures from the Jordà–Schularick–Taylor (JST) Macrohistory Database spanning 1870–2020. Generates empirical return paths that preserve time-series properties and capture dynamics across multiple economic regimes, wars, and financial crises.
>
> **Historical Markets (Modern Era)**: A block-bootstrap generator using JST data from 1950–2020. Generates return paths from a more homogeneous period potentially more relevant for contemporary investors, but with reduced diversity of economic regimes.
>
> Comparing results across these three generators reveals the sensitivity of optimal allocation to return assumptions (synthetic vs. historical) and historical context (full period vs. modern era). All three generators include asset-class-specific returns, inflation dynamics, and realistic correlations.

The advantage of synthetic return generation is that it allows us to explore a wider range of possible return dynamics and also to create a larger number of return trajectories than would be possible with historical data alone. The disadvantage is that it relies on strong assumptions about the return-generating process, which may or may not be an accurate representation of reality. For example, the easiest, and therefore most commonly used, model is the geometric Brownian motion (GBM) model, which assumes that returns are normally distributed and independent over time. In reality, asset returns exhibit features such as volatility clustering, fat tails, and serial correlation, which are not captured by the time-independent GBM model. Time series models such as GARCH can capture some of these features, but they also introduce additional complexity and make their own set of assumptions that may not hold in practice. Bootstrapping from historical returns, on the other hand, allows us to capture the empirical distribution of returns without making strong parametric assumptions. However, it is limited by the length and quality of the historical data. Underpinning both approaches is the assumption that the future will be similar to the past; in the case of synthetic return generation, this is because the model parameters are typically estimated, informed, or evaluated against historical data; in the case of bootstrapping from directly resampling historical returns. This assumption may not hold if there are structural changes in the economy or financial markets that lead to different return dynamics in the future. This limitation is inherent to this type of research, and it forms the underlying uncertainty in any conclusions we draw from our simulations.

We will generate a large number of market condition trajectories (including asset class returns and inflation) using both approaches and evaluate the performance of different asset allocation strategies under all three market condition generators. We will use three asset classes in our analysis: short-term bills (cash), fixed income (bonds), and equities. These asset classes are chosen because they represent the main building blocks of a typical retirement portfolio.

4.4.1 Geometric Brownian Motion (IID GBM) Model for Synthetic Return Generation

Consilium has historically used a simple geometric Brownian motion (GBM) model to generate synthetic returns for its analysis. Because the company already has established expectations of long-run returns and volatilities for different asset classes in its internal models, the GBM model is a convenient way to generate return trajectories that are consistent with these expectations. Consilium has used this methodology extensively for generating return paths for its internal models and GPS tool. The GBM model assumes that the returns of the underlying assets are normally distributed and independent over time, which allows for a simple and tractable way to generate return trajectories.

Consilium's GBM model does not simulate the returns of the underlying assets directly, but rather simulates the returns of the portfolio as a whole. The key inputs to the GBM model are the expected return and covariance matrix of the portfolios. Expected returns of the portfolio are derived from the expected returns obtained from the factor-analysis of the underlying assets. Covariances are obtained from back-filled historical monthly portfolio returns going back to 1995.

Our approach differs from Consilium's typical implementation of the GBM model in that we will simulate the returns of the underlying asset classes (bills, bonds, equities) rather than simulating the returns of the portfolio directly. This allows us to generate return trajectories for each asset class, which can then be combined according to different asset allocation policies to produce portfolio return trajectories. This is important for our analysis because we want to continuously vary the asset allocation across the three asset classes, which is not possible if we only simulate portfolio returns directly. By simulating the underlying asset class returns, we can evaluate how different asset allocation strategies perform under a wide range of return dynamics for each asset class. The disadvantage of this approach is that the underlying composition of the asset classes cannot vary across different risk levels. For instance, Consilium's internal models vary the allocation to domestic and international equities and bonds across the different risk profiles, while our approach will use a single aggregate return series for equities and bonds. We will base our aggregate return series for each asset class on Consilium's Classic 70/30 model portfolio allocation, which represents a fairly even split between the allocations in Consilium's more and less risky model portfolios, which will mean that the asset class return dynamics we simulate will be broadly, but not perfectly, consistent with the underlying return-generating process used in Consilium's internal models.

We derive the expected returns and covariance matrix following the same approach as Consilium, using the factor-analysis of the underlying assets to obtain expected returns for each asset class, and using back-filled historical monthly returns to estimate the covariance matrix. We will then use these inputs to simulate return trajectories for each asset class using the GBM model. Consilium has traditionally treated CPI inflation as an exogenous variable that is not part of the return-generating process, but we will include it as an additional asset class in our simulations, which allows us to capture the correlation between inflation and asset returns and to evaluate how different asset allocation strategies perform under different inflation scenarios.

The expected gross returns and covariance matrix for the three asset classes and CPI inflation are shown in Table 4. These parameters are based on Consilium's internal models and are consistent with the return dynamics that we will simulate using the GBM model.

Table 4: GBM input parameters for gross returns and covariance matrix.

	Equities	Bonds	Cash	CPI
Gross return	9.31%	4.26%	1.85%	2.50%

	Cash	Bonds	Equities	CPI
Cash	0.00074818	0.00107890	-0.00011636	0.00001394
Bonds	0.00107890	0.00327030	-0.00015073	-0.00025973
Equities	-0.00011636	-0.00015073	0.01928772	-0.00051254
CPI	0.00001394	-0.00025973	-0.00051254	0.00025696

4.4.2 Block Bootstrap Methodology

Our block bootstrap implementation follows Anarkulova et al. (2025), adapted for a different dataset and asset allocation structure. The key advantage of the block bootstrap is that by resampling contiguous blocks of historical returns rather than individual observations, we preserve the short-run time-series dependencies and cross-sectional correlations present in asset returns while generating a large and diverse set of synthetic paths for Monte Carlo simulation.

We construct return series for four variables—Equities, Bonds, Bills, and CPI inflation—using annual data from the Jordà-Schularick-Taylor (JST) Macrohistory Database (Jordà et al., 2017), which provides total returns and macroeconomic variables for 18 advanced economies from 1870 to the present. This long-run panel data allows us to capture return dynamics across multiple business cycles, market regimes, wars, and financial crises.

At each bootstrap iteration, we randomly designate one country as the "domestic" or "home" country, representing the New Zealand investor's perspective. Each asset class is then constructed as a weighted combination of domestic and international returns, reflecting realistic portfolio allocations for a small open economy. Specifically, each asset class k is characterized by two parameters: home bias $\omega^{(k)}$ specifying the allocation to the domestic market, and the hedging ratio $\eta^{(k)}$ specifying what fraction of the foreign exposure is currency-hedged. The asset class return combines domestic returns, currency-hedged international returns, and unhedged international returns according to these weights. We calibrate these parameters based on Consilium's typical 70/30 model portfolio allocations as shown in Table 5.

Asset Class	**Home Bias** (ω)	**Hedging Ratio** (η)
Equities	0.20	0.50
Bonds	0.33	1.00
Bills	1.00	1.00
CPI	1.00	0.00

Table 5: Asset class construction parameters reflecting realistic portfolio composition for New Zealand retail investors.

International equity returns are constructed as market-capitalization-weighted aggregates across eligible foreign countries, based on weights from the World Bank's World Development Indicators (World Federation of Exchanges, 2024) supplemented with historical estimates from Global Financial Data (sourced from Taylor, 2023). International bond returns use weights based on GDP and government debt-to-GDP ratios from the JST database. Currency conversions are performed via USD as the intermediary currency, and hedged returns eliminate currency risk while unhedged returns include both asset and currency components. Detailed descriptions of the eligibility criteria, weight construction, currency conversion methodology, and asset class formulas are provided in Appendices C–F.

To generate bootstrapped market condition paths, we resample contiguous blocks of years from the historical data, where each block corresponds to a continuous period within a single country's history and includes the full set of variables—asset returns, inflation (CPI), and associated market conditions. The JST database has gaps due to wars, data quality issues, and currency crises, so we first identify all continuous periods and prevent sampling across discontinuities.

For each simulated path of length $T = 40$ years, we repeatedly draw random blocks until we accumulate the required number of observations. Each block starts at a randomly selected position within a continuous country period, and has a length drawn from a Poisson distribution with mean $\lambda = 10$ years. Blocks are truncated at period boundaries to prevent artificial transitions between countries. We pre-generate approximately 500,000 paths of 40 years each by partitioning a large batch of 2 million block draws into non-overlapping segments.

This procedure ensures that each bootstrap path preserves the empirical autocorrelation structure within blocks (capturing business cycle dynamics and mean reversion), the cross-sectional correlation among asset classes, inflation, and their co-movements, and the time-varying composition of the global investment opportunity set. Critically, domestic inflation dynamics are sampled from the same country blocks as the asset returns, ensuring that inflation and return correlations observed historically are preserved in the simulated paths. The detailed sampling algorithm, period identification procedure, and implementation notes are provided in Appendix G.

We consider two historical periods for our bootstrapping: the full sample from 1870 to the present, which captures a wide range of economic regimes and return dynamics; and a more recent sample from 1950 to the present, which may be more relevant for contemporary investors but is shorter and less diverse. By comparing results across these two samples, we can assess the sensitivity of our conclusions to different historical contexts and the potential impact of structural changes in the economy and financial markets over time.

4.5 Tax and Fees

The treatment of taxes and fees is an important consideration in retirement planning, as they can have a significant impact on the net returns that investors receive and therefore on the optimal asset allocation strategy. Retirees do not typically access securities directly, but rather invest through a retirement savings vehicle. Even after the age of 65, when they can withdraw from their KiwiSaver, most retirees will continue to invest through their KiwiSaver or some other type of managed fund, which means that they will be subject to the fees charged by these funds. The tax treatment of these assets is an incredibly complex topic, and it is beyond the scope of this report to provide a comprehensive analysis of it.

For example, dividends paid from assets held in foreign markets are subject to both foreign and domestic taxes, and the interaction between these taxes can be complex. In some cases, investors may be able to claim a foreign tax credit for the taxes paid on foreign dividends, which can reduce their overall tax burden. Further complicating this are the various tax rules around FIF and PIE investment vehicles that are commonly used by New Zealand investors to access international markets. Making a full analysis of the tax implications of different asset allocation strategies would require a detailed understanding of both the domestic and international tax rules, data about distributions and tax rates for different asset classes, and a careful modelling of how these taxes would apply to the returns generated by different asset allocation strategies. This is not something that either of our models or datasets have the capability to do. For this reason, we have chosen to use a simplified approach to taxes and fees in our analysis, which is to apply a flat fee and tax rate to the returns generated by our simulations, estimated from Consilium's internal models. While, in reality, the effective tax and fee rates that investors face will vary across asset classes and over time, this simplified approach allows us to capture the general impact of taxes and fees on net returns without having to model the complex interactions between different tax rules and asset class characteristics.

We use a flat fee rate to approximate the fees and taxes that investors would face on their returns. This fee rate is estimated from Consilium's Classic 70/30 model portfolio, and includes estimates for fund management fees, taxes on dividends and interest, foreign tax credits, domestic costs, rebalancing costs, and other relevant fees tied to brokerage. The fee rate is applied to the returns generated by our simulations to obtain net returns, which are then used in our optimisation of asset allocation strategies. This approach is crude and will understate tax in good years and overstate it in bad years, but it allows us to capture the general impact of taxes and fees on net returns without having to model the complex interactions between different tax rules and asset class characteristics. These rates are shown in Table 6.

Asset Class	**Estimated Tax and Fee Rate**
Equities	1.44% per year
Bonds	1.41% per year
Bills	0.61% per year

Table 6: Estimated flat tax and fee rates applied to simulated returns for each asset class.

4.6 Representative Retiree Household and Spending Targets

For the purposes of this report, we are interested in simulating a household consisting of two retirees of the same age. We simulate the wealth trajectories of this household over a 35-year retirement period, from age 65 to age 100. Our retirees have an assumed real consumption floor equal to the "No Frills Metro" expenditure as defined by Matthews (2025). The couple also have a desired spending target equal to the "Choice Metro" expenditure, which declines by 2% in real terms each year (until it reaches the consumption floor at around age 90), consistent with the data and recommendations presented in Retirement Income Interest Group (2024). In this setup, our retiring couple know approximately how much they want to spend in retirement, and how little they absolutely need. They are primarily concerned with the risk of running out of money. Spending above the floor is valuable to them, but they are not willing to take on substantial additional risk to achieve it. This is a common situation for many retirees, who have a good sense of their desired lifestyle in retirement and the minimum they need to get by, and are primarily concerned with ensuring that they do not run out of money.

4.7 Heuristics of Risk and Success

Where Anarkulova et al. (2025) (and others) use a version of the CRRA utility function based on retiree consumption, we choose instead to focus strictly on the risk of running out of savings. In particular, we choose to minimise what we call the "bankruptcy density" of the retiree's wealth trajectory over a 35-year retirement period (from age 65 to age 100). The bankruptcy density is a measure of how frequently the retiree's wealth hits zero across all timesteps and simulation paths. Given that our retirees have expenditure that is strictly larger than their income, wealth trajectories that hit zero wealth will remain at zero for the remainder of the retirement period. The bankruptcy density therefore captures not only whether the retiree runs out of money, but also punishes strategies that hit zero wealth earlier in retirement and thus experience bankruptcy over a longer sustained period. It should be noted that our method of simulating lifetime wealth trajectories is also

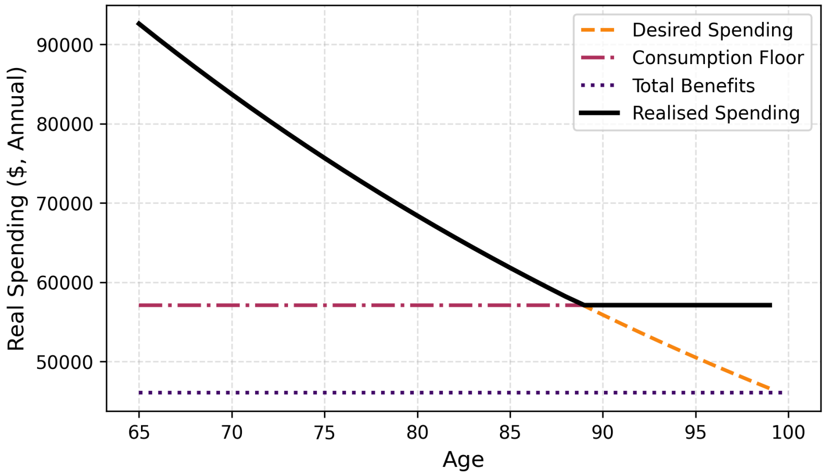

Figure 5: Desired spending targets and consumption floors for the simulated retirees, based on the expenditure guidelines from Matthews (2025) and Retirement Income Interest Group (2024). The desired spending target declines in real terms by 2% per year, while the consumption floor remains constant in real terms.

fully capable of generating consumption trajectories that could be used to optimise a CRRA utility function, which could be a useful extension of this work.

While this approach is not as common in the literature, we think it presents a more interpretable goal with less ambiguity. It is difficult, for instance, to determine what the appropriate parameterisation of a CRRA utility function should be for a given retiree, whereas simply minimising the risk of running out of money is a more straightforward goal. Additionally, where the investor is likely to prematurely deplete savings, consumption utility becomes dominated by those destitute retirees and becomes a proxy for bankruptcy density. Furthermore, because our investors have a fixed upper limit on their desired spending, the utility of consumption is bounded, which we have found tends to result in worse convergence properties for the optimisation of the CRRA utility function. While this is outside the scope of this report, we note that a potential extension of this work would be to relax this upper bound, or even allow it to be a function of wealth, which would permit a more flexible utility function that could capture the trade-offs between consumption and wealth in a more nuanced way. This would also allow for the optimisation of dynamic spending strategies in addition to asset allocation strategies, which is an important area of research in retirement planning that we have not explored in this report. We also do not consider the utility of bequests in our analysis. Based on the literature review earlier in this report, this motive is often small, and many studies have found that large bequests are often accidental.

4.7.1 Bankruptcy Density and the Cost Function for Optimisation

We define the bankruptcy density formally as follows in Equation 23, where $H(\cdot)$ is the step function defined by $H(-w) = 1$ if $w \leq 0$, and $H(-w) = 0$ otherwise. Here, N_{sims} is the total number of simulation paths, and T is the total number of time periods in each simulation path. The bankruptcy density is therefore the average number of time periods across all simulation paths in which wealth is zero, divided by the total number of time periods. A lower bankruptcy density indicates a more successful strategy, as it means that wealth is less likely to hit zero and that retirees are more likely to be able to maintain their desired spending targets throughout retirement.

$$\text{Bankruptcy Density} = \frac{1}{N_{\text{sims}} T} \sum_{i=1}^{N_{\text{sims}}} \sum_{t=1}^{T} H(-w_{i,t}). \tag{23}$$

As defined above, this metric is not differentiable, so for optimisation we use a differentiable approximation to bankruptcy density as our cost function. Specifically, we replace the step function $H(-w_{i,t})$ with a sigmoid surrogate. A smooth approximation is $H(-w_{i,t}) \approx 2(1-\varsigma(kw_{i,t}))$ if $w \geq 0$, and, up to an additive constant that does not affect optimisation, this is equivalent to maximising $\varsigma(kw_{i,t})$ or minimising $-\varsigma(kw_{i,t})$, as in Equation 24.

$$\text{Penalty}_{\text{wealth}} = -\frac{1}{N_{\text{sims}}\,T}\sum_{i=1}^{N_{\text{sims}}}\sum_{t=1}^{T}\varsigma\big(k\,w_{i,t}\big) \quad \text{where} \quad \varsigma(z) = \frac{1}{1+e^{-z}}\,. \tag{24}$$

Here, k is a scaling parameter that controls the steepness of the sigmoid function. Because the sigmoid is smooth and continuous, we can compute gradients with respect to policy parameters during optimisation. The penalty term is negative, so strategies that result in lower wealth (and therefore higher bankruptcy density) are penalised more heavily, while strategies that maintain higher wealth are rewarded. The parameter k determines the steepness of the sigmoid "knee," and therefore how strongly the penalty increases as wealth approaches zero. For large $k \to \infty$, the surrogate approaches a step-function objective based on $H(-w)$. For smaller k, the function also penalises low but positive wealth outcomes, encouraging solutions that maintain a wealth buffer. However, if k is too small, the penalty becomes ineffective and the optimisation takes on a more greedy, wealth-maximising character that can lead to riskier strategies and higher bankruptcy risk. Conversely, if k is too large, the optimisation landscape becomes ill-conditioned. In our calibration, $k = 1 \times 10^{-4}$ provides a good balance between downside protection and tractable optimisation dynamics.

4.7.2 Bankruptcy Probability

Another important metric is "bankruptcy probability" (also called "probability of ruin"), defined as the probability that a retiree's wealth hits zero at any point during retirement. This is a traditional metric in the literature, and it is calculated as the proportion of simulation paths in which wealth hits zero at least once during the retirement period:

$$\text{Bankruptcy Probability} = \frac{1}{N_{\text{sims}}}\sum_{i=1}^{N_{\text{sims}}} I\left(\min_{t=1}^{T} w_{i,t} \le 0\right) \tag{25}$$

where $I(\cdot)$ is an indicator function equal to 1 if the condition is true and 0 otherwise. In contrast to bankruptcy density, which measures the cumulative duration of bankruptcy events, bankruptcy probability captures only whether bankruptcy occurs at all, regardless of timing or persistence.

While this metric does not capture the timing or duration of bankruptcy events, it provides a useful summary measure of the overall risk of running out of money in retirement. Importantly, there is a potential trade-off between bankruptcy density and bankruptcy probability. A strategy could theoretically have a low bankruptcy density but high bankruptcy probability if bankruptcies occur late in the retirement period (near end of life). For example, a strategy might successfully preserve wealth throughout most of retirement but eventually deplete it in the final years; such a strategy would exhibit many bankruptcy paths but accumulate relatively few total timesteps of bankruptcy. Conversely, a strategy might prevent bankruptcy entirely in many paths while experiencing early and sustained bankruptcy in others, producing fewer bankruptcy paths overall but higher cumulative bankruptcy timesteps. We will use both metrics to evaluate the performance of different asset allocation strategies, and we will discuss how these metrics relate to each other and to the overall risk profile of the retiree household.

A Note on the Appropriateness of these Risk Measures for more Risk-Tolerant Investors

While the risk measures used here are appropriate for the very risk-averse retirees we are modelling, they are not the only relevant metrics for evaluating all asset allocation or spending strategies, particularly for clients who are looking to maximise the expected utility of their consumption rather than just minimise the risk of running out of money. While the scope of this report is limited, a potential extension of this work would be to evaluate the performance of different asset allocation strategies under a wider range of risk measures, including those that capture the expected utility of consumption, the variability of consumption, or other dimensions of risk and reward that may be relevant for different types of investors. This may also allow the optimisation of dynamic spending strategies in addition to asset allocation strategies, which is an important area of research in retirement planning that we have not explored in this report. Here we focus only on retirees who have relatively fixed spending targets, but a valuable future extension to this work would be to evaluate the performance of different asset allocation strategies for retirees with more flexible spending targets, who may be willing to adjust their consumption in response to changes in their wealth or market conditions. This would require a more complex modelling framework that allows for dynamic spending strategies, and it would also require the use of different risk measures that capture the trade-offs between consumption and wealth in a more nuanced way.

4.8 Optimisation Procedure

The optimisation of θ is central to our analysis, as it determines the asset-allocation strategy that maximises our chosen risk measure over simulated wealth trajectories. Mäkinen and Toivanen (2024) use an adjoint method to compute gradients of the risk measure with respect to policy parameters, enabling efficient optimisation in high-dimensional parameter spaces. This approach is powerful, but complex to implement, and it tends to restrict objective functions to those that depend only on terminal wealth. For this reason, we adopt a more direct approach based on automatic differentiation in Python using PyTorch (Paszke et al., 2019). This allows us to compute gradients without deriving adjoint equations and to use risk measures that depend on the full wealth trajectory rather than terminal wealth alone. The trade-off is lower computational efficiency than an adjoint implementation at high dimensionality, in exchange for greater flexibility and simpler implementation. Our simulation framework is fully differentiable, so we optimise policy parameters directly with a gradient-based method. We initialise each optimisation run at 100% cash (as in Mäkinen and Toivanen (2024)). We also tested alternative initialisations, including random initialisation and initialisation based on the optimal fixed-allocation strategies discussed in Section 5.1. We found that initialisation choice did not materially affect final results.

To enforce the allocation constraints in Equation 13 (non-negativity and weights summing to at most one) during gradient descent, we project policy parameters onto the feasible set after each gradient update using the simplex projection algorithm of Duchi et al. (2008). This efficient $\mathcal{O}(n \log n)$ procedure computes the Euclidean projection onto the constraint set, ensuring feasibility while staying as close as possible to the unconstrained update. A detailed description is provided in Appendix A. We use the Adam optimiser (Kingma & Ba, 2014), which adaptively adjusts the learning rate for each parameter using first- and second-moment gradient estimates. In our setting, this is especially useful in flat regions of the policy-response landscape, which are common for high-wealth retirees and for states near the end of the simulation horizon. To keep updates tractable in this constrained problem, we also project gradients onto the feasible set before applying Adam, ensuring that updates proceed in constraint-respecting directions. Without this step, Adam can become stuck at sub-optimal points where unconstrained gradients point outside the feasible region. Details of this gradient-projection step are provided in Appendix A.1.

5 Results

In this section, we apply the framework developed above to evaluate candidate retirement allocation policies in a New Zealand setting. We begin by specifying the experimental setup, including the representative retiree household and spending targets, the heuristics of risk and success, and the optimisation procedure. We then present the results of our analysis. We will proceed from zero dimensions to one-dimensional policies (time-based and wealth-based glide-paths) and finally to two-dimensional control matrices that permit adaptation along both dimensions. This progression allows us to examine how successive degrees of adaptability affect portfolio performance and risk outcomes. We will discuss the results of each of these policy classes in turn, comparing the evolution of the policy matrix as we vary experimental parameters, and evaluating the performance of the resulting policies under all three market condition generators (which include both asset returns and inflation dynamics). We will also discuss the implications of our findings for retirement planning in New Zealand, and highlight areas for future research.

To address the role of initial wealth, we employ two complementary optimisation approaches. First, we optimise *wealth-specific policies*: independent policies tailored for each wealth level in our range ($500,000 to $1,000,000 in $50,000 increments). This reveals how the optimal allocation *should vary* with wealth in an idealised setting where each retiree knows their exact starting position. Second, we optimise *unified policies*: a single policy designed to perform well across the *entire* wealth range simultaneously. This more realistic scenario represents a policy that must work well for retirees across a range of potential starting positions.

5.1 Asset Allocation in Zero Dimensions: Fixed Allocation Strategies

We begin with the simplest case: a policy with zero decision nodes, which we refer to as a fixed allocation strategy. Such a policy maintains constant weights across all assets throughout the retirement horizon, regardless of time elapsed or current wealth. It can be represented by a single parameter vector θ that specifies the allocation proportions and does not vary with time or wealth. Fixed allocation strategies serve three important roles in our analysis. First, they provide a transparent baseline against which to measure the performance gains of the more sophisticated dynamic strategies we develop later. Second, they isolate the effect of asset-class selection and return dynamics from any adaptive rebalancing benefit. Third, because the optimisation landscape is considerably simpler in the zero-dimensional case, we can achieve fast and reliable convergence and build confidence in our optimisation procedure before scaling to higher-dimensional policies.

We examine fixed allocation strategies under both optimization approaches introduced earlier: first optimising separate (wealth-specific) policies for each initial wealth level to reveal how allocations should adapt, and then optimising a single unified policy that balances performance across all wealth levels. This comparison isolates the trade-offs retirees face when unable to perfectly tailor their allocation to their true starting position.

5.1.1 Allocation Optimisation for Known Initial Wealth

Studying how the optimal allocation varies with initial wealth provides several interesting insights into both the structure of the problem and the properties of the datasets. The resulting allocations are shown in Figure 6. Historical Markets (Full Period) produces fixed 100% equity allocations for all initial wealth levels, whereas Historical Markets (Modern Era) and Synthetic Markets (IID GBM) produce allocations that vary with initial wealth. The dominance of the all-equity allocation in Historical Markets (Full Period) is likely due to the presence of several historical periods of high domestic inflation. In our asset-class construction, unhedged international equity exposure provides a natural hedge against these episodes. We discuss this further later in the report, but for now we note that these historical periods of high inflation are likely to be a key driver of the optimal allocation to equities in this case. Historical Markets (Modern Era) and Synthetic Markets (IID GBM), on the other hand, do not contain these periods of high inflation, and therefore the optimal allocation to equities is lower in these datasets.

Both Historical Markets (Modern Era) and Synthetic Markets (IID GBM) show a clear trend of decreasing equity allocation as initial wealth increases, with low-wealth retirees being optimally allocated 100% to equities. This pattern may appear counter-intuitive at first glance; one might expect that retirees with higher initial wealth would be able to take on more risk and therefore allocate more heavily to equities. While there is a reduction in safe asset allocation at the highest end of the allocation optimised on Historical Markets (Modern Era), the overall trend is still one of decreasing equity allocation with increasing initial wealth. This is likely due to the extremely conservative nature of oour cost function which is only concerned with the risk of running out of money. As it is, for low-wealth retirees the main risk to their retirement security is not market risk, but the fact that their future consumption already exceeds the capacity of their savings. In such cases, they face a high risk of running out of money regardless of market dynamics. We have termed this the "hail-mary zone" of wealth, in which the optimal strategy is to maximise expected returns at all costs and allocate entirely to the riskiest assets in the hope that favourable market returns will allow the retiree to achieve a sufficient level of wealth to accommodate their future spending. By contrast, for higher-wealth retirees, the risk of exceeding the consumptive capacity of their savings is lower, and market risk becomes a more significant factor in the optimisation. In this case, the optimal strategy is to diversify into safer asset classes to reduce market risk and protect the consumptive capacity of their wealth in the face of adverse market conditions. In this thinking, when the investor's wealth and consumption requirements place them in the hail-mary zone, the strategy is subject to *budgetary dominance* and the optimal allocation aims to maximise expected asset returns a desperate hope of increasing their consumptive capacity ahead of it being depleted by their consumption. When the investor's wealth and spending requirements place them outside this zone, the strategy is *market dominated*, and the optimal allocation increasingly aims to protect the consumptive capacity of their savings in the event of adverse market conditions.

We speculate that the wealth level at which the optimal allocation switches from a hail-mary approach to one that minimises market risk is likely to be an important threshold in the optimisation problem. A noteworthy interaction of these risks can be seen in the terminal wealth statistics. In both Synthetic Markets (IID GBM) and Historical Markets (Modern Era), mean terminal wealth decreases slightly as initial wealth increases. In Synthetic Markets (IID GBM), this occurs at approximately $700k, and in Historical Markets (Modern Era), between approximately $600k and $750k. We interpret this as evidence that the optimal strategy is trading long-term expected returns for lower market risk, resulting in a lower terminal wealth but a lower likelihood of running out of money. We also think that another reason Historical Markets (Full Period) produces a 100% equity allocation for all initial wealth levels is that the presence of historical periods of high inflation in this dataset means that the retiree's expected spending is systematically higher across that ensemble of return paths. As a result, all scenarios tested in that dataset are budgetarily dominated hail-mary scenarios.

Another interesting difference between the datasets can be observed in systematically higher allocations to bonds in Synthetic Markets (IID GBM) than in Historical Markets (Modern Era). Although both allocations have comparable allocations to equities, Synthetic Markets (IID GBM) has higher allocations to bonds and lower allocations to cash, whereas Historical Markets (Modern Era) almost never allocates to bonds. We investigate this further in a later section. Anarkulova et al. (2025) did speculate that IID return assumptions (as in Synthetic Markets IID GBM) which ignore time-series effects, tend to over-allocate to bonds relative to bootstrapped return assumptions.

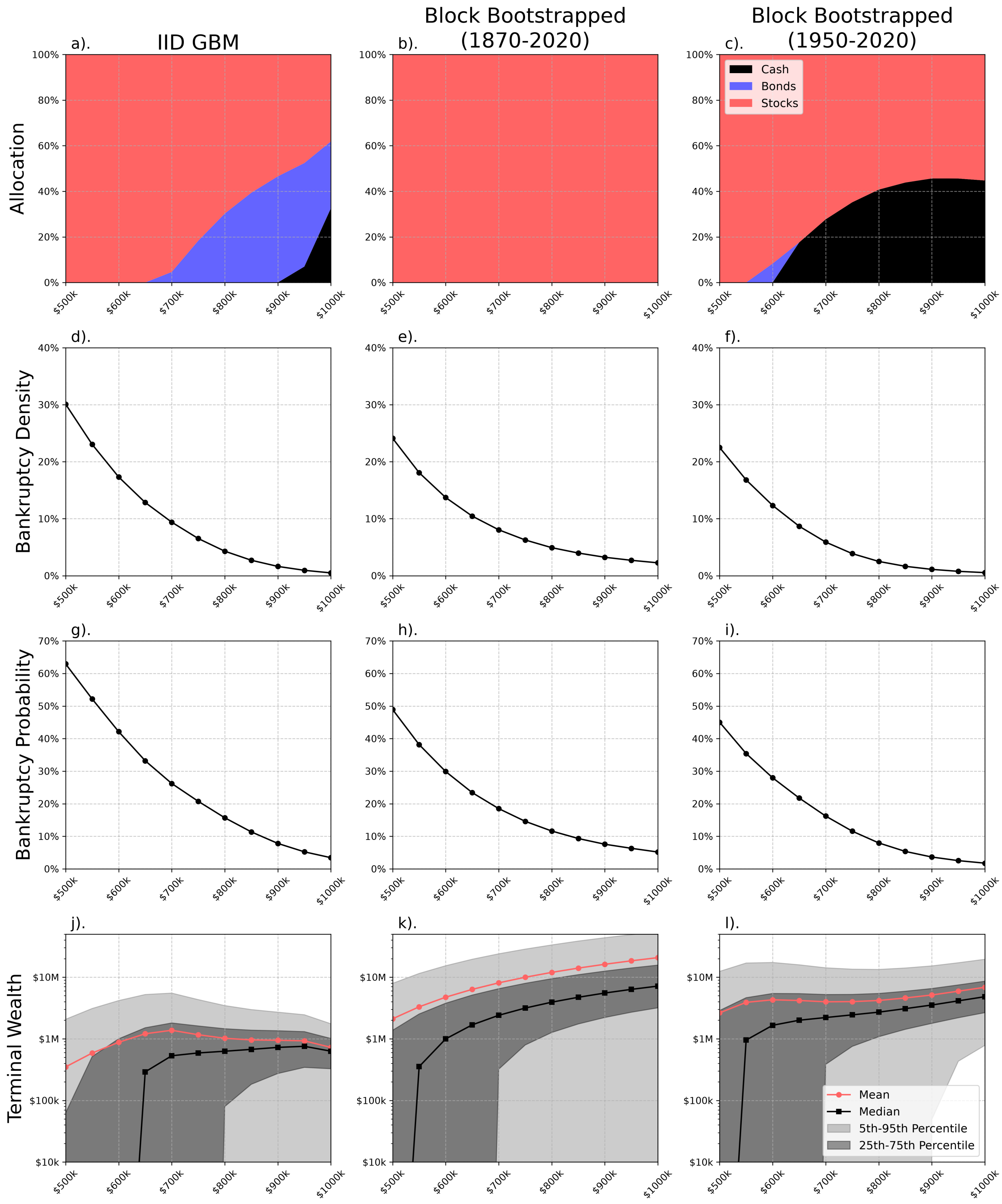

Figure 6: Optimal fixed allocation strategies and outcomes for different initial wealth levels and market condition generators. Figures (a, b, c) show the optimal allocations for the Synthetic Markets (IID GBM) generator, the Historical Markets (Full Period) generator, and the Historical Markets (Modern Era) generator, respectively. The x-axis shows the initial wealth level, and the y-axis shows the allocation to each asset class. Figures (d, e, f) show the corresponding bankruptcy densities for the optimal fixed allocation strategies under each market condition generator. Figures (g, h, i) show the bankruptcy probabilities for the optimal fixed allocation strategies under each market condition generator. Figures (j, k, l) show the corresponding expected final wealth distributions for the optimal fixed allocation strategies under each market condition generator.

5.1.2 Allocation Optimisation Across Distributed Initial Wealth Levels

Now we consider the unified case in which we optimise a single fixed allocation that performs well across simulated wealth lifetimes distributed over a range of initial wealth levels. The resulting allocations are shown in Table 7. In this case, the optimal allocation is 100% equities for both the GBM model and the 1870–2020 bootstrapped sample, whereas the 1950–2020 bootstrapped sample produces a small allocation to cash (11.05%) and a corresponding allocation to equities (88.95%). This shows that, when optimisation is performed across a range of initial wealth levels, the optimal allocation tends to be more in line with the allocations for low-wealth hail-mary scenarios. This makes sense because the optimisation is effectively trying to find a single allocation that performs well across a range of initial wealth levels and is therefore likely to favour allocations that perform better for the most at-risk retirees, namely those with lower initial wealth. In fact, it is noteworthy that only the 1950–2020 bootstrapped sample produces a non-zero allocation to any safe asset class.

Table 7: Optimised fixed allocation strategies for the distributed initial wealth case

Returns sample	Cash allocation	Bonds allocation	Stocks allocation
IID GBM	0.00%	0.00%	100.00%
Block bootstrapped (1870–2020)	0.00%	0.00%	100.00%
Block bootstrapped (1950–2020)	11.05%	0.00%	88.95%

Comparing the unified case with the wealth-specific case, we observe that bankruptcy-density figures are comparable for low-wealth retirees but substantially worse for higher-wealth retirees (see Figure 7, particularly the relative differences in bankruptcy density shown in panels (g), (h), and (i)). This is consistent with our intuition: when a single allocation must perform well across a range of wealth levels, the optimisation will tend to favour the most at-risk retirees (those with lower initial wealth) at the expense of higher-wealth retirees. Two points are worth noting. First, it is evident that the retiree's initial wealth is a significant factor in determining the optimal allocation strategy under most market condition generators. Again, it is only the Historical Markets (Full Period) generator that produces a 100% equity allocation for all initial wealth levels. Although our uniform sampling of initial wealth levels is not a realistic representation of the actual distribution of initial wealth levels in the population, it does provide a useful way to explore how the optimal allocation strategy varies with initial wealth and to understand the trade-offs involved in optimising for a range of initial wealth levels versus optimising for a known initial wealth level. From this, we may speculate that at least part of the reason Anarkulova et al. (2025) found that 100% equity allocations were almost optimal in their study is that, by modelling the accumulation period of the retiree, they were effectively optimising across a range of initial wealth levels at retirement and therefore biasing the strategy towards improving outsomes for the lower-wealth retirees. Second, it is worth noting that the strategy and outcomes for the 1950–2020 bootstrapped sample differ from the 1870–2020 bootstrapped sample. This suggests that the pre-1950 sample of the JST dataset contains asset-return dynamics that are substantially different from the post-1950 sample. We investigate what features of the data may be causing this in a later section, but it is interesting to note that this is where our findings differ somewhat from Anarkulova et al. (2025), who note that there was little difference when they restricted their sample to the "modern" era.

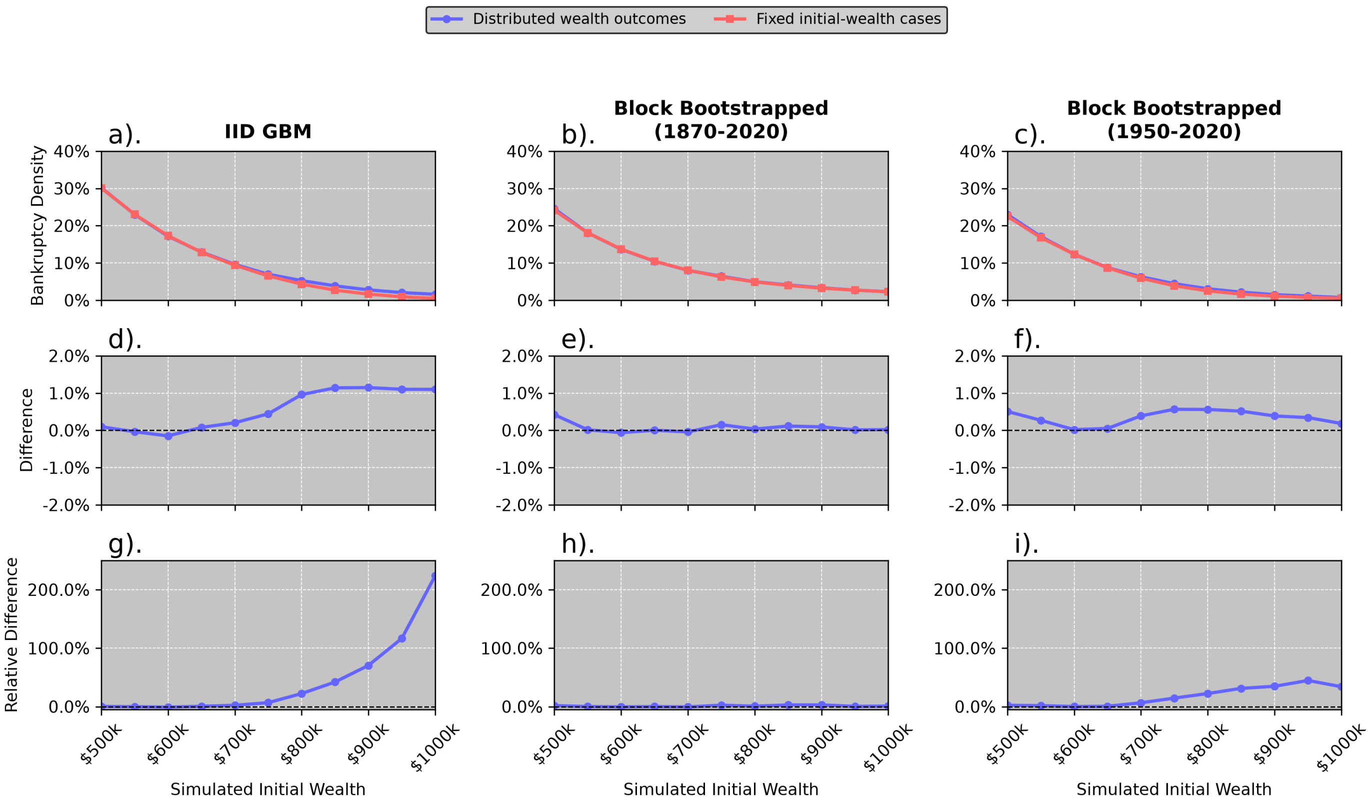

Figure 7: Comparison of bankruptcy density for fixed allocation strategies optimised for a known initial wealth level versus those optimised across a range of initial wealth levels. Panels (a), (b), and (c) show the bankruptcy density of each strategy with returns sampled from the GBM model, the 1870–2020 bootstrapped sample, and the 1950–2020 bootstrapped sample, respectively. The x-axis shows the initial wealth level, and the y-axis shows the bankruptcy density for each strategy. Panels (d), (e), and (f) show relative differences in bankruptcy density between the two strategies, with positive values indicating worse performance (higher bankruptcy density) for the distributed-initial-wealth strategy and negative values indicating better performance (lower bankruptcy density). Panels (g), (h), and (i) show the relative differences in bankruptcy density between the two strategies, with positive values indicating worse performance for the distributed-initial-wealth strategy and negative values indicating better performance.

5.2 Asset Allocation in the Time Dimension: Glide-Path Strategies

Following the same structure as the fixed-allocation analysis, we examine glide-path strategies under both wealth-specific and unified optimization scenarios. We also study the effect of varying the resolution of the glide-path by increasing the number of rows (or "nodes") in the time dimension of the policy matrix to 2, 3, 6, and 10 (with 1 as the fixed-policy baseline), which allows for more complex time-varying allocation strategies

5.2.1 Glide-Path Optimisation for Known Initial Wealth

We first present the wealth-specific glide-path results, where we optimise separate policies for each initial wealth level. The evolution of the policy matrix as we increase the number of time nodes and the initial wealth of the investor is shown in Figure 12. Across all datasets and initial wealth levels, we see that the optimal glide-paths tend to be rising-equity glide-paths or 100% equity allocations. As with the fixed strategy, low-wealth retirees are allocated entirely to equities. This frontloading of safe assets in the initial retirement period suggests that the optimal glide-path strategies are hedging against sequence risk in the early retirement period, which is consistent with the findings of Anarkulova et al. (2025), Pfau and Kitces (2014), and Estrada (2014). The ability to frontload safe assets in the early retirement period also leads to higher initial allocations to safe assets than in the wealth-specific fixed allocation case. Indeed, where the wealth-specific fixed allocation case produced a 100% equity allocation for all wealth levels in the 1870–2020 bootstrapped sample, the glide-path case produces small allocations to cash in the initial retirement period for higher-wealth retirees (see Figures 12 e, h, k, n, and q). Additionally, as the resolution of the glide-path increases, so does this initial allocation to safe assets. Outside the initial retirement period, all glide-paths allocate almost entirely to equities. This suggests that the safe asset allocation is primarily a hedge against sequence risk in the early retirement period, and that once the retiree has survived this initial period, the optimal strategy is to maximise expected returns by allocating entirely to equities. This is consistent with the findings of Anarkulova et al. (2025).

As glide-path resolution increases, the policy concentrates its hedge in the early retirement period more efficiently. This concentration enables the policy to target paths most susceptible to sequence risk without becoming overly conservative for the remainder of retirement. The effect of increasing node count on the bankruptcy density and mean terminal wealth of the resulting wealth trajectories is shown in Figures 8 and 10 respectively. As the resolution of the glide-path is increased, the bankruptcy density decreases, particularly for higher-wealth retirees. This effect is most pronounced in the GBM model, followed by the 1950–2020 bootstrapped sample, and is less pronounced in the 1870–2020 bootstrapped sample. This demonstrates that frontloading safe assets in the early retirement period effectively hedges sequence risk, and that increasing glide-path resolution enables more efficient hedging. The fact that this effect is strongest in the IID returns of the GBM model suggests that sequence risk is potentially exaggerated in IID return assumptions, and that mean-reversions and other time-series effects in the bootstrapped samples may reduce the impact of sequence risk on the optimal allocation strategy, in line with the findings of Anarkulova et al. (2025).

Increasing glide-path resolution substantially reduces the mean terminal wealth trade-off observed in the fixed allocation case. Recall that in the fixed allocation case, mean terminal wealth declined slightly as initial wealth increased for both the GBM model and 1950–2020 bootstrapped sample, reflecting the optimal strategy's choice to sacrifice expected returns for market risk reduction. In contrast, when glide-paths have sufficient resolution to frontload safe assets early in retirement, this trade-off is greatly diminished for the GBM model and largely eliminated for the 1950–2020 sample. This improvement suggests that efficiently hedging sequence risk in the early retirement period enables strategies to maintain higher expected returns throughout retirement. The 1870–2020 bootstrapped sample is an exception, with increasing glide-path resolution slightly reducing mean terminal wealth, likely because small early allocations to safe assets dampen expected returns in later periods. However, this effect is minimal and does not disrupt the monotonic relationship between initial wealth and terminal wealth.

An important but counterintuitive finding emerges in the bankruptcy probability results (Figure 9). While bankruptcy probability generally declines with both glide-path resolution and initial wealth, there is a marked local increase in bankruptcy probability in the \$550k–\$800k range, with peaks at approximately \$550k for the 1950–2020 sample and \$700k for the GBM model. Notably, this anomaly intensifies as glide-path resolution increases, which is at first puzzling. We interpret this as evidence of the density-probability trade-off discussed earlier; by concentrating safe asset hedges in the early retirement period, glide-paths successfully reduce cumulative bankruptcy duration (lower density), but at the cost of allowing bankruptcies to occur more frequently late in the retirement horizon (higher probability in this wealth band). These peaks in bankruptcy probability correlate with local reductions in mean terminal wealth at identical initial wealth levels, consistent with the

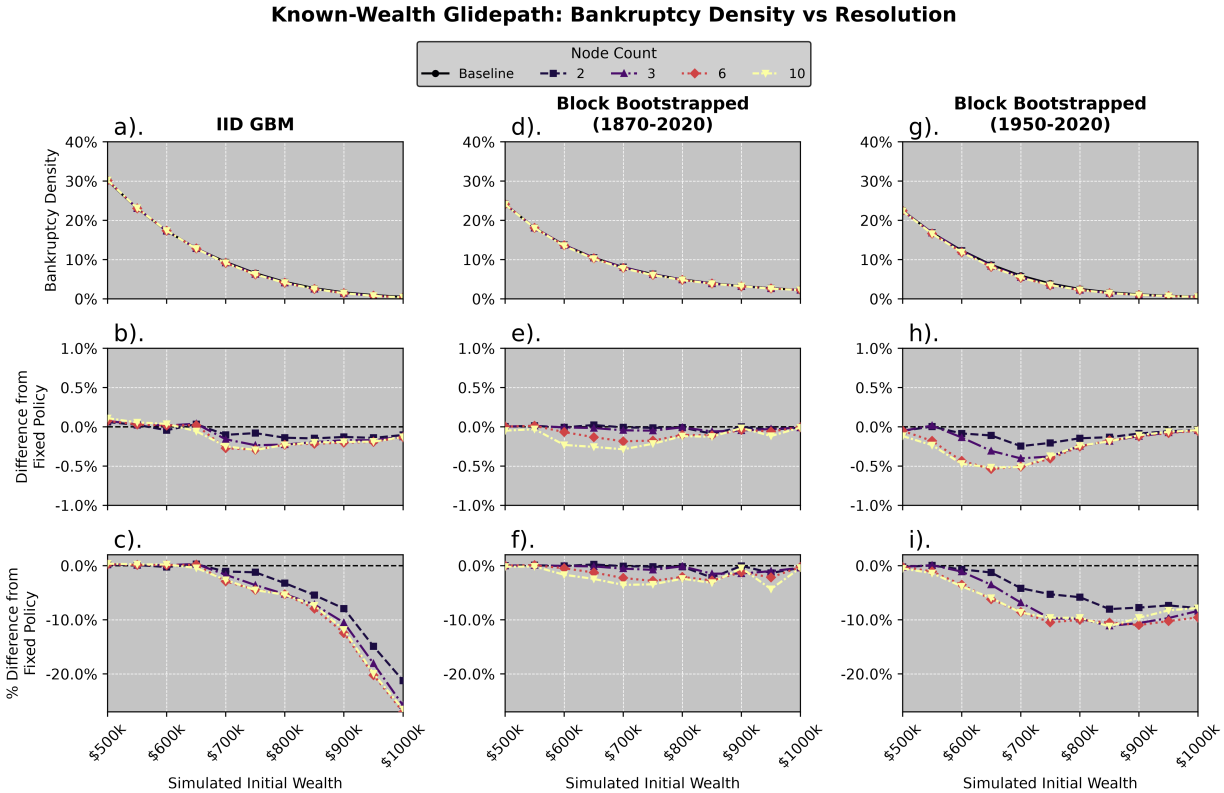

Figure 8: The bankruptcy density of the wealth trajectories resulting from the optimal glide-path policy matrix as we increase the number of time nodes and the initial wealth of the investor. The top row (figures a, b, c) shows the bankruptcy density for all policies and initial wealth levels for the three market condition generators: Synthetic Markets (IID GBM), Historical Markets (Full Period), and Historical Markets (Modern Era), respectively. The middle row (figures d, e, f) shows the difference in bankruptcy density between various glide-path policies and the fixed allocation policy for each market condition generator. The bottom row (figures g, h, i) shows relative percentage differences in bankruptcy density between various glide-path policies and the fixed allocation policy for each market condition generator.

hypothesis that early hedging sacrifices long-term growth. The key insight here is that strategies minimizing bankruptcy density may inadvertently increase the likelihood of late-life depletion, and evaluating retirement policies requires examining both metrics rather than relying on any single measure.

Significantly, no glidepath strategy produces a safe/risky asset allocation that resembles those recommended by the traditional 100-age or 120-age rules of thumb or those implemented by SuperLife (2026). Instead, all glide-path strategies allocate almost entirely to equities after the initial retirement period, with only small allocations to safe assets in the early retirement period. While we cannot comment on the shape of the glidepath during accumulation, we do not find any evidence that traditional, declining risk glide-paths are optimal during retirement.

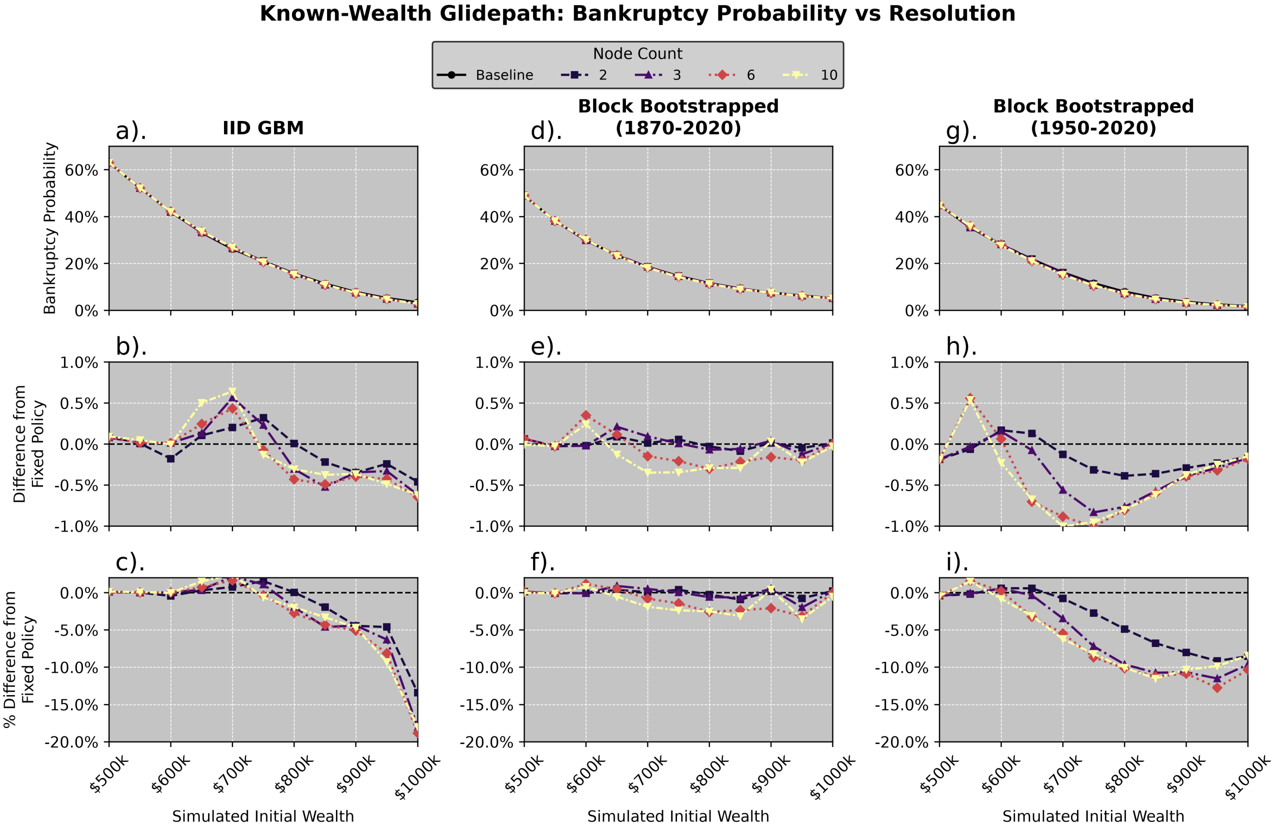

Figure 9: As in Figure 8, but showing the bankruptcy probability of the wealth trajectories resulting from the optimal glide-path policy matrix as we increase the number of time nodes and the initial wealth of the investor.

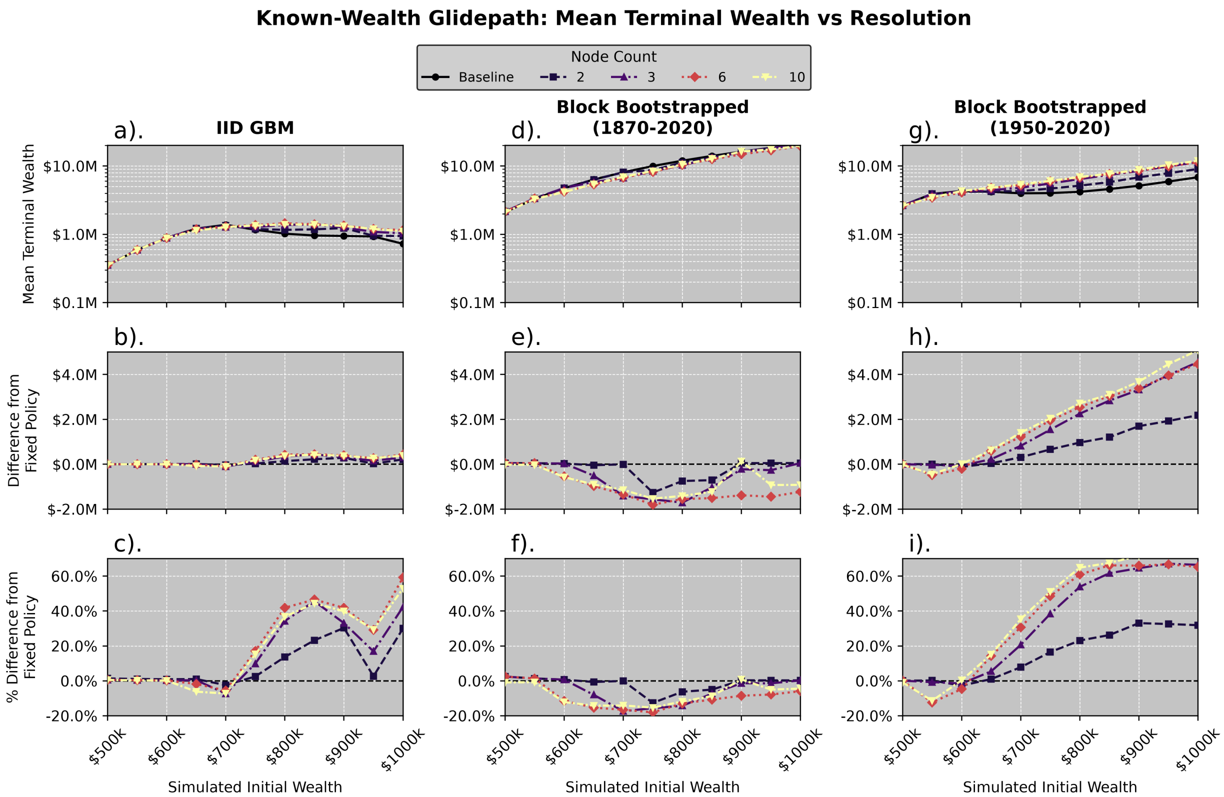

Figure 10: As in Figure 8, but showing the mean terminal wealth of the wealth trajectories resulting from the optimal glide-path policy matrix as we increase the number of time nodes and the initial wealth of the investor.

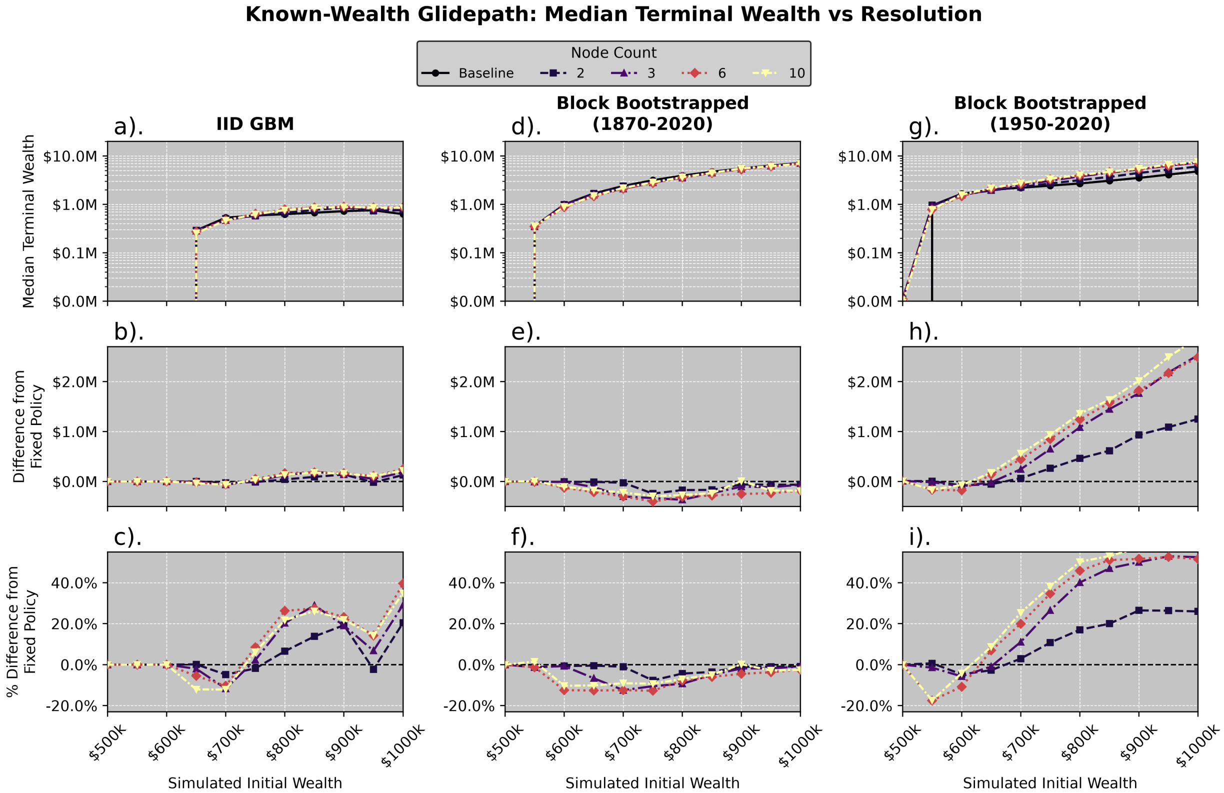

Figure 11: As in Figure 8, but showing the median terminal wealth of the wealth trajectories resulting from the optimal glide-path policy matrix as we increase the number of time nodes and the initial wealth of the investor.

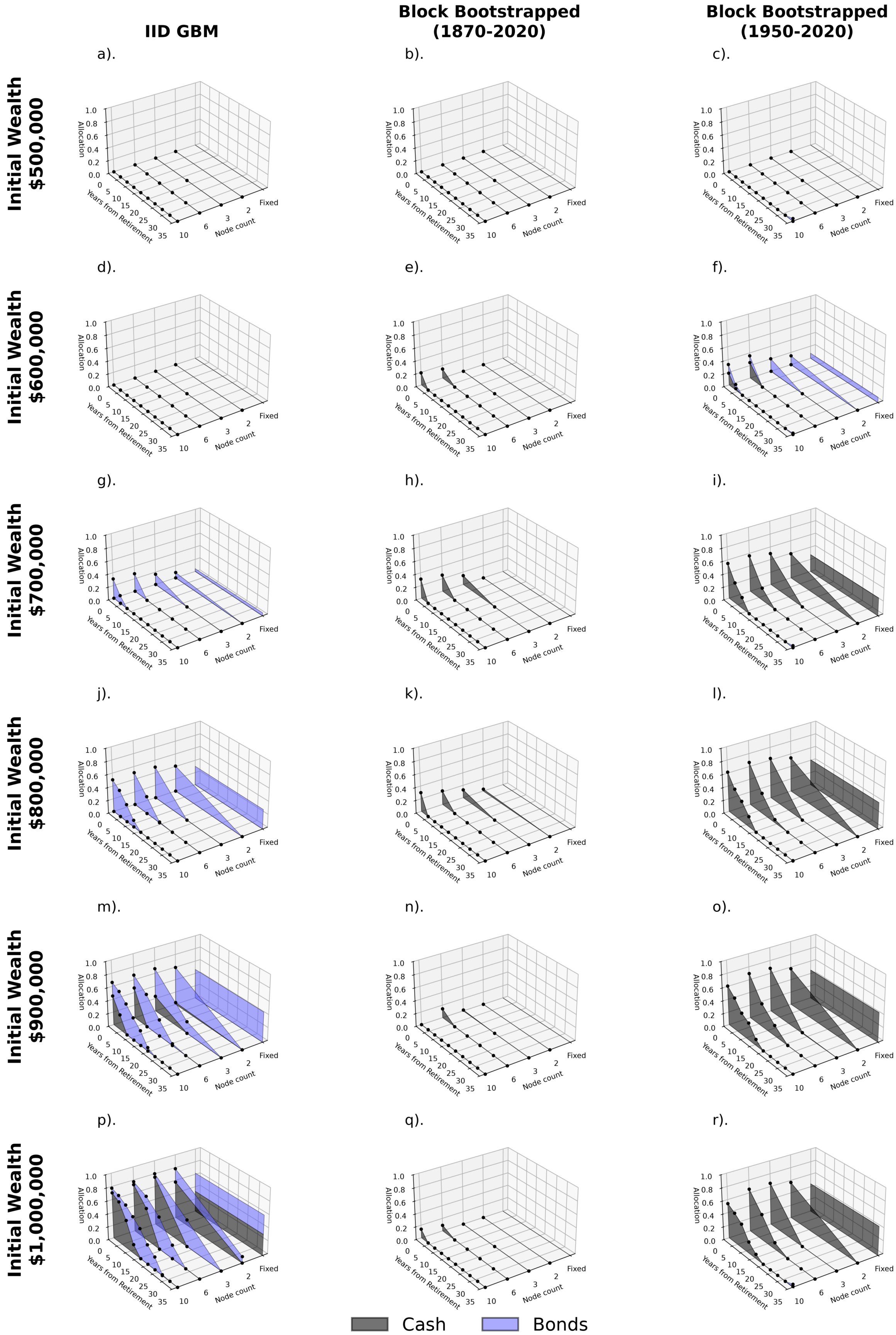

Figure 12: Evolution of the optimal glide-path policy matrix as the number of time nodes and initial wealth of the investor is increased. Each plot shows the optimal allocation to cash and bonds interpolated between nodes over time (as it is done in the simulation), with the equities allocation (not shown) being the remainder. The optimal fixed policy is also shown on the far right of each plot.

5.2.2 Glide-Path Optimisation Across Distributed Initial Wealth Levels

We now consider the unified case in which we optimise a single glide-path strategy that performs well across simulated wealth lifetimes distributed over a range of initial wealth levels. In contrast to the known-wealth case, where we optimised separate policies tailored to each specific initial wealth level, this distributed-wealth optimisation searches for a single time-varying allocation strategy that achieves good outcomes across a diverse range of starting positions. This is a more realistic and challenging problem, as real-world asset managers and robo-advisors typically must serve clients with heterogeneous initial wealth levels using unified strategy templates.

The optimal glide-paths resulting from this distributed-wealth optimisation display predominantly high equity allocations, consistent with the hail-mary phenomenon observed in the known-wealth case. The glidepath strategies that do incorperate brief allocations to safe assets concentrate this allocation in the early retirement period, serving as a targeted sequence-risk hedge. As the resolution of the glide-path increases (i.e., as the number of time nodes increases), the magnitude of safe asset allocation increases while simultaneously becoming more compressed in time. This represents a refinement rather than a fundamental shift in strategy: with more nodes, the policy can express a sharper, more temporary hedge against sequence risk. The Historical Markets (Modern Era) bootstrap produces the highest safe asset allocations in the early period, suggesting that the specific historical return dynamics of the post-1950 era provide stronger incentives for this early-retirement hedging behaviour compared to the longer historical period or IID returns.

The performance implications of this distributed-wealth optimisation are modest, as shown in Figure 13. Comparing the distributed-wealth glide-paths against the known-wealth fixed allocation baseline reveals that performance improvements are marginal and inconsistent across wealth cohorts and return samplers. The strategies perform at best only slightly better than a fixed allocation, with the most meaningful improvements appearing in the Historical Markets (Modern Era) bootstrap sample, particularly at high wealth levels. For the IID GBM and 1870–2020 bootstrapped samples, the distributed-wealth glide-paths offer negligible advantages over simple fixed allocations. This finding suggests that the ability to time allocations over the retirement horizon provides limited benefit when optimising across heterogeneous initial wealth levels, particularly when the optimisation must serve both low-wealth retirees (who benefit from aggressive growth) and high-wealth retirees (who can tolerate more defensive positioning).

These results suggest that the practical value of implementing multi-node glide-paths for heterogeneous client populations is limited. While the refined allocation timing enabled by higher node counts can improve early-retirement hedging, particularly in the Modern Era sample, the improvements over simple fixed allocations remain marginal. This finding constrains the attractiveness of time-varying strategies when a single unified policy must simultaneously serve retirees with diverse starting positions and risk tolerances.

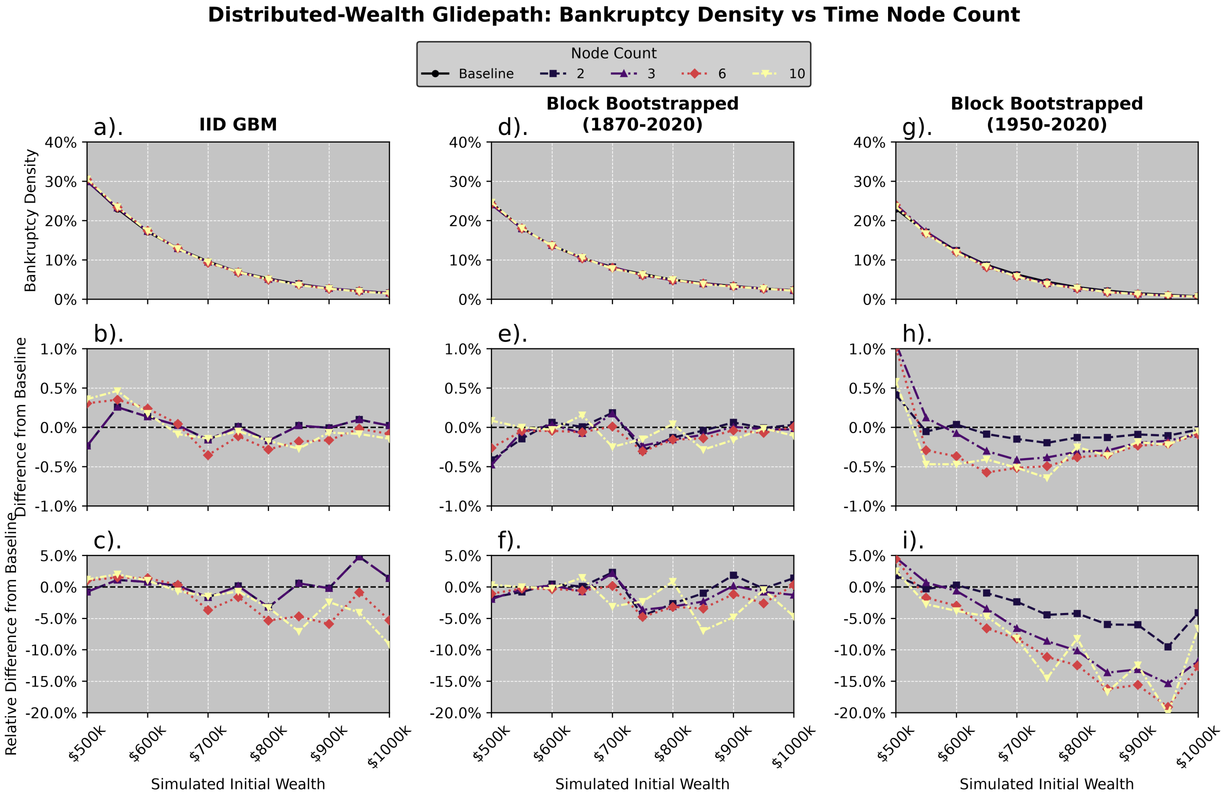

Figure 13: Bankruptcy density comparisons for distributed-wealth glide-path strategies with varying time-node resolutions (node counts: 1, 2, 3, 6, 10) against the known-wealth fixed allocation baseline. Results are stratified across three return-sampling methodologies: IID GBM (panels a–c), Block Bootstrapped (1870–2020) (panels d–f), and Block Bootstrapped (1950–2020) (panels g–i). Each panel's three subplots show: (top) absolute bankruptcy density by initial wealth level; (middle) absolute difference from the fixed allocation baseline; (bottom) percentage difference from baseline. Distributed-wealth glide-path strategies offer only marginal performance improvements over fixed allocations, with the most meaningful gains appearing in the Modern Era bootstrap (panels g–i). The IID GBM and longer historical sample show negligible advantages, indicating that time-varying allocations provide limited benefit when optimising across heterogeneous initial wealth levels.

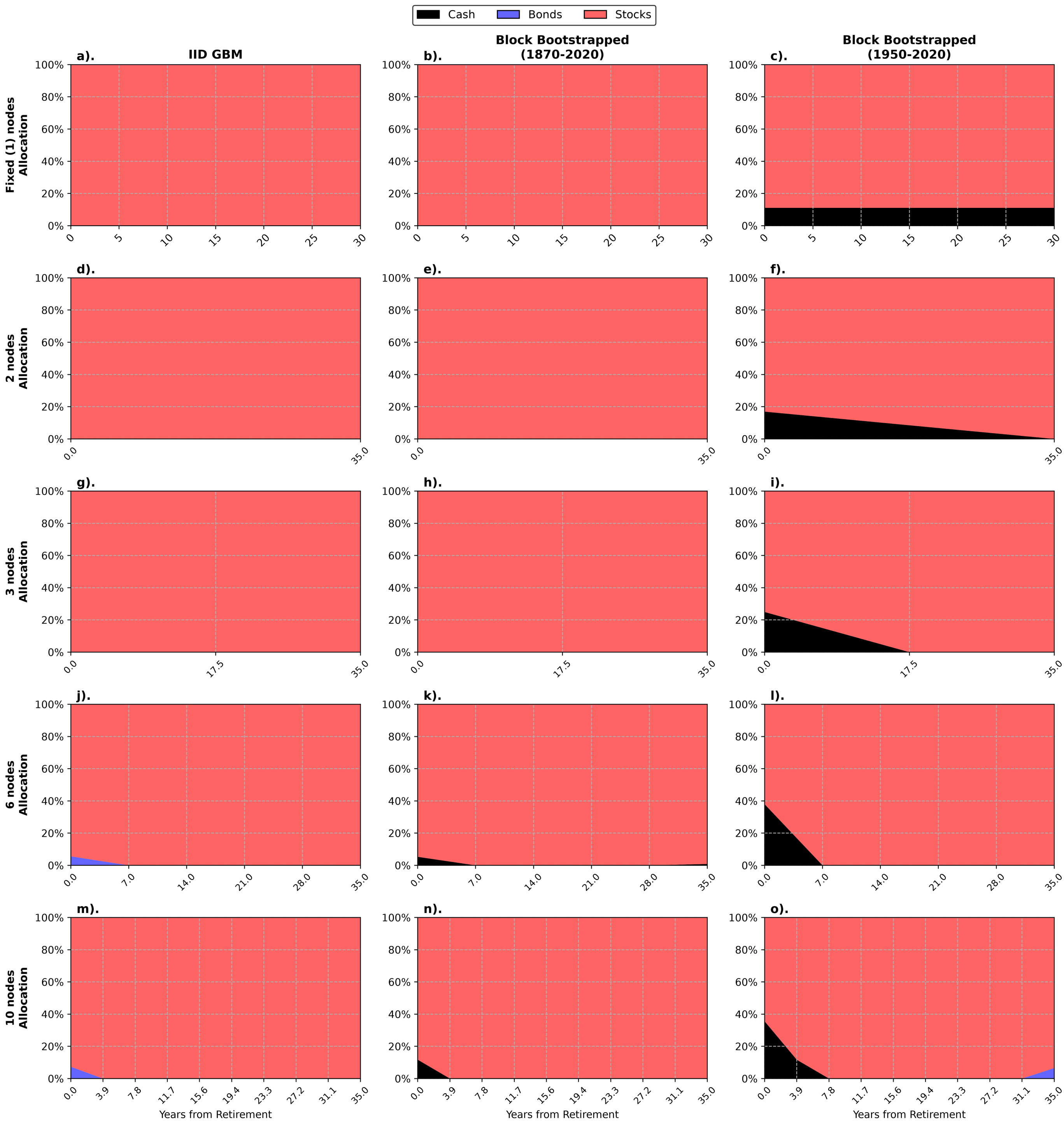

Figure 14: Evolution of the optimal glide-path policy matrix as the number of time nodes is increased, optimized for the distributed initial wealth case. Each row corresponds to a different number of time nodes (1, 2, 3, 6, and 10), while each column represents a different return-sampling methodology: IID GBM (left), Block Bootstrapped (1870–2020) (centre), and Block Bootstrapped (1950–2020) (right). Each plot shows the optimal allocation to cash (light blue) and bonds (purple) interpolated between nodes over time during the 30-year retirement horizon, with equities (orange) forming the remainder. The strategies are characterized by high equity allocations throughout retirement, with brief early-period allocations to safe assets. As node count increases, safe asset allocations increase in magnitude while becoming compressed into an increasingly narrow window at the start of retirement. The Modern Era bootstrap (right column) exhibits the most substantial early-period safe asset allocations, particularly at higher node counts, consistent with superior performance improvements observed in that sample.

5.3 Asset Allocation in the Wealth Dimension: Wealth-Responsive Allocation Strategies

Now we move on to asset allocations in the wealth dimension, which we refer to as wealth-responsive allocation strategies. In this case, retirees vary their instantaneous asset allocation based on their current wealth level while maintaining the overall structure of the strategy over time. This is an unusual approach to the glide-path problem, as most research focuses on time-based glide-paths. However, it is not without precedent. We have seen in Sections 5.1 and 5.2 that the optimal allocation strategy varies significantly with the retiree's wealth level, meaning that unified strategies optimised across a range of initial wealth levels tend to perform substantially worse than those tailored to specific wealth levels. Moreover, foundational work by Merton (1969) showed that, provided consumption is an approximately constant proportion of wealth, the optimal allocation strategy is independent of time and depends only on wealth and the consumption level. This suggests that strategies that adapt to the retiree's current wealth level may be more effective than those that adapt to time alone.

Unlike in the previous sections, we did not conduct optimisations for the known initial wealth level case here. In practice, back-propagation of gradients through the wealth dimension is more complex than through the time dimension. Moreover, we expect that varying a fixed initial wealth level for a wealth-responsive policy would yield only limited additional insight. Because the policy is already defined as a function of current wealth, the relevant variation is primarily across the realised wealth paths themselves rather than across a small set of exogenous starting values. In this setting, a fixed initial wealth sweep would largely reproduce the same economic mechanism in a more indirect way, and the resulting differences would likely be difficult to interpret clearly.

As before, we optimise the wealth-responsive allocation strategy by computing a unified policy across an evenly distributed range of initial wealth levels ($500,000 to $1,000,000 in $50,000 increments). Following Section 5.2, we also study the effect of varying the resolution of the wealth-responsive policy by increasing the number of nodes in the wealth dimension of the policy matrix from 2, 6, and 10. We arrange the nodes logarithmically over a space from $250,000 to $1,750,000. This range was chosen somewhat arbitrarily to cover the range of initial wealth levels we are studying while also allowing for some extrapolation beyond that range.

The evolution of the strategy as the number of wealth nodes is increased is shown in Figure 15. The strategies that emerge are quite interesting. In both of the bootstrapped return-sampling methodologies, the optimal allocation to safe assets is hump-shaped: the allocation to equities is high at low wealth levels, decreases to a minimum at intermediate wealth levels, and then increases again at higher wealth levels. While the high allocation to equities at low wealth levels is consistent with the "hail-mary" phenomenon observed earlier, the higher allocation to equities at high wealth levels is somewhat novel. We appear to have observed a similar pattern in the wealth-specific fixed allocation for initial wealth levels above $800k in the 1950–2020 bootstrapped sample (see figure 6 c) but it is much more pronounced here. Again, this could be due to a number of factors. We think the most likely is that at a certain wealth level inflation risk becomes a more significant concern than market risk. That is, the retiree's wealth is sufficient to meet their spending requirements even in the event of a market downturn, but a sustained period of high domestic inflation could erode the real value of their wealth and therefore their consumptive capacity. In this case, the unhedged international equity exposure in our asset-class construction provides a natural hedge against domestic inflation, and therefore the optimal allocation to equities increases.

The performance of the unified wealth-responsive strategies was compared against both the wealth-specific fixed allocation and wealth-specific glide-path strategies. A comparison of bankruptcy density against the known-wealth fixed allocation case is shown in Figure 16. Here we see that across almost all initial wealths, market condition generators, and wealth-node resolutions, the wealth-responsive strategies achieve bankruptcy densities comparable to or better than the known-wealth fixed allocation baseline. The only exception is the 2-node wealth-responsive strategy in the IID GBM model, which performs slightly worse than the known-wealth fixed allocation baseline for an initial wealth level of $1,000,000. Wealth node resolution does improve performance, but improvements from 6 to 10 nodes appear marginal. Improvements to bankruptcy density are most pronounced in the Historical Markets (Modern Era) generator and the Synthetic Markets (IID GBM) generator, and the largest improvements across all market condition generators are seen at middle-wealth levels. We speculate that this is because the wealth-responsive strategies are able to adapt to the highly heterogeneous wealth paths that emerge in these cohorts, with bad market outcomes in the early retirement period leading to low-wealth paths (and thus hail-mary scenarios) and good market outcomes leading to high-wealth paths (and thus more conservative strategies). The wealth-responsive strategies are able to adapt to these heterogeneous paths and their associated risks, whereas the wealth-specific fixed allocation baseline is unable to do so. This improvement at middle-wealth levels is also seen in the comparison against the wealth-

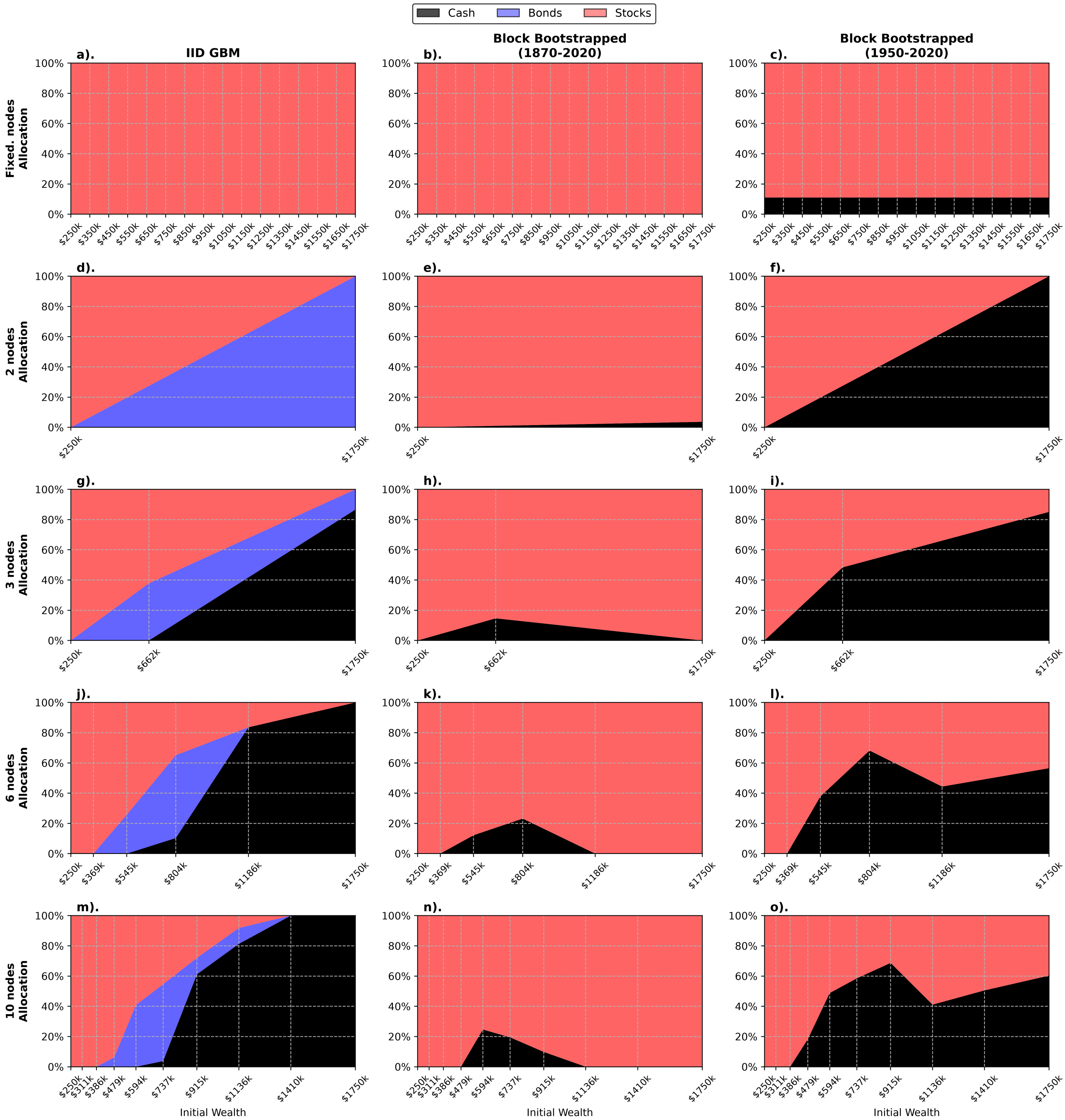

Figure 15: Evolution of the optimal wealth-responsive policy matrix as the number of wealth nodes is increased. Each row corresponds to a different number of wealth nodes (1, 2, 3, 6, and 10), while each column represents a different return sampling methodology: IID GBM (left), Block Bootstrapped 1870–2020 (centre), and Block Bootstrapped 1950–2020 (right). The coloured areas represent the optimal allocations to Cash (Black), Bonds (Blue), and Stocks (Red). The interpolation between nodes is shown as continuous curves, while black dots indicate the discrete wealth nodes where the policy is defined. The allocation surfaces reveal how the optimal strategy evolves from a fixed single-node policy to increasingly flexible multi-node policies that better adapt to varying wealth levels.

specific glide-path baseline (Figure 17), where the wealth-responsive strategies achieve bankruptcy densities comparable to or better than the known-wealth glide-path baseline across almost most middle initial wealth levels, market condition generators, and wealth-node resolutions of 6 and 10. In the IID GBM synthetic data we see around a 10% improvement in bankruptcy density for retirees with intial wealth of between $650k and $900k, and in the Modern Era bootstrapped data we see a 10-20% improvment. The IID GBM data glidepaths perform substantially better than the wealth-responsive strategies at higher wealth levels. This is likely because the time-responsive glide-path strategies are able to preemptively frontload safe assets in the early retirement period to mitigate sequence risk, whereas the wealth-responsive strategies are unable to do so, essentially relying on a drop in wealth to trigger a more conservative allocation post-hoc.

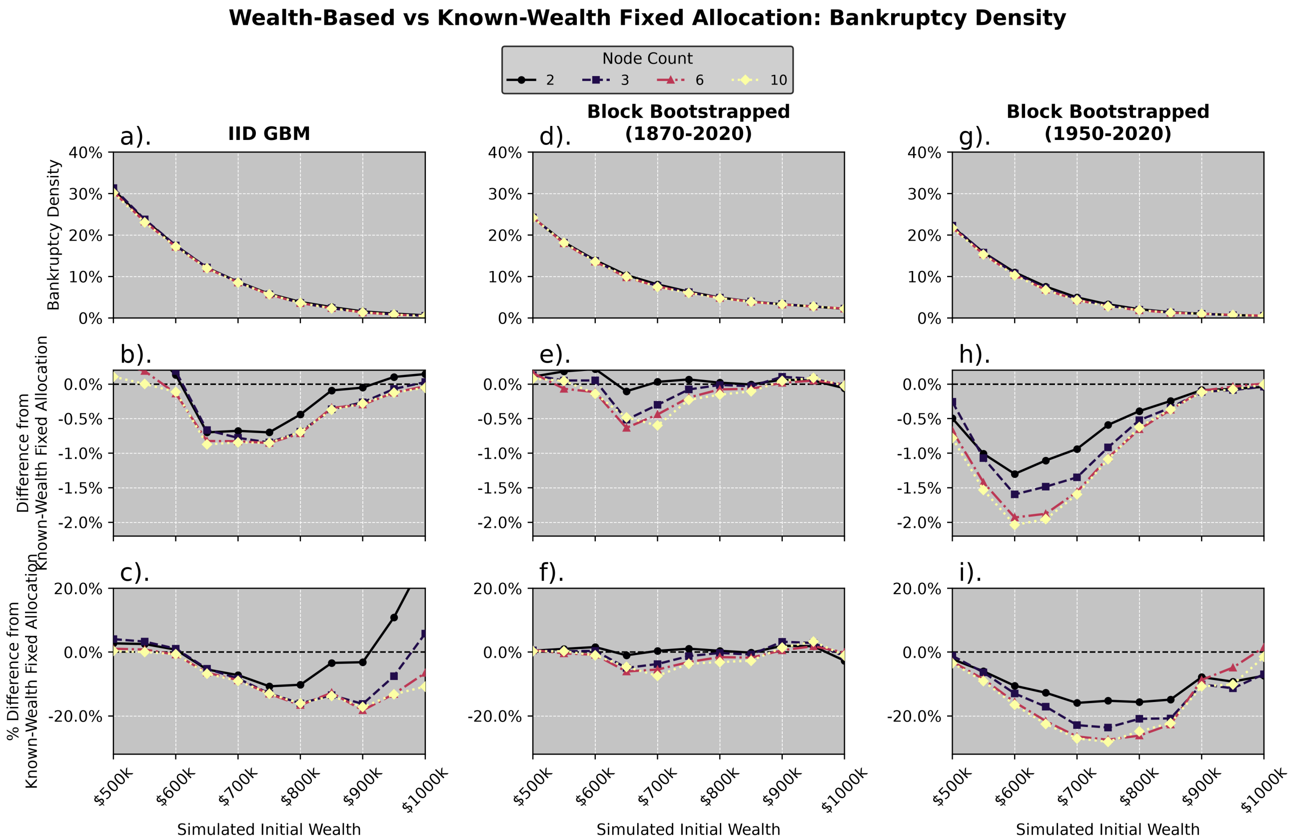

Figure 16: Bankruptcy density vs initial wealth for wealth-responsive allocation at different wealth-node resolutions compared against the wealth-specific fixed allocation baseline. Panels (a), (d), and (g) show the absolute bankruptcy density across simulated initial wealth levels for each market condition generator. Panels (b), (e), and (h) show the absolute difference in bankruptcy density relative to the wealth-specific fixed allocation baseline. Panels (c), (f), and (i) show the relative percentage difference. Wealth-responsive strategies substantially outperform the wealth-specific fixed allocation baseline, particularly at higher initial wealth levels, demonstrating the benefit of adapting allocations to current wealth rather than maintaining a single fixed allocation.

A key finding of substantial practical significance is that the unified wealth-responsive strategies achieve this competitive performance *without* knowing the retiree's initial wealth in advance. In contrast, the glide-path baselines against which we compare them were optimised with full knowledge of the initial wealth level, representing the best-possible performance for time-responsive strategies. The fact that wealth-responsive strategies (optimised jointly across the entire wealth range) largely match or exceed the performance of these wealth-informed time-responsive strategies suggests that real-time adaptation to current wealth state is a more powerful adaptive mechanism than preemptive time-based allocation in most cases. This has important implications for practical policy implementation: whereas time-based glide-paths require accurate forecasting of retirement outcomes to preempt sequence risk early in retirement, wealth-responsive strategies dynamically adjust to the actual realised wealth path and can therefore respond more flexibly to whatever economic circumstances unfold. The superiority of wealth-responsive policies at middle-wealth levels, in particular, demonstrates that the value of tracking and responding to a retiree's changing wealth position exceeds the value of age-contingent planning based on initial conditions.

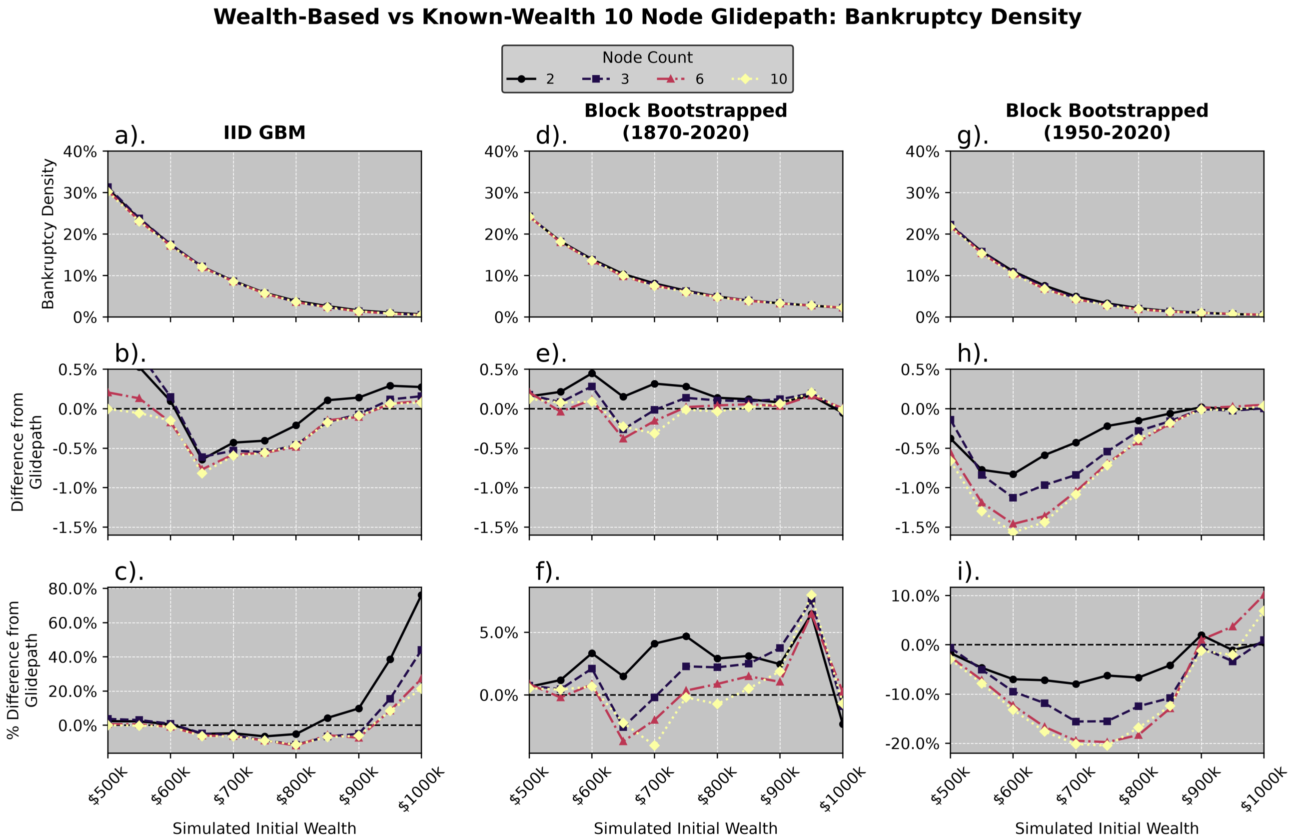

Figure 17: Bankruptcy density vs initial wealth for wealth-responsive allocation strategies at different resolutions compared against the wealth-specific 10-node glide-path baseline. Panels (a), (d), and (g) show the absolute bankruptcy density across simulated initial wealth levels for each market condition generator. Panels (b), (e), and (h) show the absolute difference in bankruptcy density relative to the wealth-specific glide-path baseline. Panels (c), (f), and (i) show the relative percentage difference. The wealth-responsive strategies with higher wealth-node resolution achieve bankruptcy densities comparable to or better than the wealth-specific glide-path baseline across most wealth levels.

5.4 Asset Allocation in Two Dimensions: Time and Wealth-Based Control Matrix

We now combine both state variables and optimise a two-dimensional control matrix, where allocations depend jointly on retirement time and current wealth. For each market condition generator, we optimise policies over distributed initial wealth buckets ($500,000 to $1,000,000 in $50,000 increments), and sweep matrix resolutions from coarse to finer grids in both dimensions, keeping the same node-spacing conventions as in the previous sections.

The resultant control matrices are visualised in Figures 18, 19, and 20. The ternary colour blend encodes the allocation to cash, bonds, and stocks, with the displayed allocations quantised to 25% increments for clarity. Like the previous examples, the optimal policy landscape depends heavily on the market condition generator. Both the bootstrapped samples exhibit systemically higher allocations to equities, and negligible allocations to bonds. Dynamics that we have observed in the previous sections are also present here. Across all samplers, control matrices switch to %100 equity allocations at low wealth levels, which is consistent with the hail-mary phenomenon observed earlier. Hedging against sequence risk in the early retirement period is also observed, with regions of low risk allocation concentrated in the very early retirement period, followed by a shift back to higher-risk allocations as time progresses. However, a new feature emerges in the control matrices in the form of a declining allocation to equities in the latter portions of the retirement period for higher-wealth retirees, such that the typical horizontal cross section observed at a fixed wealth node (somewhere in the middle of the policy matrix of a suitably high resolution, typically 6×6 or greater) shows a initially lower allocation to equities, followed by a rise in the allocation to equities, and then a decline again. This ability for the retiree to "cash-out" their risky assets when their wealth reaches a sufficient capacity is a key advantage of the control-matrix policy, especially compared to a glidepath, where the strategy is forced to adopt an allocation which best suits the most vulnerable retirees. As we have mentioned before, this causes optimal glidepaths to tend towards a rising-equity allocation as the latter portions of policy-optimisation landscape become dominated by unlucky retirees whose wealth-trajectories have fallen into the hail-mary region. Because the control-matrix policy is better able to adapt to each retiree's realised wealth path, the strategy is able to reduce risk when wealth is high enough to support the retiree's remaining consumption. "Cashing-out" is most strongly observed in the IID GBM model, where the optimal allocation to equities is reduced to 0% at high wealth levels in the latter portions of retirement. In both the bootstrapped samples, the optimal allocation to equities is but not eliminated. In the 1870–2020 bootstrapped sample, the optimal allocation is almost entirely to equities, with cash-out behaviour only appearing at the very highest wealth levels at the terminal phase of retirement. In the 1950–2020 bootstrapped sample, the cash-out behaviour is slightly longer and more pronounced at lower wealth levels, but still nowhere near as pronounced as in the IID GBM model. This is likely because the IID GBM model understates the inflation risk compared to the bootstrapped samples.

We also see a similar effect in the shape of the hail-mary region in the control matrices. While this region is harder to distinguish in the 1870–2020 bootstrapped sample, it is clearly visible in the IID GBM and 1950–2020 bootstrapped samples.Here we see a clearly defined hail-mary region in the bottom-left diagonal of the control matrix. This region of high equity slopes downwards and to the right, indicating that as time progresses and wealth increases, the optimal allocation shifts from high-risk equities to safer assets. This is essentially because as the retiree ages, the time horizon shortens, and as wealth increases, the retiree can afford to take less risk without jeopardising the risk of bankruptcy before the end of their lifetime.

The performance of a 10 time-node control-matrix at different wealth resolutions is compared against the 10 time-node glide-path in Figure 21. This glidepath was optimised for each initial wealth level, and therefore represents the best-possible performance for time-responsive strategies. The control-matrix policies achieve bankruptcy densities comparable to or better than the glide-path baseline across almost all initial wealth levels, market condition generators, and control-matrix resolutions. The only exception is for < 3 wealth nodes in the IID GBM model at low or very high initial wealth retirees. The overall bankruptcy density of the control-matrix policies at different time and wealth resolutions is shown in Figure 22. In all three market condition generators, we see that increasing the resolution of the control-matrix policy in the wealth dimension improves performance most dramatically. Interestingly, for the IID GBM model, increasing the resolution in the time dimension has a negligible effect on performance, unless the resolution in the wealth dimension is also increased. This suggests that the ability to adapt to realised wealth paths is more important than the ability to adapt to time alone, which is consistent with our findings in Section 5.4. In all cases, best performance is achieved with the highest resolution in both dimensions.

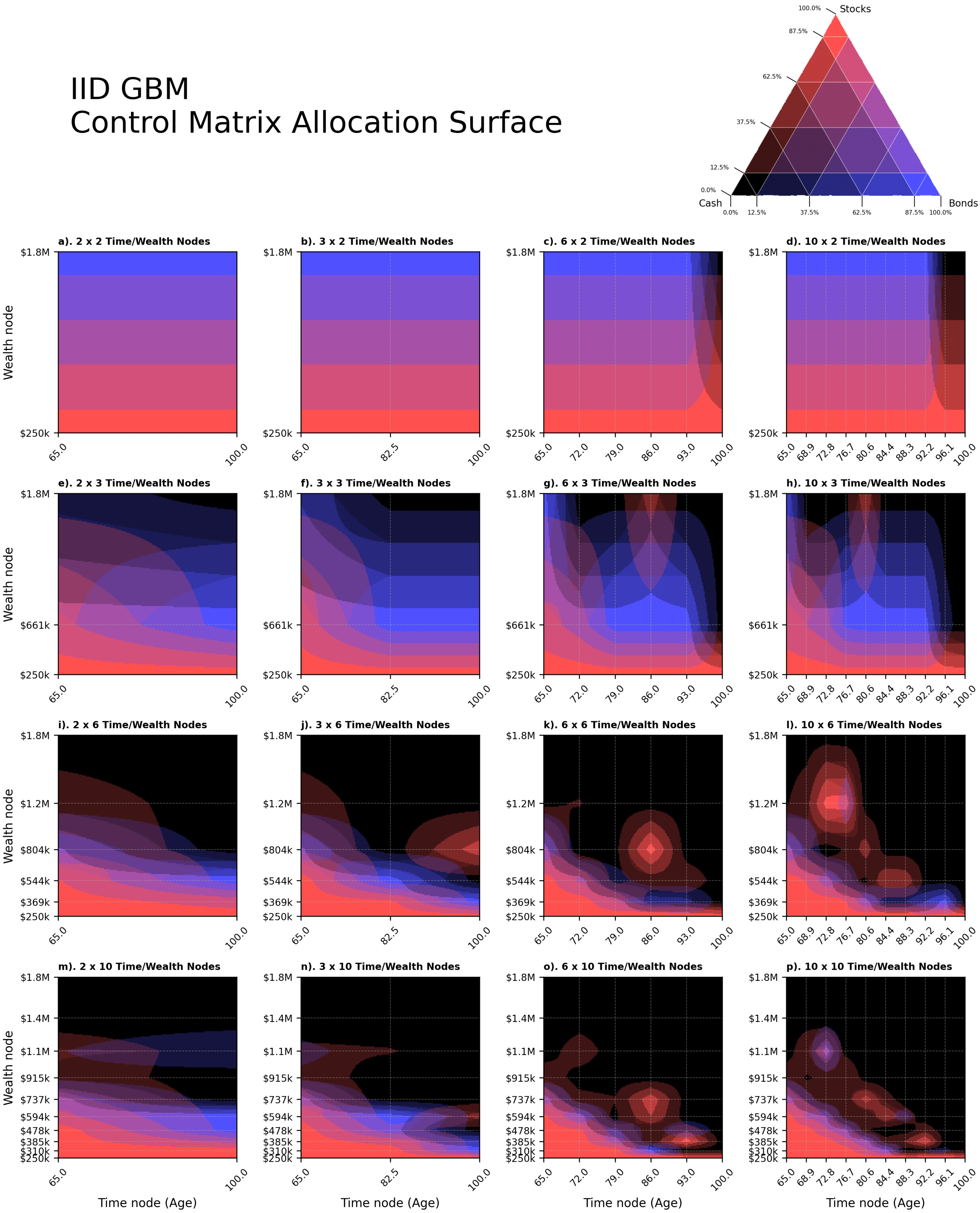

Figure 18: Control-matrix allocation surface for the Synthetic Markets (IID GBM) generator. Cash, bonds, and stocks are encoded with a ternary colour blend, and the displayed allocations are quantised to 25% increments to improve readability.

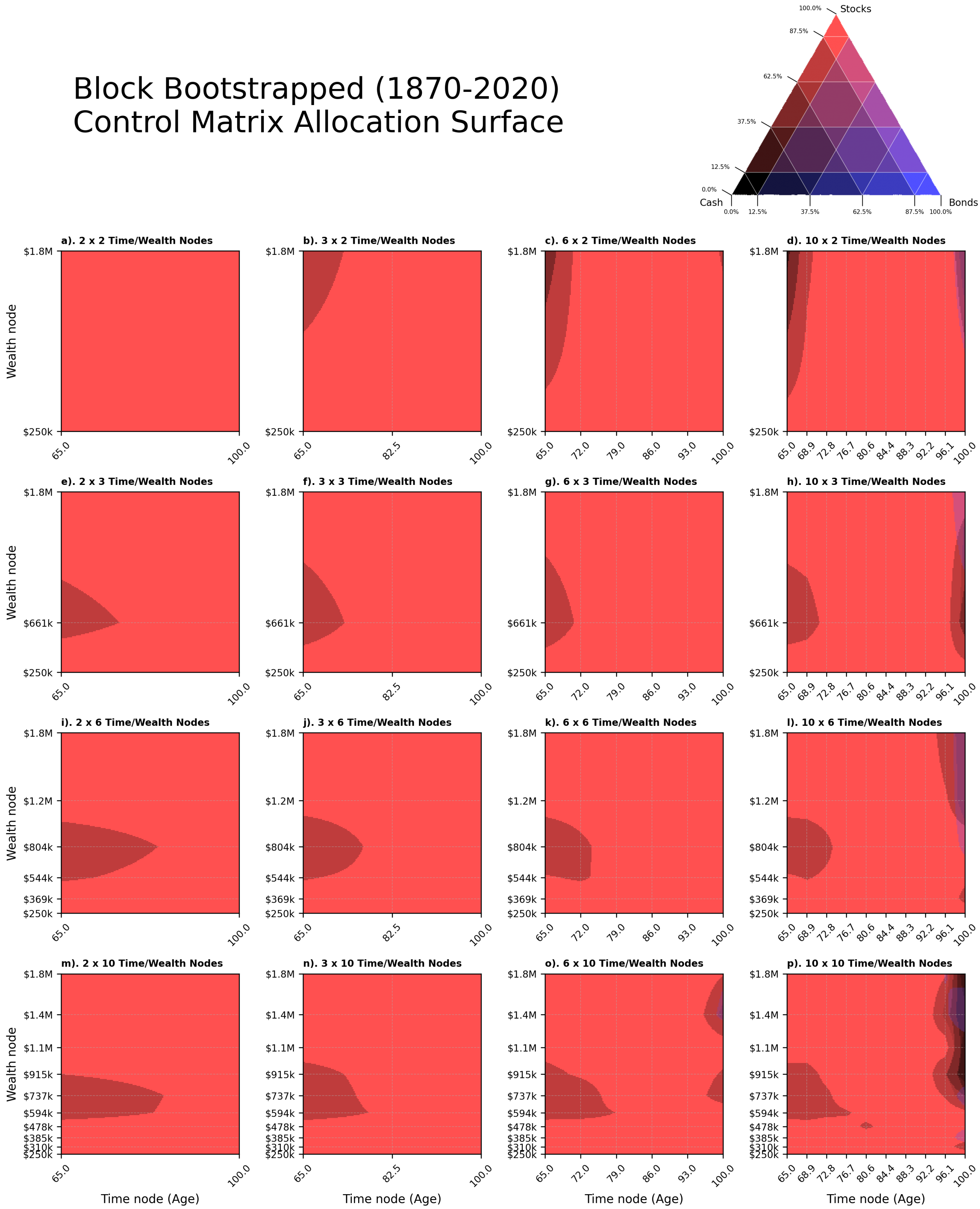

Figure 19: Control-matrix allocation surface for the Historical Markets (Full Period, 1870–2020) bootstrap generator. Cash, bonds, and stocks are encoded with a ternary colour blend, and the displayed allocations are quantised to 25% increments to improve readability.

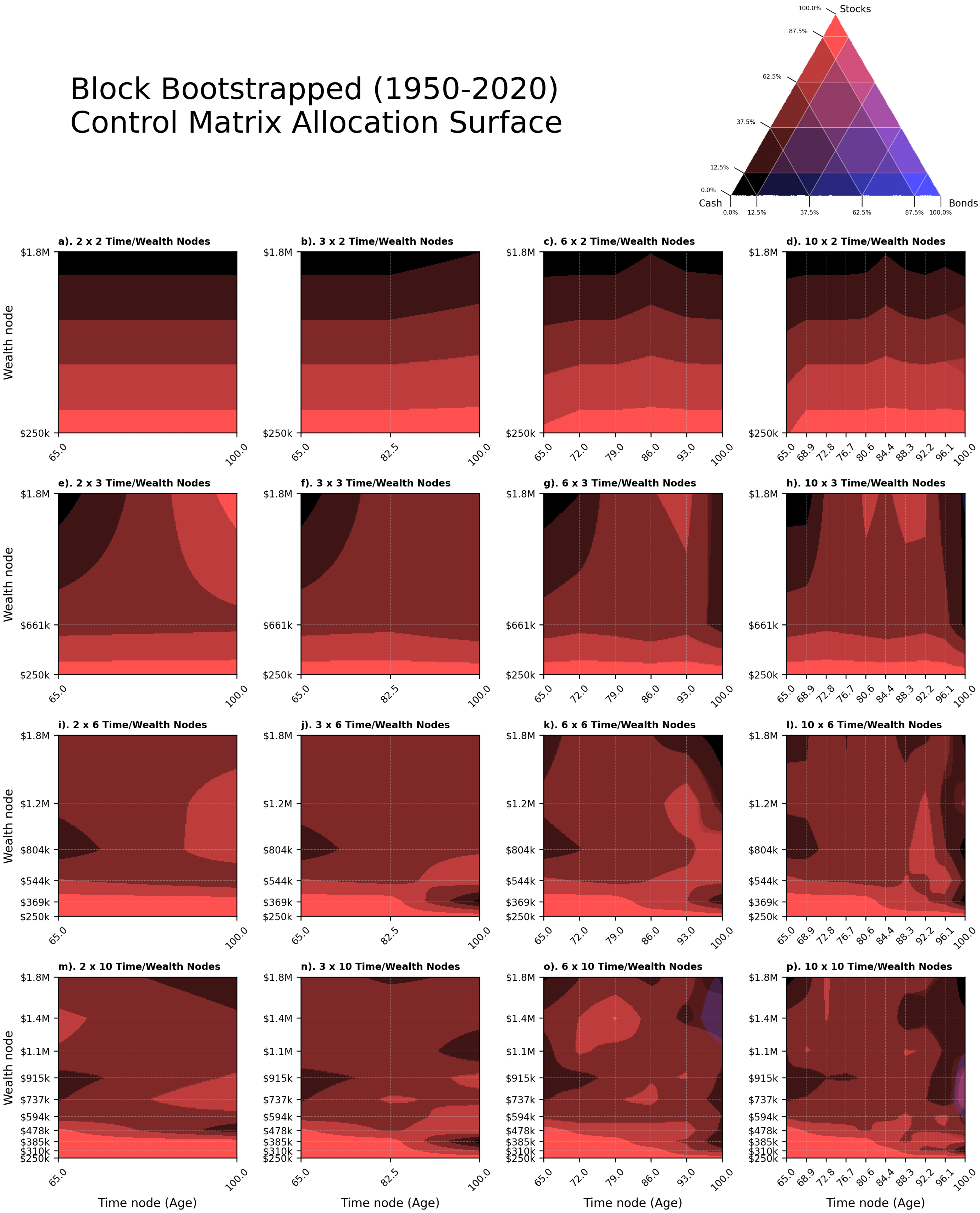

Figure 20: Control-matrix allocation surface for the Historical Markets (Modern Era, 1950–2020) bootstrap generator. Cash, bonds, and stocks are encoded with a ternary colour blend, and the displayed allocations are quantised to 25% increments to improve readability.

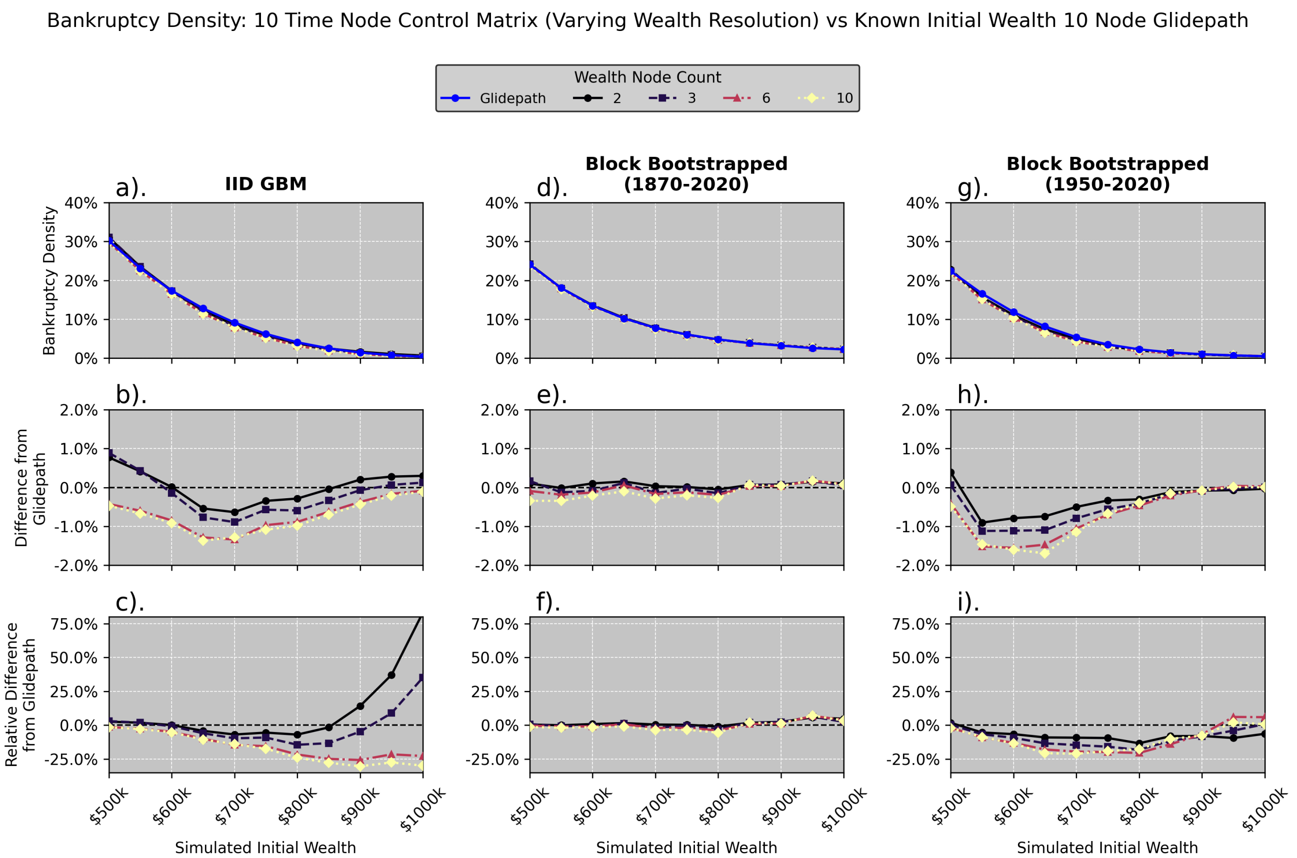

Figure 21: Bankruptcy density for distributed-wealth control-matrix policies with 10 time nodes, compared against the initial wealth-specific 10-node glidepath. Panels (a)–(c) show the absolute bankruptcy density by initial wealth level for each return-sampling methodology. Panels (d)–(f) show the absolute difference from the glidepath baseline, and panels (g)–(i) show the relative difference from the same baseline.

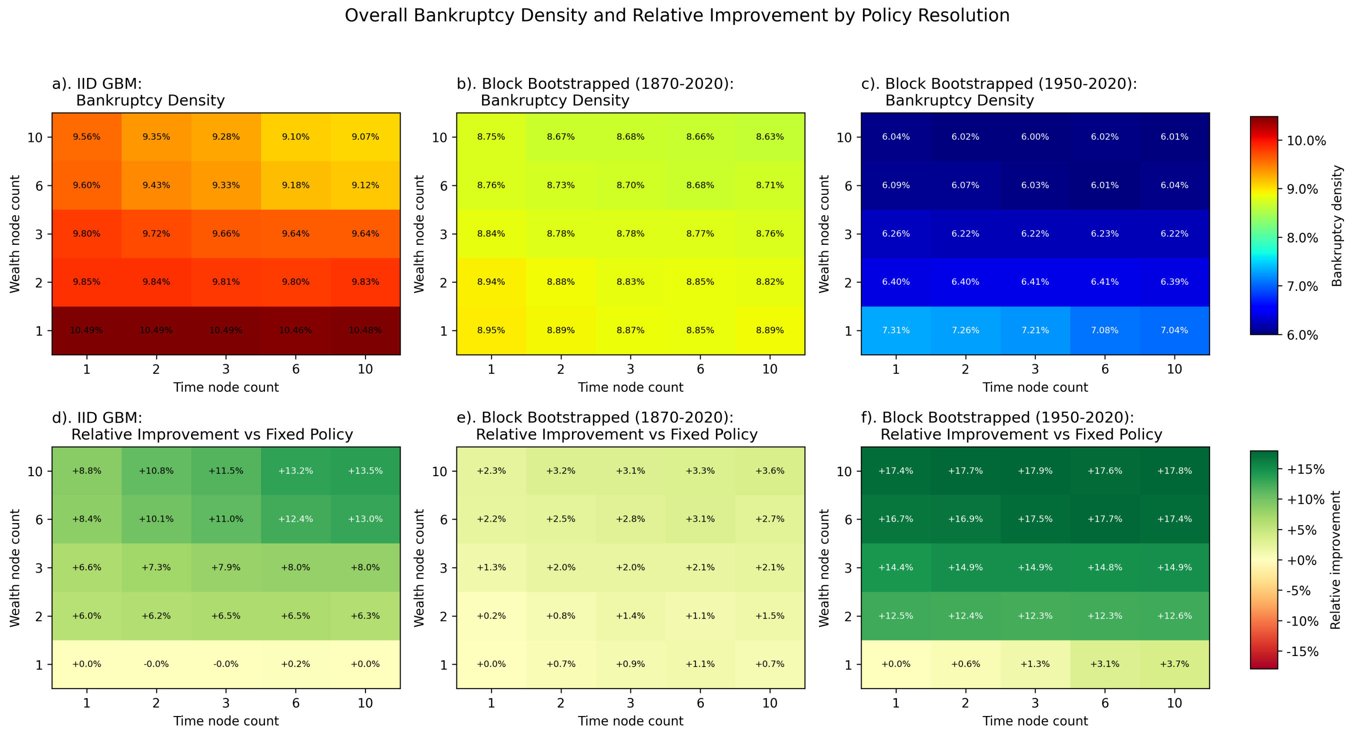

Figure 22: Simulation-weighted bankruptcy density by control-matrix resolution and return-sampling methodology. The top row reports the mean bankruptcy density for each sampler, time-node count, and wealth-node count. The bottom row reports the relative improvement against the fixed-policy baseline, with positive values indicating lower bankruptcy density than the baseline.

6 Discussion

In this section, we interpret the main mechanisms behind our empirical results and assess their practical implications for retirement advice. We first examine why rising-equity glidepaths emerge in our optimisation framework, and how this depends on sequence risk, wealth heterogeneity, and return-generation assumptions. We then evaluate the comparative performance of increasing the adaptability of the policy in wealth and time dimensions, and ask whether age-based glidepaths provide an adequate basis for allocation decisions across heterogeneous retiree populations. Finally, we discuss the role of inflation dynamics, inflation-hedging channels, and the implications of inflation modelling on asset-allocation decisions for retirees.

6.1 Why do we see Rising Equity Glide-Paths?

Across all market condition generators, we observe that the optimal glide-paths tend to be rising-equity glide-paths or 100% equity allocations (Figures 12 and 14). This is consistent with the findings of a number of previous studies (e.g. Anarkulova et al., 2025; Pfau & Kitces, 2014) and is in contrast to the traditional declining-risk glide-paths that are often recommended by popular financial rules of thumb and implemented by target-date funds. We think that this is an interesting result that warrants further discussion. While explanations for the front-loading of safe assets in the early retirement period make sense as a way to avoid sequence risk, it is not immediately clear why the optimal glide-paths tend to return to 100% equity allocations after that initial period. After all, sequence risk is not eliminated after the initial retirement period, and the retiree still faces a long retirement horizon with substantial market risk. So why then do the optimal glide-paths return to 100% equity allocations after the initial retirement period?

Pfau and Kitces (2014) provides a partial account of the mechanism we think is operating here. Their insight is that the success or failure of a retirement drawdown strategy is determined almost entirely by the sequence of returns in the first half of retirement rather than by the overall average return. If early returns are favourable, the retiree accumulates such a substantial buffer relative to their original spending goal that subsequent bear markets in the second half of retirement pose little threat to sustainability. Conversely, if early returns are poor, the portfolio is so severely stressed that any subsequent recovery is absolutely necessary to carry it through to the end. Pfau and Kitces (2014) also argue that rising equity allocations in the second half of retirement are optimal because they allow the retiree to dollar-cost-average into a recovery.

This mechanism is roughly consistent with what we observe in our results. The optimal glidepaths across all market condition generators concentrate safe-asset holdings in the early retirement period and then transition toward high equity exposure for the remainder (Figure 12). The frontloading of safe assets directly reduces the dollar-value exposure to sequence risk at the point when a bad early sequence would be most damaging. However, we think that the Pfau account is a good but not quite complete explanation of the mechanism at work. In our view, the return to high equity exposure in the second half of retirement is driven much more by the heterogeneity of the wealth paths that emerge from the early retirement period. Consistent with the observations of Anarkulova et al. (2025), the glidepaths in our study concentrate safe-asset holdings in a very narrow early window, when the variance of outcomes is lowest. Past this point, it is not so much that the sequence risk disappears (although it does reduce), but rather that a significant portion of the retiree population has already crossed into a hail-mary zone where the only way to avoid bankruptcy is to maximise expected returns regardless of risk. The cohort becomes highly heterogeneous, and the optimal strategy is to allocate to equities in the second half of retirement because it is the only lever remaining that can plausibly avert sustained bankruptcy for those who have fallen into this zone.

The rising-equity glidepath that emerges from the optimisation is therefore a population-level triage of risk over these two very different situations. Retirees who survived the early period with a strong buffer are relatively insensitive to the allocation in the second half; those who did not are in a hail-mary zone where any conservative allocation simply locks in a worse expected outcome. Those in the latter group pull the optimiser toward high equity in the second half of retirement, and the first group imposes essentially no countervailing pressure toward conservatism. This interpretation complements rather than contradicts the Pfau account: the early conservative hedge is genuinely motivated by avoiding sequence risk at the point of maximum dollar exposure, but the return to equities afterward is as much a consequence of this population bifurcation as it is of systematic dollar-cost averaging into a recovery. It also explains why the tendency toward higher equity allocations (and conversely, a lower and more concentrated sequence-risk hedge) is more pronounced when optimising policies across a range of initial wealth levels (Figure 14): including lower-wealth retirees who are more likely to be budgetarily dominated from the outset further biases the optimisation toward strategies that maximise expected returns at all stages. This is also consistent with why increasing glidepath resolution improves performance (Figure 8): finer time nodes allow the policy to concentrate the early-retirement safe-asset hedge more precisely, rather than spreading it across the entire retirement horizon and sacrificing expected returns in periods where the hedge adds little value.

We note, however, that the strength of this mechanism depends on the return-generation process. The effect is most pronounced under Synthetic Markets (IID GBM), where sequence risk is most acute because returns are serially independent and there is no mean reversion to dampen the persistence of a bad early draw (Figure 8). Under the block-bootstrapped samples, where returns preserve historical autocorrelation and mean reversion, the optimal safe-asset allocation in the early period is smaller and the performance gain from finer glidepath resolution is more modest. This is consistent with the argument of Anarkulova et al. (2025) that IID assumptions exaggerate sequence-of-returns risk relative to historically calibrated return dynamics. Our results thus provide qualified support for both the Pfau mechanism and the Anarkulova et al. critique: the sequence-risk channel is real and justifies a rising-equity structure, but its magnitude is likely overstated under IID assumptions, and the hail-mary bifurcation provides an additional and perhaps more fundamental driver of the return to high equity in the second half of retirement.

6.2 The Performance of Wealth-Responsive Strategies

A recurring theme throughout our results is that the retiree's current wealth level is a more informative state variable than elapsed retirement time. This was already implied by the structure of the fixed-allocation results (Section 5.1), where the optimal allocation varied substantially across initial wealth levels, but it is most clearly demonstrated in the performance of the wealth-responsive strategies.

Comparing unified wealth-responsive strategies against wealth-specific fixed allocations (Figure 16), we find that across almost all initial wealth levels, market condition generators, and wealth-node resolutions, the wealth-responsive strategies achieve bankruptcy densities comparable to or better than the wealth-specific fixed allocation baseline. This is a notable result on its own: a policy that has never been told the retiree's starting wealth is matching or outperforming one that was calibrated to it. The improvements are most pronounced in the Historical Markets (Modern Era) and Synthetic Markets (IID GBM) generators, and the largest relative gains consistently appear at middle initial wealth levels, roughly \$600,000–\$850,000. This is consistent with our interpretation that the wealth-responsive strategies are able to adapt to the bifurcation of wealth paths that emerges in this range: retirees who experience a poor early sequence fall into the hail-mary zone (and the policy responds by increasing equity), while those who experience favourable early returns accumulate a protective buffer (and the policy can afford to be more defensive). A fixed allocation, by contrast, must pick a single point in the risk–return space and cannot adapt to whichever of these two regimes the retiree finds themselves in.

The comparison against the wealth-specific 10-node glidepath baseline (Figure 17) is more demanding. The wealth-specific glidepaths represent the best-case benchmark for time-based strategies: optimised with full knowledge of the retiree's initial wealth and calibrated independently for each starting position. Despite this informational advantage, the unified wealth-responsive strategies match or exceed their performance at most wealth levels and under most market condition generators. At a sufficiently high wealth-node resolution, wealth-responsive strategies achieve bankruptcy densities comparable to or better than the wealth-specific glidepath baseline across all but the highest initial wealth levels.

These patterns are also consistent with the optimal asset-allocation responses in Lin et al. (2025). In their retirement investment-consumption model with inflation risk and pension default risk, low-wealth retirees tilt toward riskier assets as a compensation mechanism, while higher-wealth retirees place greater weight on inflation hedging through inflation-linked instruments. This maps closely to what we observe in our allocation surfaces: low-wealth paths are frequently pushed into the hail-mary region with high equity exposure, whereas higher-wealth paths increasingly prioritise protection of real spending capacity. Our setup differs from Lin et al. (2025) in objective function and asset menu, but the directional mechanism is similar: wealth state changes which risk is most binding, and therefore changes which allocation response is optimal.

Taken together, these results suggest that adapting to the retiree's realised wealth trajectory is a more powerful adaptive mechanism than preemptive time-based scheduling. Whereas a glidepath must commit to a particular allocation trajectory at the outset, a wealth-responsive strategy continuously re-evaluates the retiree's current position and adjusts accordingly. This dynamic responsiveness appears to be the principal source of the performance advantage.

6.3 To What Extent are Glide-Paths an Adequate Basis for Retirement Asset Allocation?

Target-date funds and age-based allocation rules are designed to serve large, heterogeneous populations through a single unified strategy. The implicit premise is that a retiree's age is a sufficiently informative signal to support this: that two investors at the same stage of retirement have similar enough financial situations that the same allocation schedule is broadly appropriate for both. This premise is most acutely tested in precisely the setting for which these products are designed, namely population-level application across retirees with very different wealth levels. Our results suggest that age is a poor conditioning variable in this setting, and that wealth-responsive strategies are a more effective foundation for allocation decisions.

A secondary observation from the literature review (Section 2.1) reinforces this point. The academic debate on glidepath shape has produced essentially no consensus: different studies, each using defensible methods, find that equity should decline with age (e.g. Estrada, 2015), rise with age (e.g. Pfau & Kitces, 2014), remain approximately constant (e.g. Anarkulova et al., 2025; Blanchett, 2007), follow a U-shape, or follow an inverted U-shape (Estrada, 2014). This profound lack of agreement is, we suggest, itself symptomatic of the problem with the glidepath framework. If age were genuinely an informative signal for allocation decisions, we would expect different empirical approaches to converge on broadly similar answers. The fact that they do not suggests that the glidepath framework is searching for a signal that is not there. Age, in isolation, is too coarse a proxy for the factors that actually drive the optimal allocation, and the disagreement in the literature reflects this: each study finds a different answer partly because each study is sampling a different slice of the heterogeneous population of retirees, each with a different distribution of wealth, spending needs, and market histories. Our results provide a concrete mechanism for why this should be the case.

The first and most direct challenge to the glidepath framework is the degree to which the optimal glidepath shape depends on initial wealth. In Figure 12, the evolution of the optimal wealth-specific glidepath as both resolution and initial wealth vary reveals a family of shapes that are highly heterogeneous across the wealth range. Low-wealth retirees are allocated entirely to equities throughout, consistent with budgetary dominance. Middle-wealth retirees show a modest early-period hedge against sequence risk that then yields to high equity exposure. Higher-wealth retirees sustain a somewhat more prolonged early safe-asset allocation before similarly returning to equity-dominated strategies. The shapes are not merely scaled versions of one another; they differ in timing, magnitude, and qualitative structure. This heterogeneity is not a minor calibration detail. It implies that a single glidepath cannot simultaneously represent the optimal allocation for retirees at different wealth levels, and that choosing a unified glidepath necessarily involves a compromise that serves some retirees poorly.

This cost is directly quantified in Figure 13. When a single glidepath is optimised to perform well across the full range of initial wealth levels, the performance improvements over a simple fixed allocation are marginal and, under the IID GBM and Historical Markets (Full Period) generators, are essentially negligible. For the Historical Markets (Modern Era) generator, there are some meaningful gains at higher wealth levels with 10-node glidepaths, but even here the improvements are modest relative to what is achievable with a wealth-responsive strategy. The implication is clear: the time dimension alone, when the policy must also contend with wealth heterogeneity, contributes very little to portfolio outcomes.

The two-dimensional control-matrix results strengthen this conclusion. In Figure 22, performance improvements are much more sensitive to resolution along the wealth dimension than along the time dimension. Across all three market condition generators, moving from low to high wealth-node counts produces the largest reductions in bankruptcy density, while increasing time-node resolution at a fixed low wealth resolution delivers comparatively small gains. Consistent with this, control-matrix policies with richer wealth resolution generally match or outperform 10-node glidepath strategies that were calibrated to a specific level of initial wealth (Figure 21). Indeed, strategies that were only responsive to wealth, and not to time at all, were able to match or outperform the best-possible glidepath strategies across almost all initial wealth levels and market condition generators (Figure 17). This is a striking result: glidepath strategies only seem to match or outperform wealth-responsive strategies when the retiree is at the very highest wealth levels and when the strategy is targeted at a retiree with a specific amount of savings and no heterogeneity in the population. For heterogeneous populations, using age as a conditioning variable offers little to no advantage over a simple fixed allocation, as shown in Figure 21. It is usually only once you add in a wealth-responsive component, either by optimising a glidepath for a specific wealth level or by using a control matrix that is responsive to wealth, that you see meaningful improvements in performance.

Taken together, these results indicate that conditioning allocation on age alone is a poor basis for retirement asset allocation. The optimal glidepath shape is highly sensitive to initial wealth, and a single unified glidepath cannot simultaneously serve retirees at different wealth levels. Wealth-responsive strategies, by contrast, can adapt to the retiree's realised wealth trajectory and achieve superior performance across a wide range of initial wealth levels and market conditions. Put simply, knowing a retiree's age just does not provide enough information to make meaningful allocation decisions, and the widespread use of age-based glidepaths as a retirement allocation vehicle is therefore questionable, especially when aimed at a broad population.

Within any population, the distribution of wealth, income, spending needs, health, and risk preference is very wide. Two 65-year-olds who appear identical from the perspective of a target-date fund may occupy very different positions in the financial landscape: one may be in a hail-mary zone where the only viable response is to maximise expected returns, while the other may have sufficient wealth that inflation hedging and sequence-risk management are the dominant concerns. Prescribing the same allocation trajectory to both, on the basis of their shared age, is a coarse approximation. We are not currently aware of any target-date fund that makes differentiated allocation decisions based on the retiree's wealth, spending needs, or other relevant financial circumstances. Our results suggest that this is a significant limitation of the glidepath framework.

It is important to note that there are genuine limitations to these conclusions. We have not modelled age-contingent healthcare costs, which may represent a meaningful expenditure shock at advanced ages and could increase the informational value of the investor's age insofar as the spending needs of the retiree are more likely to change with age. This could provide further motivation for age-dependent allocation adjustment. Moreover, our findings suggest the advantages of the wealth-responsive policies are most evident in the middle range of the wealth levels tested; the differences narrow at the extremes of the wealth distribution, where budgetary dominance or substantial wealth buffer largely determine the outcome regardless of the allocation mechanism. Target-date funds may simply be targeting these individuals. Despite these caveats, the consistent pattern across all three market condition generators is that current wealth is a more powerful conditioning variable than elapsed time, and that policies conditioned on wealth outperform their time-based counterparts across most wealth levels and return environments. We therefore find it difficult to see how a single glidepath could be an efficient allocation vehicle for a heterogeneous population of retirees.

For practical purposes, this suggests that the widespread use of target-date funds as a retirement allocation vehicle warrants scrutiny. While they appear to offer an intuitively appealing solution to the problem of life-cycle investing, they are fundamentally limited by their reliance on age as the primary conditioning variable. Moreover, the declining-risk structure that is often implemented in these products is not supported by any of the optimal strategies we have observed, and the empirical literature on glidepath shape is inconclusive at best. It is hard not to conclude that the popularity of these products is driven more by their marketability than by evidence-based financial planning.

Implications for Consilium: Should Advisers Recommend Glidepath Strategies?

The case against glidepaths developed above applies most forcefully to products such as target-date funds, which must serve large and heterogeneous populations with a single predetermined schedule and have no mechanism for incorporating individual client information. Consilium's advisers operate in a fundamentally different setting: they work closely with individual clients and will, in most cases, have a good understanding of the client's wealth at retirement and their anticipated spending needs. This changes the calculus considerably.

The strongest practical objection to glidepaths is that the optimal shape depends heavily on where the client sits relative to their spending capacity, that is, whether they are in a hail-mary zone where maximising expected returns is the only viable response, or a more comfortable position where managing market risk is the primary concern. An adviser who knows the client's starting wealth and spending targets can, in principle, select a glidepath calibrated to that specific situation rather than applying a generic schedule. In this personalised context, a glidepath can be a reasonable and communicable strategy, particularly for clients who value simplicity and a rule-based approach that reduces the temptation to make reactive allocation changes in response to short-term market movements.

However, two important caveats apply. First, the adviser must be prepared to revisit and update the strategy as the client's circumstances evolve. A glidepath calibrated to a client's situation at age 65 may become inappropriate by age 75 if their wealth has been substantially depleted by adverse market returns or has grown beyond expectations. In the former case, the client may have moved into the hail-mary zone and require a more aggressive posture than the original schedule specifies; in the latter, a more conservative approach may be warranted. In this sense, the strategy would end up being more akin to a wealth-responsive control matrix than a fixed glidepath, with the glidepath serving as a convenient shorthand for the allocation schedule at a given point in time. Even in the best case scenario, where the adviser has correctly assessed the client's initial position, a static glidepath is unlikely to remain optimal over the course of a long retirement.

What may be more valuable than a glidepath recommendation per se is guidance around the concept of a client's *safe wealth threshold*, that is, the wealth level above which projected spending is unlikely to exhaust the portfolio under plausible market outcomes, and below which more aggressive "hail-mary" strategies become necessary. This threshold is not fixed; it depends on the client's spending targets, remaining life expectancy, and the prevailing return environment. Providing advisers with a clear framework or rule of thumb for assessing where their client sits relative to this threshold, and how the appropriate allocation changes as a result, could be valuable for making an assessment of the strength of their client's current financial position.

In summary, Consilium should not recommend glidepath strategies to its investor base broadly, as the heterogeneity of client circumstances means that any single schedule will inevitably be poorly calibrated for a substantial proportion of clients. For individual advisers working closely with specific clients, a bespoke and periodically reviewed glidepath remains a reasonable tool, provided it is understood as an approximation requiring ongoing revision rather than a set-and-forget rule. The more durable contribution of this research for Consilium's advisory practice is the concept of wealth-relative risk positioning, which provides a principled and adaptable basis for ongoing allocation decisions and one that advisers are well-placed to apply given their close working relationships with clients.

6.4 Why do Bootstrapped Scenarios Produce Near-Zero Bond Allocations?

A recurring observation across our results is the near-complete absence of bonds from the optimal allocations produced under both block-bootstrapped return generators, in contrast to the non-trivial bond allocations that appear under the Synthetic Markets (IID GBM) model. We noted this divergence briefly in the fixed-allocation discussion (Section 5.1.2), where Anarkulova et al. (2025) also observed that IID assumptions tend to produce higher bond allocations than historically calibrated methods. Here we investigate the mechanism in detail.

Figure 23 compares mean annual returns and mean inflation across the three market condition generators. Two features stand out immediately. First, average inflation is substantially higher in both bootstrapped datasets (approximately 5.3% per annum for Historical Markets (Full Period) and 4.3% for Historical Markets (Modern Era)) than in the Synthetic Markets (IID GBM) model (2.5%). Second, nominal equity returns are correspondingly higher in the bootstrapped data, so that real equity returns are broadly comparable across generators. By contrast, the real bond return in Historical Markets (Full Period) collapses to near zero: the higher average inflation in the full historical sample almost entirely erodes the nominal bond premium. Under Historical Markets (Modern Era), the real bond return is positive but modest.

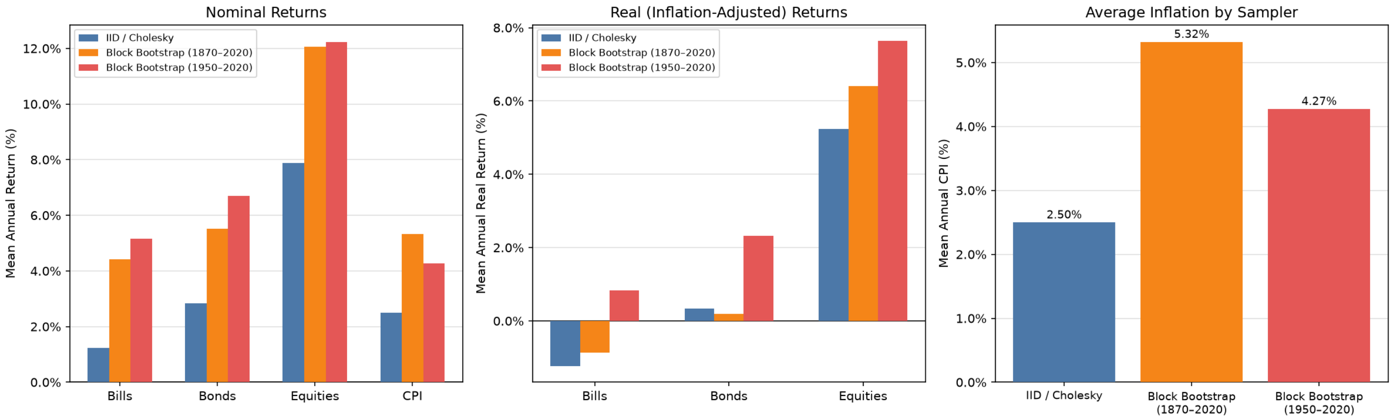

Figure 23: Mean annual nominal returns and mean inflation (CPI) across the three market condition generators (Synthetic Markets IID GBM; Historical Markets Full Period 1870–2020; Historical Markets Modern Era 1950–2020). The left panel shows nominal returns for all four asset classes including CPI; the centre panel shows real (inflation-adjusted) returns for the three investable asset classes; the right panel shows the mean annual inflation rate for each generator. The bootstrapped generators embed substantially higher average inflation than the IID model, compressing real bond returns and, in the full-period case, eliminating the bond premium over cash entirely.

Differences in average returns alone, however, do not fully explain why the optimiser assigns zero weight to bonds. Even a near-zero real bond return could justify a positive allocation if bonds provided useful diversification or reduced the downside risk of the portfolio. The deeper explanation lies in the dynamics of inflation across the two types of return generator.

A fundamental difference between the IID model and the block-bootstrapped scenarios is how inflation behaves over time. Under Synthetic Markets (IID GBM), each year's inflation draw is independent of every other; there are no inflationary regimes, only symmetric noise around a 2.5% mean. Under the block-bootstrapped generators, inflation is a persistent process: a high-inflation year is systematically followed by further high-inflation years, consistent with the clustering of inflationary episodes visible in historical macroeconomic data.

Figure 24 plots the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the annual CPI series for both bootstrapped generators and the underlying JST historical panel. The ACF decays slowly from a substantial lag-1 value, and the PACF is dominated by the first lag, consistent with an AR(1)-like process for the inflation rate. The IID model, by construction, has zero autocorrelation at all lags.

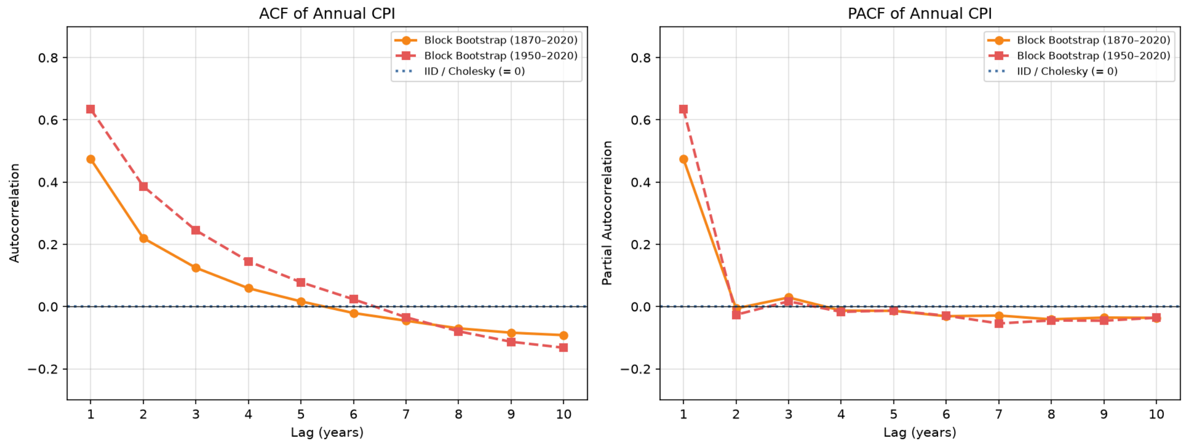

Figure 24: Autocorrelation function (ACF, left) and partial autocorrelation function (PACF, right) of the annual CPI inflation rate for the two block-bootstrapped generators. The grey shaded band indicates the approximate 95% white-noise confidence interval ($\pm 1.96/\sqrt{T}$). The dotted blue line marks the IID reference of zero autocorrelation. Both bootstrapped series exhibit a substantial positive lag-1 ACF and a PACF that drops sharply after lag 1, consistent with an AR(1) inflation process. This persistence implies that a single bad inflation year predicts several further years of elevated inflation.

This persistence matters enormously for a 35-year retirement simulation. In the IID world a bad inflation year is isolated, whereas in the bootstrapped world inflationary episodes cluster. Additionally, the bootstrapped generators embed a higher average inflation rate than the IID model, so inflation mitigation is a more pressing concern.

Another reason the optimiser retains cash (bills) while discarding bonds is not simply that both are "safe" assets. Bills and bonds respond to inflationary regimes in fundamentally different ways. Figure 25 plots the Pearson correlation of each asset's multi-year compounded return with the corresponding multi-year compounded inflation, across horizons of one to ten years. Under Historical Markets (Modern Era), the bill–inflation correlation is large and positive at all horizons (approximately 0.5–0.7), while the bond–inflation correlation is much lower. This pattern may reflect central bank policy that has increasingly moved to adjusting short-term policy rates in response to inflation, amplifying the positive correlation between bill yields and the inflation rate. The implication is that, over a 35-year retirement horizon, bills function as a substantially better inflation hedge than nominal bonds under historically calibrated return dynamics. Under Historical Markets (Full Period), where the pre-1950 period includes episodes without active central bank management of short rates, this effect is substantially weaker but still present.

The most direct evidence for why the optimiser drops bonds is the distribution of 35-year compounded real returns across the full ensemble of bootstrap scenarios. Table 8 summarises key percentiles and tail probabilities for each asset class. Several features of this table explain the optimiser's behaviour. For equities, the 5th percentile of the 35-year real return is +60% under Historical Markets (Full Period) and +109% under Historical Markets (Modern Era), both positive and substantial. For bonds, the corresponding figure is −87% and −37% respectively; in the worst decile of scenarios, bonds lose the large majority of their real value over a 35-year retirement. Under Historical Markets (Full Period), the probability that bonds deliver a negative 35-year real return is 26%; under Historical Markets (Modern Era) it is 14%. In approximately one in four or one in five scenarios, holding bonds generates a negative real return over the entire retirement horizon.

Across both datasets, the 5th percentile of equity real returns over 35 years is positive and the probability of a negative outcome is below 2.5%. This is not because equities are less volatile on a year-to-year basis; they are substantially more volatile. Rather, it reflects the long-run behaviour of equities versus bonds under historically calibrated return dynamics: over sufficiently long horizons, the equity risk premium and the compounding of

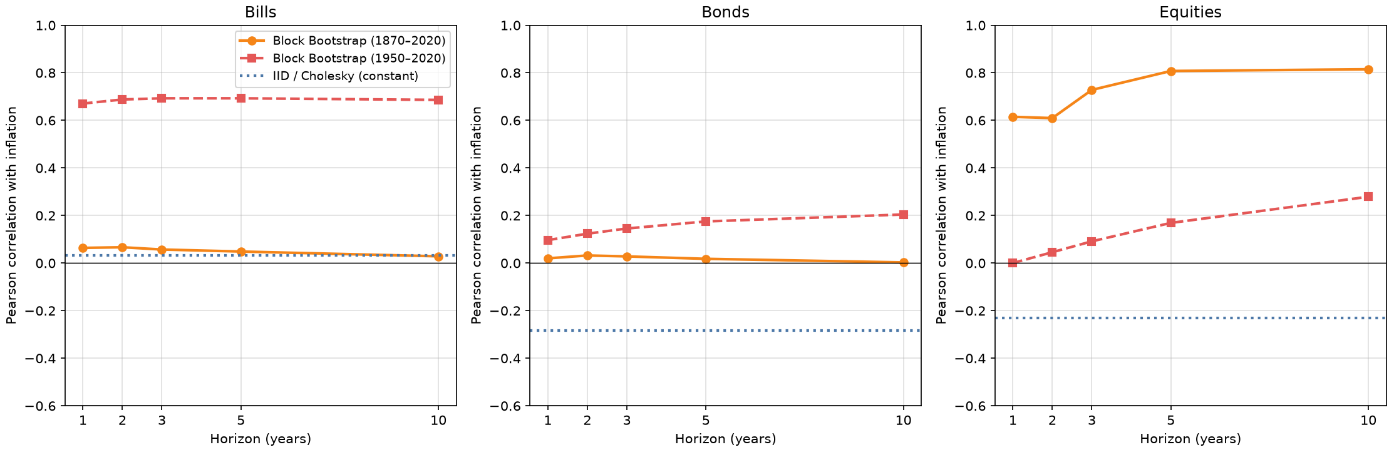

Figure 25: Pearson correlation of each asset class's k-year compounded return with k-year compounded inflation, for $k \in \{1, 2, 3, 5, 10\}$ years, for both block-bootstrapped generators. The dotted lines show the corresponding IID correlation, which is constant across horizons by construction. Under Historical Markets (Modern Era), bills exhibit strongly positive correlation with inflation at all horizons (≈0.5–0.7), indicating that bill returns tend to rise with inflation and thus preserve real purchasing power. Bond correlation with inflation remains low and near zero under both generators. Equity correlation with inflation is modestly negative in the modern era.

Table 8: Distribution of 35-year compounded real returns across bootstrap scenarios, for each investable asset class. Statistics are computed over all scenarios in each bootstrapped dataset. The retirement horizon of 35 years is used here to match the 65–100 specification used in the optimisations. P(< 0) is the probability of a negative 35-year compounded real return; P(<Bills) is the probability that the asset's real return falls below that of bills in the same scenario.

Dataset	Asset	p5	p25	p50	p75	p95	P(< 0)	P(<Bills)
Historical Markets Full Period	Bills	−89.0%	−16.8%	39.4%	109.0%	253.5%	32.0%	—
	Bonds	−86.6%	−5.0%	91.3%	233.4%	559.7%	26.3%	24.2%
	Equities	60.3%	341.7%	708.6%	1317.5%	2946.9%	2.2%	0.6%
Historical Markets Modern Era	Bills	−33.6%	2.6%	34.9%	78.1%	167.7%	23.0%	—
	Bonds	−37.4%	37.3%	123.6%	251.4%	514.0%	13.9%	19.6%
	Equities	108.8%	413.9%	853.4%	1625.6%	3681.6%	0.7%	0.5%

positive real returns tends to dominate the variance of the return distribution, while the persistent inflation risk of nominal bonds creates an asymmetric left tail that does not diminish in the same way. The finding is consistent with the argument of Siegel (1994) that equities are the dominant long-run asset when the holding period is long and inflation is uncertain, and with the empirical evidence of Dimson et al. (2002) that the long-run real return on equities has exceeded that of bonds in virtually every developed market over the twentieth century.

The comparative position of bills is more nuanced. Under both bootstrapped generators, bills also suffer substantial negative real returns in bad scenarios: their 5th-percentile outcome is approximately −89% under Historical Markets (Full Period) and −34% under Historical Markets (Modern Era). However, from the optimiser's perspective, the relevant comparison is not between bills and some absolute standard of safety, but between bills and the available alternatives. In the scenarios where bills perform poorly, bonds perform similarly (the bill and bond 5th percentiles are almost identical), so substituting from bills to bonds provides no left-tail improvement while forfeiting the bill's inflation-hedging benefit in moderate inflationary scenarios. In approximately 24% of Historical Markets (Full Period) scenarios and 20% of Historical Markets (Modern Era) scenarios, bonds underperform even bills in the same scenario. The optimiser therefore finds bills weakly dominant over bonds: they provide no better expected return, no better downside protection, and a meaningfully better inflation correlation. The dominant alternative for generating wealth over the full retirement horizon is equities, which under both bootstrapped generators exhibit a much better left tail and substantially higher median real return. Thus the optimiser concentrates there.

6.4.1 Caveats to the No-Bonds Allocations Based on the Bootstrapped Data

Several important caveats should accompany these conclusions. Firstly, the bond return series in the Synthetic Markets (IID GBM) model was calibrated using Consilium's forward-looking capital market assumptions, which likely reflect a blend of government and corporate bonds and are therefore informed by recent market data and current yield levels rather than the long historical average. The bootstrapped bond series, derived from the JST Macrohistory dataset, is based on long-term government bond yields in developed markets. Corporate bonds, and particularly investment-grade credit, typically offer a spread over government yields that reflects default and liquidity risk. This means the Synthetic Markets (IID GBM) model may be incorporating a higher expected bond return than is embedded in the bootstrapped data, making the comparison of bond allocations across generators partly an artefact of this calibration mismatch rather than a purely methodological one. All else equal, a higher expected bond return would tend to increase the optimal bond allocation in the IID model; the bootstrapped model does not benefit from this premium.

Secondly, the Historical Markets (Full Period) generator draws from return data beginning in 1870, a period that encompasses world wars, hyperinflationary episodes in continental Europe, the Great Depression, and the early twentieth century gold standard regime. Many financial economists would argue that these experiences are not representative of the future investment environment for a New Zealand retiree: institutional arrangements, central bank mandates, and global capital markets are fundamentally different today. In particular, the hyperinflationary scenarios drawn from the early twentieth century may represent tail risks that are not plausibly comparable to those faced in modern developed economies with inflation-targeting central banks. The near-zero bond allocations under Historical Markets (Full Period) should therefore be interpreted cautiously; they may reflect a degree of historical pessimism about inflation regimes that overstates the realistic risk to a contemporary investor.

The bootstrapped asset class does not include inflation-linked bonds (such as New Zealand government inflation-indexed bonds or TIPS in the US context). Inflation-linked bonds directly address the inflation sensitivity that drives bond losses in the bootstrapped scenarios: by design, their principal and coupon payments adjust with the price level, so their real return is insulated from inflationary shocks. The absence of this asset class from the investable universe means the model may be understating the potential role of fixed income in a well-structured retirement portfolio. In practice, a meaningful allocation to inflation-linked bonds could partially substitute for the inflation-hedging function currently fulfilled by bills.

Finally, there is a broader question of whether the historical return data provided by JST Macrohistory is of high enough quality to support the conclusions drawn from it and if it is appropriate to use it as the basis for a long-horizon retirement planning going forward. Data quality issues, particularly in the early part of the sample, may introduce biases or inaccuracies. Comparing results derived from JST data to other historical datasets, such as those compiled by Dimson et al. (2002) or GFD as used by Anarkulova et al. (2025), may provide a robustness check on the findings. Additionally, the historical period covered by JST may not be representative of future market conditions, particularly given structural changes in global financial markets, monetary policy frameworks, and demographic trends.

Taken together, these caveats do not overturn the main finding but do suggest that the optimal bond allocation under historically calibrated return dynamics is sensitive to assumptions that deserve careful scrutiny. The appropriate conclusion for Consilium is not that bonds are categorically bad investments, but that their role in a retirement portfolio depends critically on the return-generation methodology, the specific bond instruments considered, and the historical period used to calibrate expectations. Potentially, the most important insight is that inflation needs to be explicitly modeled and considered in the asset-allocation process. Consilium's traditional modelling of inflation as a fixed 2.5% per annum rate is not sufficient to capture the true risk of inflationary regimes. Even the model used in this report, which incorporates a correlated inflation process into the Brownian motion of the Synthetic Markets (IID GBM) generator, is a simplification that does not capture the autocorrelation behaviour of inflation observed in the historical data. A more sophisticated treatment of inflation dynamics, potentially including stochastic volatility or regime-switching models, may be warranted to better inform asset-allocation decisions. This is particularly relevant to retirees, who are more exposed to the long-term effects of inflation on their purchasing power and retirement income, as we discuss in the next section.

6.5 Why does Inflation Matter for Retirement Asset Allocation in Particular?

Retirees are structurally more exposed to sustained real-income erosion than working-age households. Workers may receive partial inflation compensation through nominal wage growth, either through explicit indexation or through labour-market adjustment. By contrast, retirees finance consumption primarily from accumulated nominal claims, drawdown portfolios, and pension streams that are often imperfectly indexed. When inflation remains elevated for multiple years, this asymmetry compounds, workers can absorb part of the shock through income adjustment, whereas retirees must absorb more of it through real wealth depletion.

This distinction is consistent with recent household-level evidence from the euro-area inflation surge. Pallotti et al. (2024) estimate sizable and heterogeneous welfare losses, and show that retirees were among the largest losers, while many younger working-age households experienced neutral or positive effects. Their interpretation emphasises nominal balance-sheet positions and heterogeneous income channels, which aligns with our mechanism. In our framework, the key channel is not only the level of inflation, but its persistence over a long retirement horizon, because persistent inflation repeatedly raises real spending pressure while eroding the real value of nominally exposed wealth.

The wage channel itself is also state-dependent and incomplete. The Phillips-curve relationship documented by Phillips (1958) links labour-market tightness to money-wage growth, and modern inflation episodes often include feedback between wage inflation and price inflation. This channel cushions workers but retirees cannot benefit from this mechanism.

Implications for Consilium's Strategic Asset Allocation

A practical implication for Consilium is that strategies calibrated primarily to maximise expected nominal returns can understate a central retirement risk, persistent inflation erosion of real spending capacity. This matters most for clients outside accumulation, where labour-income adjustment is limited and portfolio drawdown is the primary funding source for consumption.

Recent evidence in Lin et al. (2025) is informative here. In their model, low-wealth retirees increase risky exposure as a compensation response, while higher-wealth retirees place greater weight on inflation-hedging assets, particularly inflation-indexed bonds. We observe a related mechanism in our own results, once wealth is sufficiently high that immediate bankruptcy risk is lower, inflation protection in the form of exposure to unhedged equities causes equity exposures to persist. Moreover, any autocorrelation and thus the persistence of inflationary shocks further increases the risk faced by retirees, as it prolongs periods of high inflation and exacerbates the erosion of real wealth. Taken together, these factors suggest that a proper treatment of inflation risk is critical for designing retirement asset allocation strategies.

This suggests that Consilium's current inflation treatment is likely too weak for retirement decumulation analysis. A constant-rate framework cannot represent inflation persistence, regime shifts, or clustered inflation shocks, and therefore cannot properly value inflation-hedging instruments or identify when they should become more prominent in higher-wealth retirement portfolios. Moreover, Consilium does not currently have any inflation-mitigation instruments in its conservative or moderate portfolios, which may be a gap in the current strategic asset allocation framework. If Anarkulova et al. (2025) are correct that bonds display a negative correlation with inflation at long horizons (such that they lose value due to inflation), then the current SAA framework may be systematically exposing retirees to inflation risk.

7 Conclusion

This research was motivated by the apparent lack of empirical support for glidepath frameworks that are so widely used and recommended in retirement planning. Where the literature does exist, it often disagreed with the declining-risk intuition that underpins glidepath design. We therefore set out to investigate the optimality of glidepath strategies in a retirement context, and to compare them with alternative allocation mechanisms that are responsive to wealth and time. In doing so, we believe that we have provided a more complete and empirically grounded answer to the question of whether glidepaths are an appropriate allocation vehicle for retirees. In addition, the research has highlighted the importance of other modelling assumptions, and has raised potentially fruitful avenues for future research.

On balance, our results suggest that the declining-risk glidepaths recommended by target date funds and popular rules of thumb (such as age-in-bonds rules) are largely ineffectual as financial advice when applied to broad and heterogeneous populations of investors. Our results show that optimal asset-allocation policies are highly sensitive to the retiree's wealth and consumption needs, and that a single glidepath cannot simultaneously serve retirees across a wide range of individual circumstances. We show that policies with greater responsiveness to wealth, or to time and wealth jointly, consistently outperform glidepath strategies. Glidepaths perform well only when they are tailored to an individual whose circumstances are known in advance. Moreover, the glidepath shapes recommended by the industry do not resemble the rising-equity glidepaths that emerge in our analysis. These findings were consistent across all modelling methodologies. While we cannot comment on their effectiveness during the accumulation period (when the household is still working), we would question any recommendation to use a glidepath as a retirement allocation vehicle, especially when it is applied uncritically to a broad audience.

Our results also highlight the sensitivity of the strategies to modelling assumptions. In our report, we have examined three methodologies used to create simulations of different market conditions (asset returns and inflation) for our retirees. The first was a synthetic market generator based on a geometric Brownian motion model with parameters calibrated to Consilium's forward-looking capital market assumptions. The second and third were block-bootstrapped generators based on historical return data from the JST Macrohistory dataset, one using the full 1870–2020 period and the other using only the 1950–2020 modern era. Major differences emerged across all three generators, particularly in the relative allocation to safe and risky assets, as well as in the optimal allocation to bonds as a safe asset, with both of the bootstrapped samples producing trivial bond allocations. The substantial disagreements arising from the choice of modelling methodology suggest that the optimality of any given allocation policy is highly sensitive to the underlying assumptions about market conditions, and that care should be taken in interpreting results from any single model. While we have already stated that the goal of this research is not to make any recommendation in particular for a retirement allocation strategy, we further highlight the risk of making any decision based on the results presented in this paper.

However, while the specifics of the strategies do vary across the three generators, underlying features emerge consistently. The first is that there is consistently a level of wealth relative to one's future consumption where the optimal strategy tips towards increasing expected return irrespective of the additional market risk. We have termed this the "hail-mary" zone, as the strategy essentially takes the form of a "nothing-to-lose" gambit where the small chance of a large positive outcome that "miraculously" sustains the retiree's consumption needs is preferred to a more conservative strategy that will fail anyway. Studying where this zone forms relative to a retiree's savings and consumption could be an informative area for future research. We note that this region is likely the reason that, when they have been properly studied, the optimal glidepaths that emerge from the literature are often rising-equity rather than declining-risk. This is because the optimal strategy becomes more influenced by retirees who have fallen into the hail-mary zone as the simulated cohort ages, which overwhelms the preference for conservative high-wealth retirees to reduce risk once their wealth is sufficient to meet their consumption needs.

With this in mind, we also note that papers which conclude that all-equity allocations perform similarly to glidepaths in terms of expected outcomes are likely observing the same phenomenon. When the optimisation landscape is dominated by the low-wealth retirees who are in the hail-mary zone, the overall allocation is heavily weighted towards equities. Commenting on the findings of Anarkulova et al. (2025), Felix (2025) argues that the "all-equity" allocation is a reasonable approximation of the optimal strategy, and that it is therefore a reasonable recommendation for retirees. We would caution against applying this interpretation to an individual's retirement allocation strategy, as it is based on the average of a heterogeneous population and ignores the fact that the optimal strategy sensitive to an individuals wealth and consumption needs. Depending on modelling assumptions, we find that the optimal allocation for a retiree with sufficient wealth to meet their consumption needs is often a more conservative allocation. Again, the specific modelling assumptions and asset menu, particularly concerning treatment of inflation and the inclusion of inflation-hedging products, will influence the optimal allocation.

The modelling of inflation, particularly as it correlates with asset returns over long horizons, is a key area for future research. Our results appear to show that inflation hedging is a key driver of the systematically higher equity allocations observed in the bootstrapped samples. While the combined hedged/unhedged asset class makes it difficult to say for sure, and future research might increase the granularity of the asset menu to better understand the role of inflation hedging, we note that inflation hedging was not a feature of the synthetic market generator, which may explain the lower equity allocations observed in that generator. The role of inflation hedging is particularly important for retirees, who are more exposed to the long-term effects of inflation on their purchasing power and retirement income. For any work that seeks to make recommendations for retirement allocation strategies, we would recommend that the modelling of inflation risk be a key consideration, and that the inclusion of inflation-hedging products in the asset menu be considered.

Overall, we think that this research demonstrates that the framework developed in this report is a powerful tool for testing the optimality of retirement allocation strategies across a wide range of modelling environments. We have demonstrated that the framework can find robust optimal strategies across a range of modelling assumptions, including with consumption needs that vary dynamically and different asset-return generators.

Using the control-matrix approach pioneered by Mäkinen and Toivanen (2024), we have evaluated glidepaths and alternative allocation rules in a way that is both flexible and interpretable, while remaining explicit about the trade-offs implied by each objective function and return generator. Because the framework is largely agnostic to specific modelling assumptions, it can be used to ask a much broader class of policy-design questions than glidepath shape alone.

A natural next step is to extend the framework to the full life-cycle problem by incorporating the accumulation phase and a richer model of labour-income risk. A second extension is to broaden the policy class beyond static time–wealth controls to include strategies that can adapt to realised market states, spending adjustments, and potentially additional retirement risks such as health-related expenditure shocks. Additionally, allowing the investor to adjust their consumption in response to realised market conditions would provide a more realistic and flexible framework for retirement planning. This would allow for the exploration of dynamic strategies that can respond to changing circumstances, rather than being locked into a predetermined glidepath.

We therefore see this framework not only as a way to evaluate existing retirement rules, but also as a practical platform for developing and stress-testing new ones. For Consilium, it could form the basis of a new generation of retirement-planning tools that produce personalised allocation strategies for Consilium's clients. Compared with Consilium's current offering (the GPS), which mainly relies on iterative trial-and-error adjustments, this framework could provide a more systematic and transparent decision engine for advisers and clients. Provided the modelling assumptions are carefully specified, validated, and periodically reviewed, such a tool could support more consistent client outcomes while remaining interpretable in practice. At the same time, because the optimisation can be run across broad distributions of retiree circumstances, it also has the potential to function as a scalable and more effective alternative to conventional target-date fund designs.

We hope this work provides a useful foundation for future research, and contributes to the design of retirement allocation strategies that are more adaptive, more transparent, and better aligned with retirees' real financial constraints.

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A Simplex Projection Algorithm for Constrained Optimisation

To enforce the allocation constraints in Equation 13 during gradient-based optimisation, we use the simplex projection algorithm of Duchi et al. (2008). This algorithm solves the Euclidean projection problem:

$$\text{proj}_{\Theta}(\mathbf{x}) = \arg\min_{\mathbf{y}\in\Theta} \|\mathbf{y} - \mathbf{x}\|^2 \tag{26}$$

where $\Theta = \{\mathbf{y} \in \mathbb{R}^{k-1} : y_j \geq 0, \sum_j y_j \leq 1\}$ is the feasible set for risky asset allocations (with the safe asset allocation determined residually).

The algorithm proceeds as follows:

1. Sort the elements of $\mathbf{x}$ in descending order to obtain $x_{(1)} \geq x_{(2)} \geq \cdots \geq x_{(k-1)}$.
2. Find the threshold λ such that the projection is given by:
$$y_j = \max(0, x_j - \lambda) \tag{27}$$
3. The threshold λ is chosen such that $\sum_j y_j = 1 - \epsilon$ for some small $\epsilon > 0$. We use $\epsilon = 10^{-8}$ to ensure the sum is strictly less than one, which ensures the safe asset receives a non-negative residual allocation: $1 - \sum_j y_j \geq 0$.
4. To find λ, identify the largest index k^* such that:
$$x_{(k^*)} - \frac{1}{k^*}\left(\sum_{j=1}^{k^*} x_{(j)} - (1-\epsilon)\right) > 0 \tag{28}$$
5. Then set:
$$\lambda = \frac{1}{k^*}\left(\sum_{j=1}^{k^*} x_{(j)} - (1-\epsilon)\right) \tag{29}$$

A.1 Projecting Gradients onto the Tangent Cone Before Adam Updates

For the control-matrix optimisation runs that use Adam, we found it useful to project the raw gradient before calling the optimiser step. Let $\boldsymbol{\theta}_t \in \Theta$ denote the risky-asset weights at iteration t, and let $\mathbf{g}_t = \nabla_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta}_t)$ be the raw gradient of the loss.

The feasible set is defined by the non-negativity constraints $\theta_{t,j} \geq 0$, together with the simplex inequality $\sum_{j=1}^{k-1}\theta_{t,j} \leq 1$. We project gradients onto the tangent cone of this feasible set so that infinitesimal gradient-descent steps remain feasible whenever a constraint is active.

First, directions that would immediately violate an active lower-bound constraint are removed. Define the active set

$$\mathcal{A}_{\text{low}} = \{j : \theta_{t,j} \leq \epsilon,\ g_{t,j} > 0\}, \tag{30}$$

where $g_{t,j} > 0$ would move $\theta_{t,j}$ below zero under a gradient-descent update. The masked gradient is

$$\tilde{g}_{t,j} = \begin{cases} 0, & j \in \mathcal{A}_{\text{low}}, \\ g_{t,j}, & \text{otherwise.} \end{cases} \tag{31}$$

Next, we consider the simplex budget constraint. Unlike an equality-constrained simplex, the budget constraint is inactive whenever $\sum_{j=1}^{k-1}\theta_{t,j} < 1-\epsilon$. In this case no further modification is made and the optimiser is free to increase or decrease total risky-asset exposure.

When the budget constraint is active, $\sum_{j=1}^{k-1}\theta_{t,j} \geq 1-\epsilon$, we check whether the gradient-descent step would increase total risky exposure. Since the update takes the form $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\mathbf{g}_t$, this occurs whenever $\mathbf{1}^\top\tilde{\mathbf{g}}_t < 0$.

Only in this case do we project the gradient onto the tangent space of the active budget constraint by removing its component in the $\mathbf{1}$ direction,

$$\bar{\mathbf{g}}_t = \tilde{\mathbf{g}}_t - \frac{1}{|\mathcal{F}|}\left(\sum_{j\in\mathcal{F}}\tilde{g}_{t,j}\right)\mathbf{1}_{\mathcal{F}}, \tag{32}$$

where $\mathcal{F}$ denotes the coordinates not fixed by active lower-bound constraints. If the budget constraint is inactive, or if the descent direction reduces total risky exposure, we simply set $\bar{\mathbf{g}}_t = \tilde{\mathbf{g}}_t$. The resulting projected gradient $\bar{\mathbf{g}}_t$ replaces the raw gradient in the Adam update. After the optimiser step, we still apply the Euclidean simplex projection of Equation 26 as a feasibility safeguard. This combination allows Adam to adjust total risky exposure whenever the simplex inequality is inactive, while preventing updates that would move outside the feasible region when a boundary constraint is active.

B Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)

Mäkinen and Toivanen (2024) formulate their Monte Carlo optimisation in a way that is similar to the framework described in the main text. As in our setup, they use a control matrix that allows asset allocation to respond jointly to age and wealth. Their objective function is based on terminal wealth moments, specifically variance or semi-variance. They then compute gradients of this objective with respect to control-matrix parameters using adjoint methods, which enables efficient gradient-based optimisation (in their case, a quasi-Newton method with BFGS updates).

Because this method is mathematically elegant and computationally efficient, we have attempted to replicate it here. The code for this replication can be found in the accompanying repository, and the results are shown in Figure 26 for the mean-variance and mean-semi-variance cases.

In line with the setup described by Mäkinen and Toivanen (2024), the replication uses a Monte Carlo horizon of $T = 20$ years with time step $\Delta t = 0.25$, giving $N = 80$ time steps. The policy is represented on a time-wealth control grid with $M = 31$ wealth nodes, upper wealth bound $W_{\max} = 30$, and bilinear interpolation between nodes. In the notebook, wealth is clipped to $[0, W_{\max}]$ before interpolation. Optimisation is performed with L-BFGS-B (a quasi-Newton method) with box constraints.

For the one-risky-asset, no-leverage case used in Figure 26, the full parameterisation is summarised in Table 9.

Table 9: Parameter setup for the one-risky-asset mean-variance replication.

Parameter	Value
Risk-free rate (r)	0.03
Risky-asset drift (μ_1)	0.0795
Risky-asset volatility (σ_1)	0.15
Contribution rate (π)	0.1
Initial wealth (w_0)	1
Minimum risky allocation ($p_{\min}$)	0
Maximum risky allocation ($p_{\max}$)	1
Number of simulation paths (K)	10^5
Risk-aversion weight (λ)	0.1

The replication results are broadly consistent with those reported by Mäkinen and Toivanen (2024). However, they also highlight important limitations. First, as an objective for a retirement investor, terminal-wealth variance is not especially intuitive. In our replication, the terminal-wealth distribution becomes bimodal because the optimiser shifts to a highly conservative policy at high wealth levels, truncating the upper tail that would normally appear under a log-normal-like process. Although this behaviour is mathematically consistent with variance minimisation, it is not obviously aligned with investor welfare. A similar pattern appears under semi-variance, albeit at higher wealth levels. Because semi-variance is computed relative to the mean, higher-return strategies can increase the penalty by raising the mean itself, again encouraging upper-tail truncation. This reinforces the need for risk measures that are both interpretable and behaviourally relevant.

Additionally, the control matrix has limited effective resolution in early periods: many simulated paths map to only a few low-wealth nodes because initial wealth is far below the fixed upper wealth bound. This reduces the optimiser's ability to distinguish trajectories precisely when policy differences likely matter most due to compounding. It also creates many redundant parameters at wealth levels that are never reached (upper-left region in Figure 26). For this reason, we propose calibrating the control-matrix upper bound to the simulated wealth distribution so that resolution is concentrated where paths actually lie.

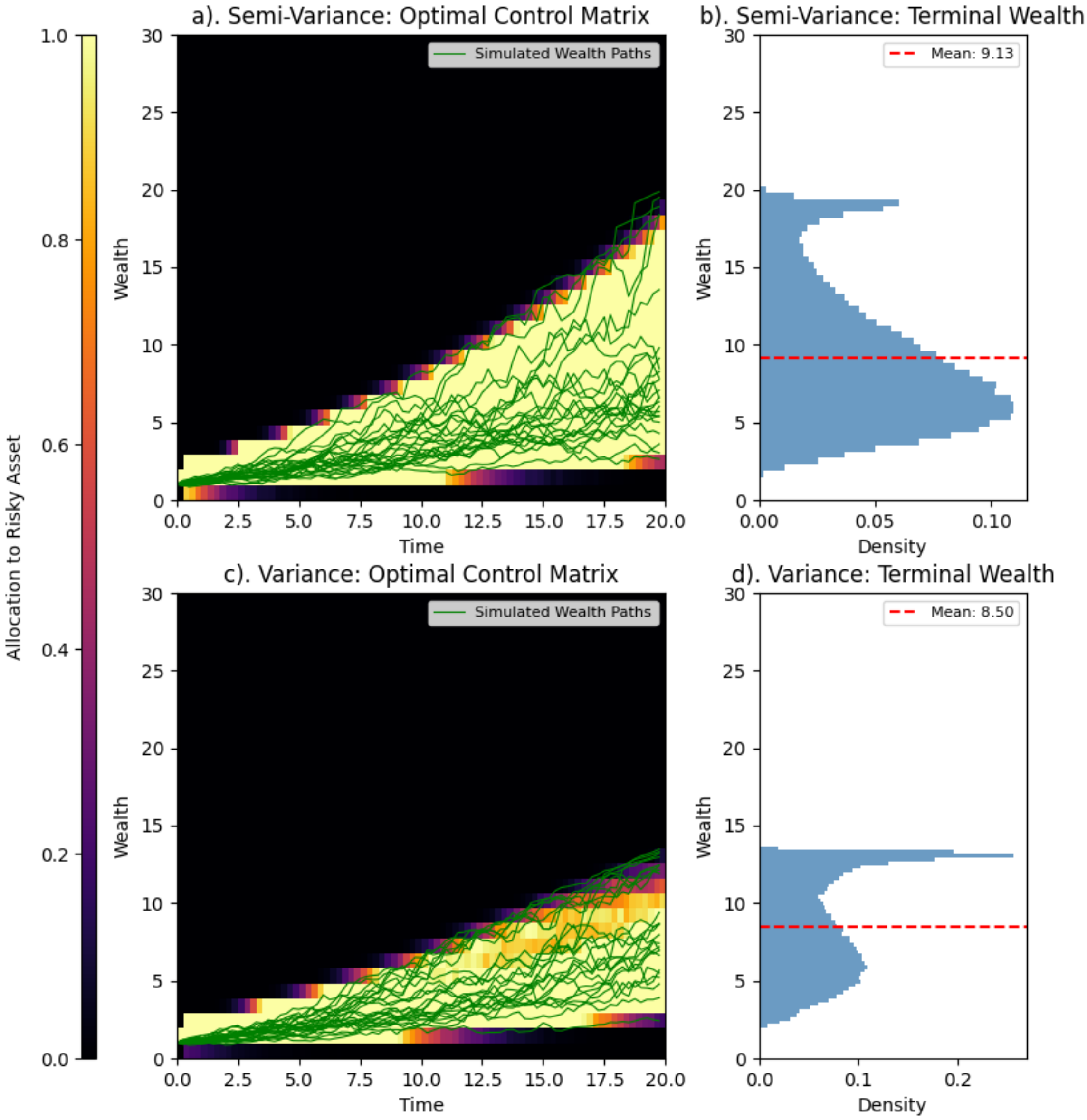

Figure 26: Replication of the control matrix optimization method proposed by Mäkinen and Toivanen (2024) for one risky asset and no leverage. The left column shows the optimal control matrix for the mean-semi-variance case (top) and the mean-variance case (bottom). The right column shows the corresponding histograms of terminal wealth for each risk measure respectively.

C Block-Bootstrapping: Investable Universes

To implement the block bootstrap methodology for New Zealand retirees, we must first construct appropriate investable universes from the JST database (Jordà et al., 2017). A critical insight from Anarkulova et al. (2025) is that the bootstrap procedure should reflect the investment opportunity set available to the domestic investor, which differs from simply using aggregate global indices. Specifically, at each bootstrap iteration, we designate one country as the "home" or "domestic" country, representing the investor's perspective, and construct return series for three asset classes relative to that country's viewpoint: domestic short-term government bills (cash), domestic intermediate-term government bonds, and international equities.

This approach requires us to define, for each year in the historical data, which countries can serve as domestic markets and which can serve as sources of foreign returns. These definitions are not uniform across time, as data availability and quality vary significantly across countries and periods. The following sections describe how we construct these investable universes and define the corresponding asset classes.

C.1 Data Quality Requirements and Eligibility Criteria

For a country to be eligible as a domestic market in a given year, we require complete and reliable data for all key return series needed to construct a domestic investor's portfolio. Specifically, a country can serve as "domestic" only if the following conditions are met:

1. **Complete asset return data:** The country must have non-missing values for equity total returns (`eq_tr`), bond total returns (`bond_tr`), and short-term bill rates (`bill_rate`) in that year.

2. **Inflation data:** The country must have a valid inflation rate (computed from CPI) to allow for real return calculations.

3. **Exchange rate data:** The country must have a valid exchange rate against the US dollar (`xrusd`) and its percentage change (`xrusd_return`), which is necessary for converting foreign returns to the domestic investor's currency (and vice versa).

4. **Market size constraint:** The country's equity market capitalization must not exceed 20% of the total market capitalization across all JST countries in that year. This threshold prevents the United States and United Kingdom being selected as domestic markets, as their inclusion would significantly reduce the diversity of bootstrap samples and make the simulated paths less representative of smaller developed economies like New Zealand.

For a country to be eligible as a source of foreign returns (either equity or bonds), we require non-missing asset return data (equity total returns `eq_tr` for the foreign equity universe, or bond total returns `bond_tr` for the foreign bond universe), valid exchange rate data (`xrusd_return`), and three-year rolling cumulative exchange rate depreciation not exceeding 50%.

The 50% cumulative depreciation threshold over a three-year rolling window serves to exclude protracted currency-crisis episodes that, while historically occurred, create irregularities in the return distribution that are not representative of normal market dynamics. Currency crises often unfold over multiple years rather than single-year shocks, so a multi-year filter is more effective at identifying and excluding crisis periods. It is unlikely that a typical investor would maintain exposure to a foreign market experiencing such extreme sustained currency depreciation, so excluding these episodes makes the simulated return paths more realistic for retirement planning purposes. However, we choose to retain these episodes in the domestic return series when they occur in the home country, as they are part of the historical experience that a domestic investor would have faced and cannot be avoided through diversification.

Figure 27 displays the time periods during which each JST country satisfies the eligibility criteria for domestic, foreign equity, and foreign bond status. Panel (a) shows that most countries have continuous domestic eligibility from the mid-20th century onward. There are some gaps in eligibility for certain countries, the USA, Japan and the UK get excluded at times due to the market-cap constraint, and some countries have shorter data histories that limit their eligibility in earlier years. Germany loses foreign equity eligibility during the hyperinflation episode in the early 1920s, and many other gaps are present due to wars, regime changes, or data quality issues.

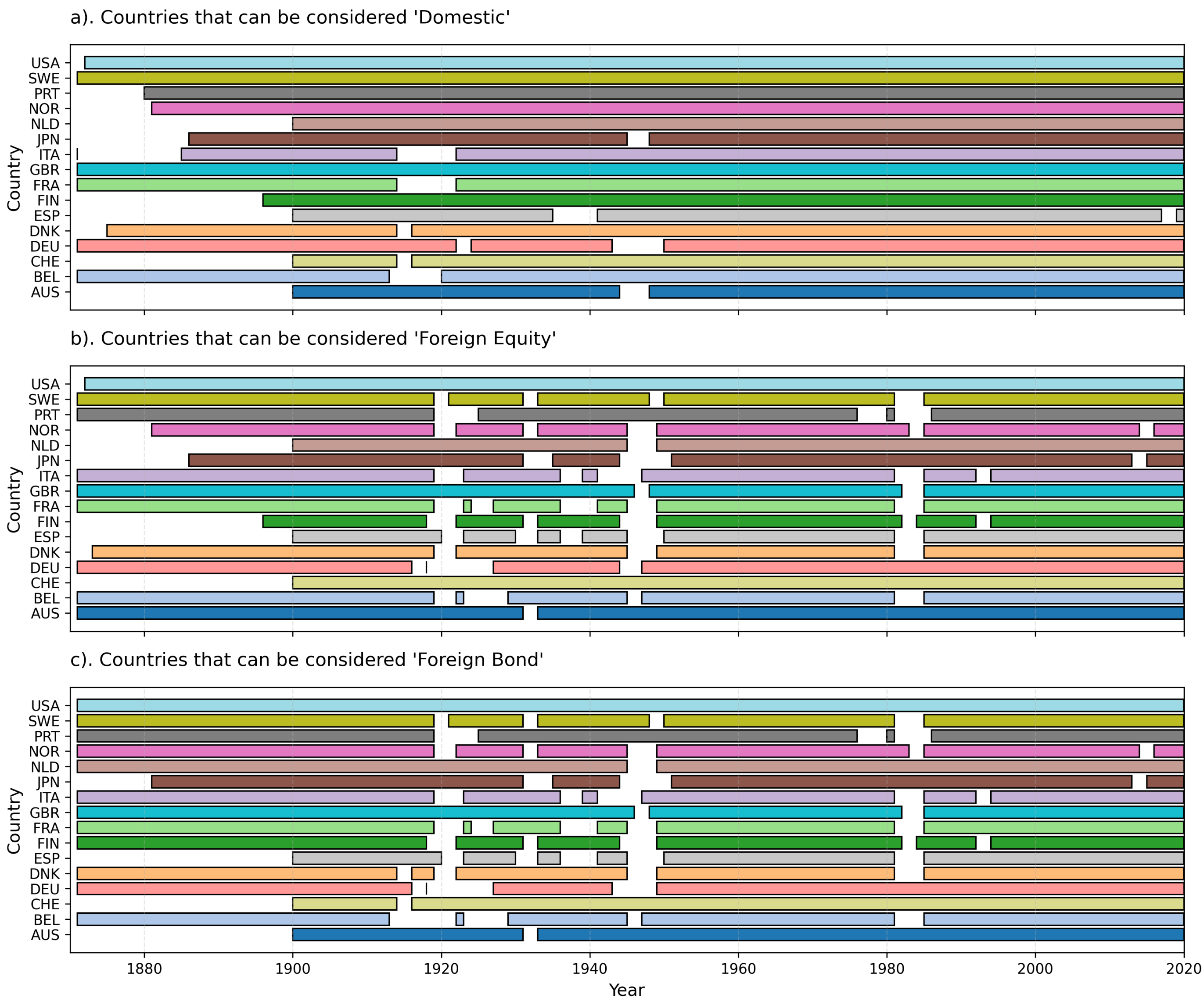

Figure 27: Eligibility of JST countries as domestic markets (a), foreign equity sources (b), and foreign bond sources (c) over time.

C.2 Construction of Investable Universes

Given the eligibility criteria defined above, we construct two investable "universes" that define the valid constituents for the international equity and bond return series in the block bootstrap simulations:

1. **Equities universe:** For each year t, this universe consists of all ordered pairs (i, j) where country i is eligible as domestic and country j is eligible as foreign equity in year t, subject to the constraint that $i \neq j$ (i.e., a country cannot be both the domestic and foreign market in the same bootstrap draw). This universe defines the feasible combinations for constructing the international equity return series from the perspective of domestic country i.

2. **Bonds universe:** Analogously, for each year t, this universe consists of all ordered pairs (i, j) where country i is eligible as domestic and country j is eligible as foreign bonds in year t, with $i \neq j$. This universe defines the feasible combinations for constructing the international bond return series.

At each iteration of the block bootstrap, we randomly select a domestic country i from those eligible in the starting year of the block, then construct foreign equity and bond return series as weighted averages of the returns from all countries j in the corresponding universe for that domestic country and year. This approach ensures that the simulated return paths reflect the diversification opportunities and correlation structure available to the domestic investor, while also preserving the time-varying composition of the global investment opportunity set.

D Block-Bootstrapping: Equity Market Weights

Following Anarkulova et al. (2025), our implementation of the block bootstrap method for simulating returns relies on constructing a market-capitalisation-weighted aggregate return series for international securities. To do this, we need to construct a timeseries of market capitalization weights for the countries in the JST database. This is necessary because the JST database does not include equity market capitalization weights.

We define equity market capitalization weights as the ratio of a country's total market capitalization to the total market capitalization of the country cohort. To construct the baseline capitalization weights, we combine data from the World Bank's World Development Indicators (WDI) database and historical estimates from Global Financial Data (GFD). The World Bank provides market capitalization of listed domestic companies (in inflation-adjusted US Dollars) from the World Federation of Exchanges database (World Federation of Exchanges, 2024). This dataset covers 1975–2024 with good coverage for most developed economies. The data is publicly available under a CC BY-4.0 license and represents the most authoritative source of the two used in our analysis for market capitalization figures. However, the WDI dataset does not cover the historical period of interest for our analysis (1870–1975), so we supplement it with estimates from GFD, which provides long-run historical estimates of market capitalization weights for major economies. This is the dataset leveraged by Anarkulova et al. (2025) to construct their international equity return series. However, we do not have direct access to the GFD database, so we rely on a published chart from Finaeon's website (sourced from Taylor, 2023) that displays market capitalization weights a number of countries from 1782 to 2022. We extract approximate values from this chart using an image processing method that counts pixels of different colors corresponding to each country. This method is subject to measurement error and should be interpreted with caution, but it provides a rough estimate of the historical market capitalization weights for the countries in the JST database.

The World Bank data processing is straightforward and follows the following wrangling procedures:

1. Load the raw CSV file containing market capitalization values in current US$ for all countries
2. Filter to include only the JST countries
3. For each year, calculate the total market capitalization across all JST countries
4. Compute each country's weight as its market cap divided by the total
5. Apply a 2% minimum weight floor to ensure diversification
6. Re-normalize weights to sum to 1.0 after applying the floor

Again, with no direct access to the GFD database, we rely on a published chart from Taylor (2023) that displays market cap percentages as a stacked area graph. The extraction procedure involves the following steps:

1. Crop the chart image to isolate the plotting area
2. Map pixel x-coordinates to years using the known chart range (1782–2022)
3. For each year, identify pixels corresponding to each country color
4. Calculate the vertical percentage occupied by each color
5. Apply a centered 3-year rolling average to smooth pixel-counting noise
6. Distribute the unclassified "Other" percentage evenly across all the JST countries.
7. Apply forward-fill for missing values

This introduces several sources of uncertainty and measurement error. Because we extract percentages by analyzing the pixel colors in the chart image, each country is represented by a distinct color, and we count pixels of each color at each year's x-coordinate position. This approach is fundamentally limited by the chart resolution and image quality, interpolation between discrete x-axis positions, anti-aliasing and edge effects at color boundaries, and potential color bleeding or compression artifacts. For these reasons, the pixel-counting method produces noisy estimates. To mitigate this, we apply a 3-year centered rolling average to reduce this noise, but which may over-smooth genuine historical variation.

Furthermore, the original chart displays only five countries relevant to the JST data (USA, UK, Germany, France, and Japan), with all other countries aggregated into an "Other" category. To construct weights for all JST countries, we distribute this unclassified mass evenly across the remaining 13 countries. This is a strong and likely unrealistic assumption, as it ignores the actual variation in market capitalizations among these countries.

Because we do not have access to the original GFD database, we also cannot validate our extracted values against the true underlying data. The accuracy of our historical estimates is therefore unknown, but what data we have extracted roughly aligns with the World Bank data in the overlapping period (see Figure 28). The extracted GFD data shows reasonable aggregate patterns (e.g., U.S. dominance in the mid-20th century, rising Japanese share in the 1980s), but individual country estimates, particularly for the "Other" countries, should be interpreted with considerable caution. However, because the goal is to capture broad historical trends rather than precise country-level weights, we consider this approach acceptable for our purposes.

The final, combined dataset of weights prioritizes the more reliable World Bank data where available and fills historical gaps with the GFD estimates. Figure 28 displays the three datasets side-by-side: GFD extracted estimates (panel a), World Bank data (panel b), and the combined final dataset (panel c). The overlap period (1988–1989) shows reasonable consistency between the two sources.

Users of this dataset should be aware of several important caveats. The market cap weights extracted from the GFD data should not be interpreted as precise measurements. They are sufficient for characterizing broad historical trends and the construction of a proxy index, but are unsuitable for applications requiring exact country-level weights. The even redistribution of the unclassified percentage across all JST countries is a strong simplifying assumption. In reality, countries like Australia, Canada, and Switzerland likely had substantially different market capitalizations than smaller European economies. The GFD and World Bank datasets may differ in their treatment of market entries, exits, and index composition. These methodological differences are absorbed into our combined series but are not explicitly documented. Ideally, historical market cap weights would be constructed from direct access to the GFD database or from alternative sources such as the Dimson-Marsh-Staunton Global Returns Database (as used by Estrada (2014, 2015)). However, these sources were not available for this study.

Despite these limitations, we believe the combined dataset provides a reasonable approximation of historical market capitalization weights for the JST countries and is suitable for the portfolio construction and simulation exercises undertaken in this research. The primary analyses focus on aggregate portfolio behavior rather than individual country dynamics, which reduces the impact of country-level measurement error. Nevertheless, readers should interpret historical results (particularly those prior to 1988) with appropriate caution regarding the data quality constraints described above.

All data processing code, quality checks, and intermediate outputs are available in the project repository to facilitate transparency and replication.

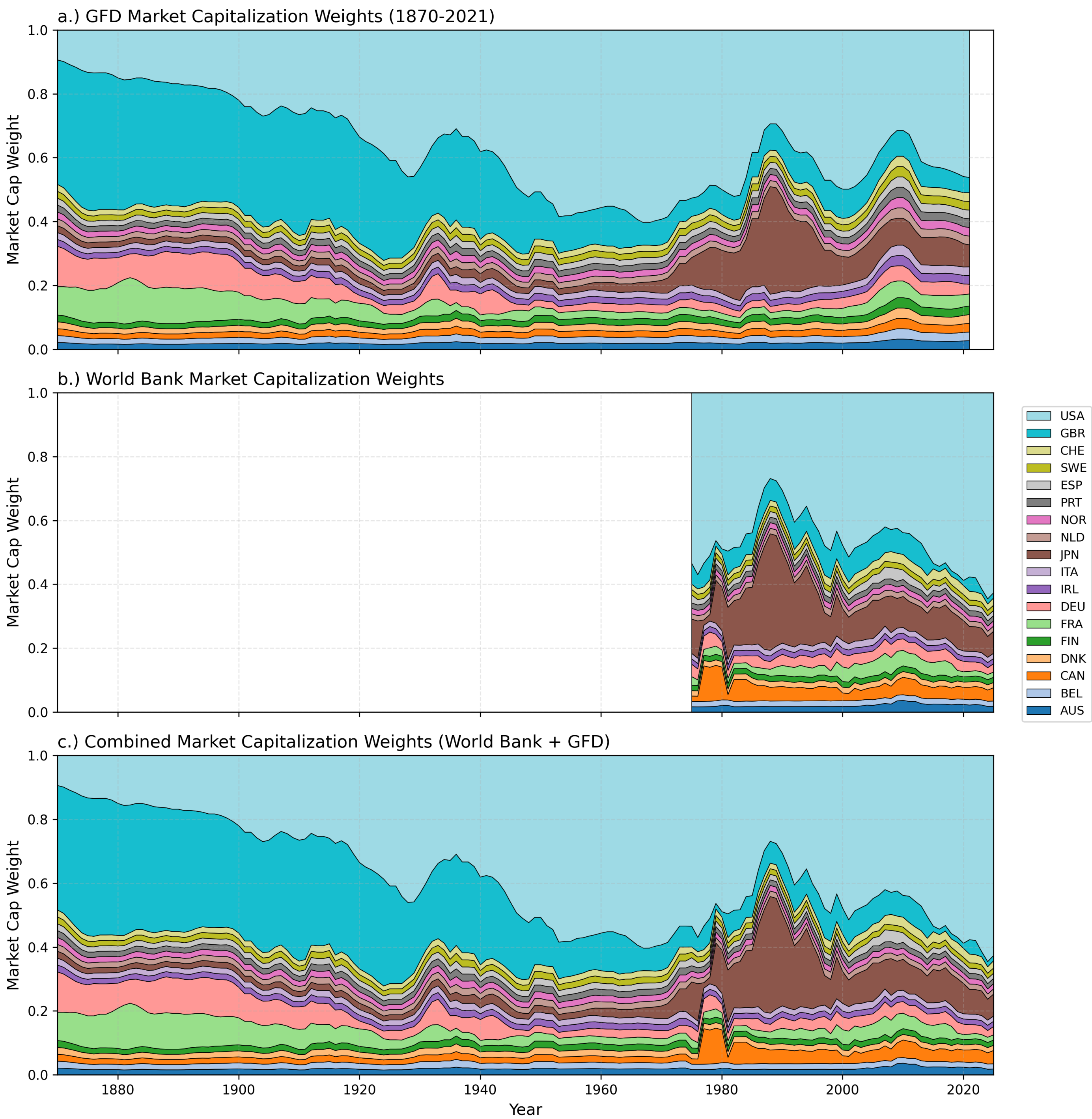

Figure 28: Market capitalization weights for JST countries from three sources. Panel (a) shows estimates extracted from the GFD historical chart (1870–2021), panel (b) shows World Bank data (1988–2024), and panel (c) shows the combined dataset using World Bank data where available and GFD estimates for earlier periods. Colors are consistent across panels to facilitate comparison.

E Block Bootstrapping: Debt Market Weights

Unlike Anarkulova et al. (2025), investors in our analysis have access to a domestic and international bond market. To construct the return series for the international bond component, we need to determine the appropriate weights for the bonds of the different countries in the JST database. Fortunately, the JST database includes a GDP and debt-to-GDP ratio for each country, which allows us to estimate the total market capitalization of each country's bond market. We then compute the weights for the international bond component as the ratio of each country's bond market capitalization (converted to USD using exchange rates from the JST database) to the total bond market capitalization across all countries. This method is a simplification, as it assumes that the bond market capitalization is directly proportional to GDP and debt levels, which may not hold in practice due to differences in financial development, government borrowing needs, and investor preferences. However, it provides a reasonable approximation for our purposes given the data constraints. The resulting bond market capitalization weights are shown in Figure 29.

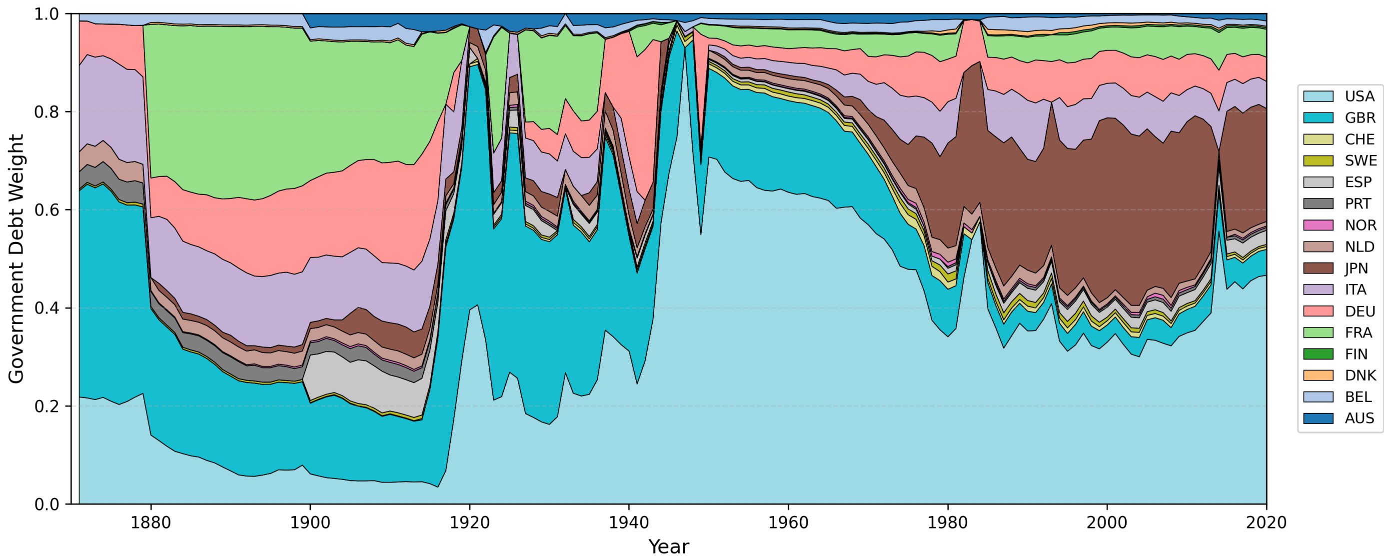

Figure 29: Estimated bond market capitalization weights for JST countries. We compute these weights using GDP and debt-to-GDP ratios from the JST database, converting to USD using the provided exchange rates.

F Block Bootstrapping: Asset Class Returns

To construct the return series for the three asset classes (Cash, Bonds, Equities) and the inflation rate for each country, we use the following variables from the JST database:

1. `bill_rate`: This variable represents the short-term government bill rate for each country. We use this as the return series for the Cash asset class.

2. `bond_tr`: This variable represents the total return on government bonds for each country (in its local currency). It is calculated by JST as $r_t = \frac{p_t + c_t}{p_{t-1}} - 1$, where p_t is the price of the bond at time t and c_t is the coupon payment received at time t. We use this as the return series for the Bonds asset class.

3. `eq_tr`: This variable represents the total return on equities for each country (in its local currency). It is calculated by JST as $r_t = \frac{p_t + d_t}{p_{t-1}} - 1$, where p_t is the price of the equity index at time t and d_t is the dividend payment received at time t. We use this as the return series for the Equities asset class.

4. `cpi`: This variable represents the annual inflation rate for each country, calculated as the percentage change in the Consumer Price Index (CPI) from one year to the next. We use this variable to compute real returns for each asset class by adjusting the nominal returns for inflation.

5. `xrusd`: This variable represents the exchange rate of the country's currency against the US dollar.

These asset class returns are not totally analogous to the asset classes used in the Consilium model portfolios. Bonds in the JST database are government bonds, whereas the Consilium portfolios include a mix of government and corporate bonds. However, given the data constraints and the focus on capturing broad return dynamics rather than precise asset class replication, we consider the JST bond returns to be a reasonable proxy for the fixed-income component of the portfolios. Similarly, the equity returns in the JST database are based on national stock market indices, which may differ from the global equity exposure in the Consilium portfolios. Additionally, Consilium's international equity exposure includes emerging markets, which are not represented in the JST database. While at some points countries in the JST database may have had emerging market characteristics, we acknowledge that our international equity return series may not fully capture the risk-return profile of a global equity portfolio with emerging market exposure. However, the JST equity returns still provide valuable insights into the historical performance of equities across different countries and time periods, which is relevant for our analysis of retirement portfolio strategies.

F.1 Foreign Return Construction and Currency Considerations

While Cash will comprise entirely of domestic bills, the Bonds and Equities asset classes will be constructed as weighted averages of the returns from eligible foreign countries, with both currency hedged and unhedged components (for equities). All returns in the JST database are reported in nominal local currency terms. To construct return series from the perspective of a domestic investor in country i, we must account for exchange rate movements when investing in foreign assets.

We introduce the following notation to clarify the currency conversions:

- $r^j_{j,t}$: Total return on an asset in country j at time t, denominated in country j's local currency (as reported in the JST database)

- $x_{j,t}$: Exchange rate return for country j's currency relative to USD at time t

- $r^{\text{USD}}_{j,t}$: Total return on an asset in country j, converted to USD

- $r^i_{j,t}$: Total return on an asset in country j, from the perspective of an investor in country i (denominated in country i's currency)

The exchange rate variable `xrusd` in the JST database represents the exchange rate from USD to local currency. We compute the exchange rate return for country j as the percentage change in this exchange rate:

$$x_{j,t} = \frac{\texttt{xrusd}_{j,t}}{\texttt{xrusd}_{j,t-1}} - 1 \tag{33}$$

This represents the return (or loss) from holding USD relative to country j's local currency. A positive $x_{j,t}$ indicates depreciation of country j's currency against the USD (this is what we use in Appendix C to exclude extreme depreciation episodes), while a negative $x_{j,t}$ indicates appreciation.

To convert a foreign asset's local currency return to USD terms, we account for both the asset return and the exchange rate movement. The total return in USD for an asset in country j is:

$$r_{j,t}^{\mathrm{USD}} = (1 + r_{j,t}^{j})(1 + x_{j,t})^{-1} - 1 \tag{34}$$

This formulation shows that if country j's currency depreciates against the USD (positive $x_{j,t}$), the USD-denominated return will be lower than the local currency return, as the investor experiences a currency loss when converting back to USD.

From the perspective of a domestic investor in country i, we must further convert the USD return to country i's local currency. The unhedged return on a foreign asset in country j from country i's perspective is:

$$r_{j,t}^{i} = (1 + r_{j,t}^{\mathrm{USD}})(1 + x_{i,t})^{-1} - 1 = \frac{(1 + r_{j,t}^{j})}{(1 + x_{j,t})(1 + x_{i,t})} - 1 \tag{35}$$

This expression captures the complete currency conversion chain: starting with country j's local currency, converting to USD, and then converting to country i's local currency. The investor experiences both the asset return and the net exchange rate movement between the two currencies.

For the currency-hedged return, we assume perfect hedging that eliminates all currency risk. In this case, the investor receives only the local currency return, expressed in their domestic currency terms:

$$r_{j,t}^{i,\mathrm{hedged}} = r_{j,t}^{j} \tag{36}$$

In practice, currency-hedged returns would also include the cost of hedging (typically related to the interest rate differential between countries). For simplicity, and given data constraints, we assume this cost is negligible and that hedging is perfectly efficient in removing currency risk. This assumption is reasonable for developed markets with liquid forward currency markets, which characterizes most of the countries in the JST database during the period of our analysis.

F.2 Foreign Return Weighting and Aggregation

To construct the international component of each asset class, we aggregate returns from all eligible foreign countries using market-capitalization-based weights. For equities, we use the equity market capitalization weights described in Appendix D; for bonds, we use the government debt weights from Appendix E.

Because the eligible universe varies across time (as countries gain or lose eligibility) and across domestic perspectives (due to the constraint that $i \neq j$), we must renormalize the weights at each time step for each domestic country. For a domestic investor in country i investing in asset class k at time t, let $\mathcal{F}_{i,t}^{(k)}$ denote the set of countries eligible as foreign sources for that asset class. The renormalized weight for foreign country j is then:

$$w_{i,j,t}^{(k)} = \frac{w_{j,t}}{\sum_{\ell \in \mathcal{F}_{i,t}^{(k)}} w_{\ell,t}} \tag{37}$$

Where $w_{j,t}$ is the baseline market capitalization weight for country j at time t (before filtering for eligibility), and the denominator sums over all countries ℓ that are eligible foreign sources for domestic country i and asset class k at time t. This renormalization ensures that the weights sum to exactly 1.0 across the eligible universe.

The aggregated international return for asset class k from the perspective of a domestic investor in country i is then the weighted average:

$$r_{k,t}^{i,\mathrm{intl}} = \sum_{j \in \mathcal{F}_{i,t}^{(k)}} w_{i,j,t}^{(k)} \cdot r_{j,t}^{i} \tag{38}$$

Where $r_{j,t}^{i}$ is the return on asset class k in country j, converted to country i's currency as described in equations 34 and 35. This aggregation preserves the empirical correlation structure among foreign markets while reflecting the time-varying composition of the global investment opportunity set.

F.3 Constructing Asset Class Returns

In our block-bootstrapping framework, each asset class k is characterized by two parameters: a Home Bias $\omega^{(k)} \in [0,1]$ and a Hedging Ratio $\eta^{(k)} \in [0,1]$. The home bias determines the allocation between domestic and foreign markets within that asset class, while the hedging ratio determines what fraction of the foreign exposure is currency-hedged versus unhedged. These parameters allow us to construct asset class returns that reflect realistic portfolio compositions for New Zealand investors.

The hedged international return is computed analogously to the unhedged return in equation 38, but using local currency returns without currency conversion:

$$r_{k,t}^{i,\text{hedged}} = \sum_{j \in \mathcal{F}_{i,t}^{(k)}} w_{i,j,t}^{(k)} \cdot r_{j,t}^{j} \tag{39}$$

Where $r_{j,t}^{j}$ is the asset return in country j's local currency (i.e., the same return that appears in equation 36). This represents the return from a perfectly currency-hedged foreign investment.

The final return for asset class k from the perspective of a domestic investor in country i at time t combines domestic, hedged foreign, and unhedged foreign components:

$$r_{k,t}^{i} = \omega^{(k)} \cdot r_{k,t}^{i,\text{home}} + (1 - \omega^{(k)}) \cdot \left[\eta^{(k)} \cdot r_{k,t}^{i,\text{hedged}} + (1 - \eta^{(k)}) \cdot r_{k,t}^{i,\text{intl}}\right] \tag{40}$$

Where $r_{k,t}^{i,\text{home}}$ is the return on asset class k in the domestic country i (e.g., `bill_rate` for Cash, `bond_tr` for Bonds, `eq_tr` for Equities), $r_{k,t}^{i,\text{hedged}}$ is the currency-hedged international return from equation 39, and $r_{k,t}^{i,\text{intl}}$ is the unhedged international return from equation 38 which includes currency effects. For the purposes of block bootstrapping, we treat CPI inflation as an asset class in its own right with a 100% home bias and no foreign component.

G Block Bootstrapping: Sampling Procedure

Having constructed the asset class returns for each eligible domestic country and year in Appendix F, we now describe the procedure for generating bootstrapped return paths. Our implementation follows in the spirit of Anarkulova et al. (2025) but includes some adjustments. The key idea is to sample blocks of consecutive years from the historical data, where each block corresponds to a continuous period of data for a single country. By concatenating these blocks, we create synthetic return paths that preserve the short and medium-term autocorrelation structure of returns while also capturing the cross-sectional correlation across countries and asset classes.

The first step is to identify continuous time periods within each country's history. As shown in Figure 27, many countries have gaps in their eligibility due to wars, data quality issues, or currency crises. We cannot simply sample random year ranges, as this would risk drawing blocks that span discontinuities where no data exists.

We identify continuous periods by sorting the domestic mask by country and year, then flagging years where the gap from the previous year exceeds one. A cumulative sum of these flags creates period groups within each country. For each continuous period p, we record the country identifier (iso_p), start year (y_p^{start}), end year (y_p^{end}), and period length ($L_p = y_p^{\mathrm{end}} - y_p^{\mathrm{start}} + 1$).

We then create a global index that maps each year of data to a single sequential index $s \in \{1, 2, \ldots, S\}$, where $S = \sum_p L_p$ is the total number of country-year observations across all periods. Each period p occupies indices from s_p^{start} to s_p^{end} in this global sequence.

To generate a single bootstrap path of length $T = 40$ years, we repeatedly draw random blocks and concatenate them until we have accumulated the required number of observations. For each block b, we:

1. Draw a random starting index $s_b \sim \mathrm{Uniform}\{1, 2, \ldots, S\}$ from the global index
2. Draw a block length $\ell_b \sim \mathrm{Poisson}(\lambda = 10)$ where the Poisson parameter $\lambda = 10$ sets the mean block length to 10 years
3. Identify which period p contains index s_b
4. Compute the actual block end as $s_b^{\mathrm{end}} = \min(s_b + \ell_b - 1, s_p^{\mathrm{end}})$ to prevent the block from extending beyond the period boundary
5. Extract the corresponding years from period p: if s_b corresponds to year y within period p, then the block spans years $y, y+1, \ldots, y + (s_b^{\mathrm{end}} - s_b)$

This process ensures that each block respects the continuous history of a single country, preventing artificial discontinuities in the bootstrapped series.

Our implementation uses a Poisson distribution for block lengths, whereas Anarkulova et al. (2025) use a geometric distribution. Both choices target a mean block length of approximately 10 years, but they differ in their theoretical properties and practical implications. The practical impact on our analysis is minimal. Both distributions preserve the medium-run autocorrelation structure that is the primary motivation for block bootstrapping. We chose the Poisson distribution primarily for computational convenience: it is straightforward to implement and produces non-negative integer block lengths directly without requiring rejection sampling or other adjustments. This pragmatic choice does not materially affect the properties of the bootstrap procedure or the substantive conclusions of our analysis.

Blocks are concatenated sequentially until we reach T years. If a block would cause the total to exceed T, we truncate it. The result is a sequence of (i_t, y_t) pairs for $t = 1, \ldots, T$, where i_t is the domestic country and y_t is the year for observation t in the bootstrapped path.

For each (i_t, y_t) pair, we retrieve the corresponding asset class returns from the processed data created in Appendix F. This gives us a $T \times 4$ matrix of returns for the path, with columns for Equities, Bonds, Bills, and CPI (in that order).

To generate a large number of paths efficiently, we pre-generate N block draws, where $N = 2{,}000{,}000$ in our implementation. We then partition this sequence into non-overlapping segments of length T to create approximately $N/T \approx 450{,}000$ independent paths. Each path represents a plausible 40-year sequence of annual returns that preserves the short-run autocorrelation and cross-sectional correlation structure of the historical data.

The final output is a three-dimensional array of shape ($N_{\mathrm{paths}} \times T \times 4$) where:

- Dimension 1 indexes the path (simulation)
- Dimension 2 indexes the year within the path
- Dimension 3 indexes the asset class (Equities, Bonds, Bills, CPI)

This array is saved to disk for use in the Monte Carlo simulations described in the main analysis.